AGR 304 PRINCIPLES OF ORGANIC FARMING (1+1)



B.Sc., (Hons) Agriculture – VI semester (2020 batch)

Theory Study Material

Course Teacher

Dr. P. KISHORE KUMAR,
Assistant Professor (Agronomy)

2023

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Theory:

Unit - I: Components and Principles of Organic Cotton

Organic farming: Definition - Scope - principles and concepts - history of organic farming - global scenario - biodiversity: importance and measure to preserve biodiversity - pre requisites for Organic farming:- Soil organic carbon: status and improvement strategies.

Unit - II: Organic sources of Nutrients

Organic sources of nutrients - manures and other inputs - on farm and off farm sources - organic waste recycling - methods - Soil and crop management - inter cropping, crop rotation, green manures, cover crops, mulching - bio fertilizers.

Unit - III: Non - Chemical weed and Pest disease management

Non-chemical weed management methods: preventive, physical, cultural, mechanical and biological measures - Bio-intensive pest and disease management.

Unit - IV: Indigenous Technical Knowledge (ITK)

Indigenous Technical Knowledge (ITK) in organic agriculture - scientific rationale - soil, nutrient, weed, water, management - prospects and problems in organic farming.

Unit - V: Certification of label

Organic certification - NPOP guidelines - Certification agencies in India - crop production standards - Quality considerations - labeling and accreditation process - marketing and export opportunities.

Lecture Schedule:

1. Organic farming; definition - prospects - principles and concepts - History and genesis of organic farming in World and India: Present status in World, India and Tamil Nadu.
2. Introduction to bio - diversity; importance and measures to preserve bio - diversity.
3. Pre-requisites and basic steps for organic farming; conversation to organic farming - planning and processes in practices - IFS approach - Integration of animal components.
4. Organic carbon; status and improvement strategies - conservative tillage systems.
5. Sources of organic manures - plant, animal and microbial origin - on - farm resources; FYM, green manures, crop residues, poultry manure, sheep and goat manures, biogas slurry and vermicompost.
6. Off-farm resources; coir pith, press mud, oilcakes, flyash, bio compost, minerals, bone meal, bio fertilizers, traditional preparations.
7. Organic waste recycling methods and techniques - composting, vermicomposting, *in situ* composting - system approach.
8. Soil and crop management in organic farming; Inter cropping and companion planting, crop rotation green manures and cover crops, mulching.
9. **Mid semester examination**
10. Weeds - Ecology - habitat management of weeds - Non - chemical weed management methods; preventive, physical, cultural, use of tools and implements and biological measures - good crop husbandry practices.
11. Integrated pest and diseases management - bio control agents, bio rational pesticides; minerals, botanicals, soaps, trap crops, bird perches, and traditional preparations - sanitation.

12. Indigenous technical knowledge (ITK) in organic agriculture - rationale and principles - general, indigenous practices for soil, nutrient, weed, water pest and disease management in farming - ITK's in farmers practice.
13. Benefits and problems in organic farming.
14. Organic farming; Promotional activities; role of government and NGO's - action plan - policy considerations.
15. Economic evaluation of organic production systems - cost - benefit analysis and comparison with conventional systems.
16. Organic certification - procedures - certification agencies in India - labeling, marketing and export opportunities.
17. Crop production standards - NPOP guidelines - principles, recommendations and standards - Quality considerations - assessment methods - premium and export opportunities.

Practical Schedule:

1. Resource inventory of organic farm- Soil sampling and analysis for organic carbon and pesticide residues / contaminants.
2. Raising of green manures (Sunnhemp / Daincha / Fodder cowpea).
3. Incorporation of green manure - seed treatment and raising of field crop (Rice / Maize / Cowpea / Cotton / Gingelly).
4. Hands on practice on preparatory cultivation; soil and water conservation methods.
5. Hands on experience on recycling techniques; bio-composting and vermicomposting.
6. Quantification of nutrients from organic sources and application of manures and bio-fertilizers.
7. Exposure visit to an organic farm to learn ITK based preparations.
8. Organic crop production and weed management.
9. Skill development in composting farm residues.
10. Organic crop production and pest management.
11. Exposure visit to bio-control agent (*Pseudomonas, Trichoderma* etc.,) production units.
12. Organic crop production and diseases management.
13. Skill development in vermicompost preparation.
14. Hands on training on grading, packaging and post-harvest management.
15. Exposure visit to organic market out lets.
16. Exposure visit to organic certification agencies / Directorate of Organic Certification, Tamil Nadu.
17. **Practical Examination**

References:

Dahama, A.K.2009. Organic farming for sustainable agriculture, Agrobros publishers.
SP. Palaniappan and K Annadurai. 2008. Organic Farming: Theory and Practice. 2008. Scientific Publishers.
Panda, S.C. 2012. Principles and Practices of Organic Farming. Agribios (India), Jodhpur.
Gehlot, D. 2010. Organic Farming- Components and Management. Agribios (India), Jodhpur.
Dushyant Gehlot . 2010. Organic farming: Components and management. Agrobios (India), Jodhpur.
Ranjan Kumar Biswas.2014. Organic farming in India. N.D.Publishers.New Delhi.

E:References:

1. www.ifoam.org
2. www.apeda.org
3. www.cowindia.org
4. www.ncof.org
5. www.earthfooda.co.uk,
6. www.newfarm.org/training

Lecture-1

Organic farming- Definition-Prospects-Principles and concepts- Genesis and History of organic farming in the world and India: Present status in world, India and Tamil Nadu

Introduction

Green revolution technologies such as greater use of synthetic agro chemicals like fertilizers and pesticides, adoption of nutrient responsive, high-yielding varieties of crops, greater exploitation of irrigation potentials etc. has boosted the production output in most of cases. Without proper choice and continues use of these high energy inputs is leading to decline in production and productivity of various crops as well as deterioration of soil health and environments. The most unfortunate impact on Green Revaluation Technology (GRT) not only on Indian Agriculture but also the whole world is as follows:

1. Change in soil reaction
2. Development of nutrient imbalance/deficiencies
3. Damage the soil flora and fauna
4. Reduce the earth worm activity
5. Reduction in soil humus/organic matter
6. Change in atmospheric composition
7. Reduction in productivity
8. Reduction in quality of the produce
9. Destruction of soil structure, aeration and water holding capacity

All these problems led to reduction in productivity, deterioration of soil health as well as natural eco-system. Moreover, today the rural economy is now facing a challenge of over dependence on synthetic inputs and day by day change in the price of these inputs. Further the world will face the market competition due to globalization of trade as per World Trade Organization (WTO). Thus apart from quantity, quality will be the important factor. Such as Agriculture gave birth to various new concepts of farming such as organic farming, natural farming, bio-dynamic Agriculture, do-nothing agriculture, eco-farming etc.

The essential concept of the practices is "Give back to nature", where the philosophy is to feed the soil rather them the crop to maintain the soil health. Therefore, for sustaining healthy ecosystem, there is need for adoption of an alternatives farming system like organic farming.

DEFINITIONS

As per the definition of the USDA (1980) study team on organic farming "*organic farming is system which avoids or largely excludes the use of synthetic inputs (such as fertilizers, pesticides, hormones, feed additives etc) and to the maximum extent feasible rely upon crop rotations, crop residues, animal manures, off-farm organic waste, mineral grade rock additives and biological system of nutrient mobilization and plant protection*".

In another definition FAO(1999) suggested that "*Organic agriculture is a unique production management system which promotes and enhances agro-ecosystem health, including biodiversity, biological cycles and soil biological activity, and this is accomplished by using on-farm agronomic, biological and mechanical methods in exclusion of all synthetic off-farm inputs*".

Organic agricultural methods are internationally regulated and legally enforced by many nations, based in large part on the standards set by the **International Federation of Organic Agriculture Movements (IFOAM)**, an international umbrella organization for organic organizations established in 1972.

IFOAM describes "Organic Agriculture is a production system that sustains the health of **soils, ecosystems** and **people**. It **relies on ecological processes, biodiversity and cycles adapted to local conditions**, rather than the use of inputs with adverse effects. Organic Agriculture combines **tradition, innovation** and **science** to benefit the shared environment and promote **fair relationships** and a good **quality of life** for all involved."

The concept of the soil as a living system which must be"fed" in a way that does not restrict the activities of beneficial organisms necessary for recycling nutrients and producing humus is central to this definition.

Principles of Organic farming

The International Federation for Organic Agriculture Movement's (IFOAM) definition of Organic agriculture is based on:

- ✓ The principle of health
- ✓ The principle of ecology
- ✓ The principle of fairness and
- ✓ The principle of care

The principles are to be used as a whole. They are composed as ethical principles to inspire action.

1.Principle of health

Organic Agriculture should sustain and enhance the health of soil, plant, animal, human and planet as one and indivisible.

2.Principle of ecology

Organic Agriculture should be based on living ecological systems and cycles, work with them, emulate them and help sustain them.

3.Principle of fairness

Organic Agriculture should build on relationships that ensure fairness with regard to the common environment and life opportunities. Fairness is characterized by equity, respect, justice and stewardship of the shared world, both among people and in their relations to other living beings.

4.Principle of care

Organic Agriculture should be managed in a precautionary and responsible manner to protect the health and well-being of current and future generations and the environment.

Organic agriculture is a living and dynamic system that responds to internal and external demands and conditions. Practitioners of organic agriculture can enhance efficiency and increase productivity, but this should not be at the risk of jeopardizing health and well-being. Consequently, new technologies need to be assessed and existing methods reviewed. Given the incomplete understanding of ecosystems and agriculture, care must be taken.

Concepts of organic farming

- Use of on- farm resources and avoiding / minimizing the use of off – farm resources.
- Integrating farm resources for reducing the use of external inputs
- Soil health management using farm waste (crop, weed and animal wastes)
- Generating feed resources in the farm: by growing forage crops and fodder (shrub and trees)
- Green biomass generation for soil fertility management by growing high foliage producing plants and trees.
- Composting of crop residues and livestock manures: with the help of earth worms converting them into value added vermi compost.

- Preparing plant growth promoting and disease tolerant/ controlling substances (like Panchagavya, Coconut milk slurry, Amudhakaraisal *etc.*,)
- Use of eco friendly pest management / crop protection measures (like use of predators and parasites, herbal extracts, crop diversification etc.,)

Objectives of organic farming

1. To produce food of high nutritional quality in sufficient quantity
2. To interact in a constructive and life enhancing way with all natural systems and cycles
3. To encourage and biological cycles with in the farming system, involving micro-organisms, soil flora and fauna, plants and animals and careful mechanical intervention
4. To maintain and increase long-term fertility of soils
5. To promote the healthy use and proper care of water, water resources and all life therein
6. To help in the conservation of soil and water
7. To use, as far as is possible, renewable resources in locally organized agricultural systems
8. To work, as far as possible, within a closed system with regard to organic matter and nutrient elements
9. To work, as far as possible, with materials and substances which can be reused or recycled, either on the farm or elsewhere
10. To give all livestock conditions of life which allow them to perform the basic aspects of their innate behaviour
11. To maintain all forms of pollution that may result from agricultural practices
12. To maintain the genetic diversity of the production system and its surroundings including the protection of wild life habitats
13. To allow everyone involved in organic production and processing a quality of life confirming to the UN Human Rights Charter, to cover their basic needs and obtain an adequate return and satisfaction from their work, including a safe working environment
14. To consider the wider social and ecological impact of the farming system
15. To produce non-food products from renewable resources, which are fully degradable

16. Weed, disease and pest control relaying primarily on croprotation, natural predators, diversity, organic manuring, resistant varieties, and limited (preferably minimal)thermal, biological and chemical intervention
17. To create harmonious balance between crop production and animal husbandry
18. To encourage organic agriculture associations to function along democratic lines and the principle of division of powers
19. To progress towards an entire production, processing and distribution chain which is both socially just and ecologically responsible

Components of organic farming

Essential components of organic farming are keeping the soil alive through effective management natural resources. They are as follows

- **Enrichment of soil:** Abandon use of chemicals, use crop residue as mulch, use organic and biological fertilizers, adopt crop rotation and multiple cropping, avoid excessive tilling and keep soil covered with green cover or biological mulch.
- **Management of temperature**: Keep soil covered, plant trees and bushes on bund
- **Conservation of soil and rain water:** Dig percolation tanks, maintain contour bunds in sloppy land & adopt contour row cultivation, dig farm ponds, maintain low height plantation on bunds.
- **Harvesting of sun energy:** Maintain green stand throughout the year through combination of different crops and plantation schedules.
- **Self-reliance in inputs:** Develop your own seed, on-farm production of compost, vermicompost, vermiwash, liquid manures and botanical extracts. Preparing plant growth promoting and disease tolerant/ controlling substances (like Panchagavya, Coconut milk slurry, Amudhakaraisal *etc*.,)
- **Maintenance of life forms**: Develop habitat for sustenance of life forms, never use pesticides and create enough diversity.
- **Integration of animals**: Animals are important components of organic management and not only provide animal products but also provide enough dung and urine for use in soil.
- **Use of renewable energy**: Use solar energy, bio-gas and other eco-friendly machines.

History of Organic Farming

Pre-World War II

- The first 40 years of the 20th century saw simultaneous advances in biochemistry and engineering that rapidly and profoundly changed farming. The introduction of the gasoline-powered internal combustion engine ushered in the era of the tractor and made possible hundreds of mechanized farm implements. Research in plant breeding led to the commercialization of hybrid seed. And a new manufacturing process made nitrogen fertilizer — first synthesized in the mid-19th century - affordably abundant. These factors changed the labor equation.
- Consciously organic agriculture began more or less simultaneously in Central Europe and India. The British Botanist **Sir Albert Howard** is often referred to as the **father of modern organic agriculture**. From 1905 to 1924, he worked as an agricultural adviser in Pusa, Bengal, where he documented traditional Indian farming practices and came to regard them as superior to his conventional agriculture science. His research and further development of these methods is recorded in his writings, notably, his 1940 book, *An Agricultural Testament*, which influenced many scientists and farmers of the day.
- In Germany, **Rudolf Steiner's** biodynamic agriculture, was probably the **first comprehensive organic farming system**. Steiner emphasized the farmer's role in guiding and balancing the interaction of the animals, plants and soil. Healthy animals depended upon healthy plants (for their food), healthy plants upon healthy soil, healthy soil upon healthy animals (for the manure).
- In 1909, American agronomist F.H. King toured China, Korea, and Japan, studying traditional fertilization, tillage, and general farming practices. He published his findings in *Farmers of Forty Centuries* (1911, Courier Dover Publication). King foresaw a "world movement for the introduction of new and improved methods" of agriculture and in later years his book became an important organic reference.
- **The term *organic farming* was coined by Lord Northbourne** in his book *Look to the Land* (written in 1939, published 1940). From his conception of "the farm as organism," he described a holistic, ecologically balanced approach to farming.
- In 1939, influenced by Sir Albert Howard's work, **Lady Eve Balfour launched the Haughley Experiment** on farmland in England. It was the first scientific, side-by-side comparison of organic and conventional farming. Four years later, she published

The Living Soil, based on the initial findings of the Haughley Experiment. Widely read, it led to the formation of a key international organic advocacy group, the Soil Association.

- In Japan, Masanobu Fukuoka, a microbiologist working in soil science and plant pathology, began to doubt the modern agricultural movement. In 1937, he quit his job as a research scientist, returned to his family's farm in 1938, and devoted the next 60 years to developing a radical no-till organic method for growing grain and many other crops, now known as **Nature Farming (Natural Farming)**, **'do–nothing' farming or Fukuoka farming.**

Post-World War II

- Technological advances during World War II accelerated post-war innovation in all aspects of agriculture, resulting in large advances in mechanization (including large-scale irrigation), fertilization, and pesticides. In particular, two chemicals that had been produced in quantity for warfare, were repurposed for peace-time agricultural uses.
- Ammonium nitrate, used in munitions, became an abundantly cheap source of nitrogen. And a range of new pesticides appeared: DDT, which had been used to control disease-carrying insects around troops, became a general insecticide, launching the era of widespread pesticide use. At the same time, increasingly powerful and sophisticated farm machinery allowed a single farmer to work larger areas of land and fields grew bigger.
- In 1944, an international campaign called the Green Revolution was launched in Mexico with private funding from the US. It encouraged the development of hybrid plants, chemical controls, large-scale irrigation, and heavy mechanization in agriculture around the world. During the 1950s, sustainable agriculture was a topic of scientific interest, but research tended to concentrate on developing the new chemical approaches. In the US, **J.I. Rodale began to popularize the term and methods of organic growing**, particularly to consumers through promotion of organic gardening. In 1962, **Rachel Carson**, a prominent scientist and naturalist, published *Silent Spring*, chronicling the effects of DDT and other pesticides on the environment. A bestseller in many countries, including the US, and widely read around the world, ***Silent Spring*** is widely considered as being a key factor in the US government's 1972 banning of DDT**.**

- The book and its author are often credited with launching the worldwide environmental movement. In the 1970s, global movements concerned with pollution and the environment increased their focus on organic farming. As the distinction between organic and conventional food became clearer, one goal of the organic movement was to encourage consumption of locally grown food, which was promoted through slogans like "Know Your Farmer, Know Your Food".
- In 1972, the **International Federation of Organic Agriculture Movements (IFOAM)** was founded in Versailles, France and dedicated to the diffusion and exchange of information on the principles and practices of organic agriculture of all schools and across national and linguistic boundaries.
- In 1975, **Fukuoka** released his first book, *The **One-Straw Revolution***, with a strong impact in certain areas of the agricultural world. His approach to small-scale grain production emphasized a meticulous balance of the local farming ecosystem, and a minimum of human interference and labor.
- In the 1980s, around the world, farming and consumer groups began seriously pressuring for **government regulation of organic production**. This led to legislation and certification standards being enacted through the 1990s and to date. Since the early 1990s, the retail market for organic farming in developed economies has been growing by about 20% annually due to increasing consumer demand. Concern for the quality and safety of food, and the potential for environmental damage from conventional agriculture, are apparently responsible for this trend.

Twenty-first century

- Throughout this history, the focus of agricultural research and the majority of publicized scientific findings have been on chemical, not organic farming. This emphasis has continued to biotechnologies like genetic engineering. One recent survey of the UK's leading government funding agency for bioscience research and training indicated GM crop projects, and only one related to organic agriculture. This imbalance is largely driven by agribusiness in general, which, through research funding and government lobbying, continues to have a predominating effect on agriculture-related science and policy.
- Agribusiness is also changing the rules of the organic market. The rise of organic farming was driven by small, independent producers and by consumers. In recent years, explosive organic market growth has encouraged the articipation of

agribusiness interests. As the volume and variety of "organic" products increases, the viability of the small-scale organic farm is at risk, and the meaning of organic farming as an agricultural method is separated areas of organic food and organic certification.

- In Havana, Cuba, a unique situation has made as organic food production a necessity. Since the collapse of the Soviet Union in 1989 and its economic support, Cuba has had to produce food in creative ways like instituting the world's only state-supported infrastructure to support urban food production. Called **organopónicos**, the city is able to provide an ever increasing amount of its produce organically.

World Scenario of organic farming

Organic agriculture is developing rapidly and today at least 187 countries produce organic food commercially. A total of 72.3 million hectares were organically managed at the end of 2019. The global sales of organic food and drink reached more than 106 billion euros. The region with the most organic agricultural land is Oceania, with 35.9 million hectares, followed by Europe with 16.5 million hectares, Latin America (8.3 million hectares), Asia (5.9 million hectares), North America (3.6 million hectares) and Africa (2.0 million hectares). Oceania has half of the global organic agricultural land (35.9 million hectares). Europe has the second largest area (23 percent; 16.5 million hectares), followed by Latin America (11.5 percent; 8.2 million hectares). The organic area increased in all continents. In 2019, 3.1 million organic producers were reported. India continues to be the country with the highest number of producers (1,366,226), followed by Uganda (210,352), and Ethiopia (203,602). (Source: FiBL survey 2021)

Marketing chances for organic produce in the world

Organic agricultural produce has a world over market of $106 billion in 2019. India has produced 1.7 million tonnes of certified organic produce during 2017-18. There are a number of uncertified organic products which the small farmers produce for the use of their family and friends. The country has exported 458,000 tonnes of organic produce worth Rs 3,625 crores. It only shows the growing reception for organic farming.

- Cuba has totally shifted to organic farming
- 181 countries practise organic farming, but only 11 countries are using more than 10 per cent of their land mass for organic farming
- Only 1.5 per cent of the total land mass in the world is being used for organic farming

- Japanese farmer and philosopher Masanobu Fukuoka is hailed as the father of organic farming
- Radhelal Herlal Richharia was the first one to raise a voice against chemical fertilisers, hybrid crops and seeds. He was the Director of the Central Rice Research Institute. He preserved 22,972 traditional paddy varieties. Now less than 200 varieties exist

Organic Agriculture in India

In the last 20 years, attention towards organic farming has increased, thanks to relentless campaigning by Subhash Palekar in North India and Nammalvar in South India.

Certified lands under organic cultivation have increased to 14.9 lakh hectares now from 73,000 hectares in 2003. Then only two lakh farmers were involved in organic farming but today 1,599,010 farmers are into organic farming. Sikkim practises cent percent organic farming. Madhya Pradesh with 540993 hectares of land, Maharashtra (219,659 hectares) and Rajasthan (177,599 ha) are the top three States in organic farming. Kerala, Mizoram, Goa, Rajasthan, Meghalaya, Chattisgarh and Gujarat have declared that they would be shifting to organic farming totally.

Tamil Nadu has 14,086 hectares of land certified under organic farming. At present, there are 5,000 farmers who practice certified organic farming. Many of the organic farmers of the state are, however, functioning without certification.

State-wise Cultivated farm area for the Year 2020-21 (Source:APEDA, 2021)				
S. No.	**State Name**	**Organic Area (in ha)**	**Conversion area (in ha)**	**Total area (in ha)**
1	Madhya Pradesh	540993.98	479024.00	1020017.98
2	Maharashtra	219659.41	152063.21	371722.62
3	Rajasthan	177599.56	121086.74	298686.29
4	Gujarat	72318.02	75548.39	147866.41
5	Karnataka	61115.97	33934.11	95050.08
6	Odisha	78148.00	14546.81	92694.81
7	Sikkim	74647.31	1082.34	75729.66
8	Uttarakhand	31556.80	43269.61	74826.40
9	Uttar Pradesh	53194.79	14247.83	67442.61
10	Jharkhand	0.00	53261.70	53261.70
11	Kerala	25656.60	19413.78	45070.38
12	Meghalaya	34816.30	3560.09	38376.39
13	Andhra Pradesh	22343.07	14458.29	36801.36
14	Tamil Nadu	14086.32	17542.74	31629.06
15	Jammu & Kashmir	19028.70	11591.12	30619.82
16	Bihar	4.90	29897.64	29902.54
17	Chhattisgarh	14744.46	8465.06	23209.52
18	Assam	6719.27	11751.57	18470.84
19	Nagaland	7384.96	7405.42	14790.38
20	Arunachal Pradesh	265.37	12848.74	13114.12
21	Mizoram	40.45	12998.44	13038.89
22	Manipur	4419.25	8305.67	12724.92
23	Goa	9243.79	3388.53	12632.32
24	Himachal Pradesh	9108.87	2745.12	11854.00
25	Telangana	4723.74	2141.82	6865.56
26	Tripura	203.56	6317.75	6521.31
27	West Bengal	5462.79	839.82	6302.61
28	Haryana	3345.85	1557.22	4903.06

S. No.	State Name	Organic Area (in ha)	Conversion area (in ha)	Total area (in ha)
29	Punjab	879.87	1141.63	2021.50
30	Lakshadweep	895.51	0.00	895.51
31	LADAKH	0.00	817.85	817.85
32	Pondicherry	2.84	20.81	23.65
33	New Delhi	0.72	4.45	5.17
Total		**1492611.02**	**1165278.31**	**2657889.33**

Organic versus conventional farming

Organic Farming Vs Conventional Farming

Conventional farming	**Organic farming**
It is based on *economical* orientation, heavy mechanization, specialization and misappropriates development of enterprises with unstable market oriented programme	It is based on *ecological* orientation, efficient input use efficiency, diversification and balanced enterprise combination with stability
Supplementing nutrients through fertilizers, weed control by herbicides, plant protection measures by chemicals and rarely combination with livestock	Cycle of nutrients within the farm, weed control by crop rotation and cultural practices, plant protection by non-polluting substances and better combination of livestock
Based on philosophy of to feed the crop/ plants	" Feed the soil not to the plant" is the watch word and slogan of organic farming
Production is not integrated into environment but extract more through technical manipulation, excessive fertilization and no correction of nutrient imbalances	Production is integrated into environment, balanced conditions for plants and animals and deficiencies need to be corrected
Low input : output ratio with considerable pollution	High input : output ratio with no pollution
Economic motivation of natural resources without considering principles of natural up gradation	Maximum consideration of all natural resources through adopting holistic approaches

Lecture-2

Introduction to bio-diversity, importance and measures to preserve bio-diversity

What is biodiversity?

Biodiversity is the shortened form of two words "biological" and "diversity." It refers to all the variety of life that can be found on Earth (plants, animals, fungi and micro-organisms) as well as to the communities that they form and the habitats in which they live.

The term biodiversity was coined as a contraction of biological diversity by **E.O. Wilson in 1985.**

"Biological diversity means the variability among living organisms from all sources including, inter alia, terrestrial, marine and other aquatic ecosystems and the ecological complexes of which they are part; this includes diversity within species, between species and of ecosystems".

Biodiversity is not only the sum of all ecosystems, species and genetic material. Rather, it represents the variability within and among them. It can be distinguished from the expression "biological resources", which refer to the tangible components of ecosystems. Biological resources are real entities (a particular species of bird, a wheat variety growing in a field, oak wood, etc.) while biological diversity is rather an attribute of life (the variety of bird species, the genetic variability of wheat around the world, forest types, etc.).

There are three interrelated hierarchical levels of biodiversity namely, genetic diversity, species diversity and community or ecosystem diversity.

1. Genetic diversity:

- It describes the variation in the number and types of genes as well as chromosomes present in different species. The magnitude of variation in genes of a species increases with increase in size and environmental parameters of the habitat.
- The genetic variation arises by gene and chromosome mutation in individuals and in sexually reproducing organisms and it is spread in the population by recombination of genetic materials during cell division after sexual reproduction.

  Genetic diversity has the following importance:

  (i) It helps in speciation or evolution of new species;

  (ii) It is useful in adaptation to changes in environmental conditions;

  (iii)It is important for agricultural productivity and development.

2. Species diversity:

- It describes the variety in the number and richness of the spices with in a region. The species richness may be defined as the number of species per unit area. The richness of a species tells about the extent of biodiversity of a site and provides a means for comparing different sites.
- The species richness depends largely on climatic conditions. The number of individuals of different species with in a region represents species evenness or species equitability. The product species richness and species evenness give species diversity

of a region. When a species is confined entirely to a particular area, it is termed as endemic species.

3. Ecosystem diversity:

- It describes the assemblage and Interaction of spices living together and the physical environment a given area. It relates varieties of habitats, biotic community's ecological processes in biosphere. It also tells about the diversity within the ecosystem. It is referred as Land escape diversity because it includes placement and size of various ecosystems.
- For example, the landscapes like grass lands, deserts, mountains etc. show ecosystem diversity. The ecosystem diversity is due to diversity of niches, trophic levels and ecological processes like nutrient cycling, food webs, energy flow, role of dominant species and various related biotic interactions. Such type of diversity can generate more productive and stable ecosystems or communities capable of tolerating various types of stresses e.g. drought, flood etc.

According to Whittaker (1965), the community diversities are of three types:

(i) α-Diversity:

- It tells the species diversity in a given community.
- It depends upon species richness and evenness.

(ii) β-Diversity:

- It describes a range of communities due to replacement of species which arises due to the presence of different microhabitats, niches and environmental conditions.

(iii) γ -Diversity:

- It describes diversity of habitat over a total land escape or geographical area.

Agro-biodiversity

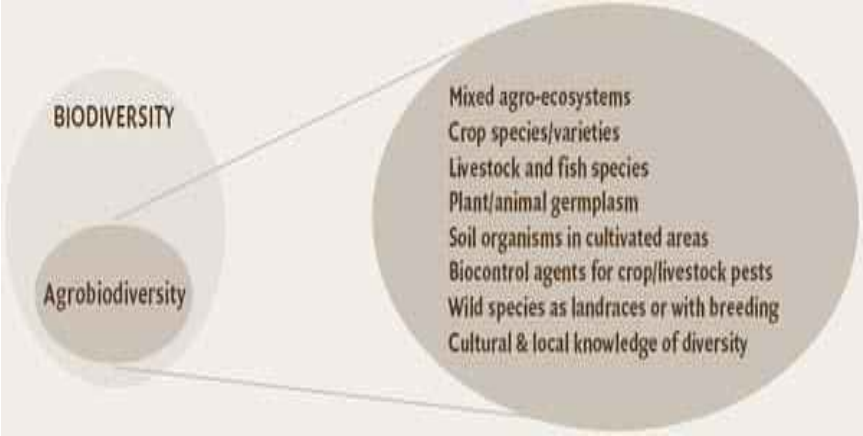
Agro biodiversity is a vital sub-set of biodiversity. Many people's food and livelihood security depend on the sustained management of various biological resources that are important for food and agriculture. Agricultural biodiversity, also known as agro-biodiversity includes:

- Harvested crop varieties, livestock breeds, fish species and non - domesticated (wild) resources within field, forest, rangeland including tree products, wild animals hunted for food and in aquatic ecosystems (e.g. wild fish);

- Non-harvested species in production ecosystems that support food provision, including soil micro-biota, pollinators and other insects such as bees, butterflies, earthworms, greenflies; and

- Non-harvested species in the wider environment that support food production ecosystems (agricultural, pastoral, forest and aquatic ecosystems).

AGROBIODIVERSITY IS CENTRAL TO OVERALL BIODIVERSITY



Agro-biodiversity is the result of the interaction between the environment, genetic resources and management systems and practices used by culturally diverse peoples, and therefore land and water resources are used for production in different ways. Thus, agro-biodiversity encompasses the variety and variability of animals, plants and micro-organisms that are necessary for sustaining key functions of the agro-ecosystem, including its structure and processes for, and in support of, food production and food security.

DEFINITION OF AGROBIODIVERSITY

The variety and variability of animals, plants and micro-organisms that are used directly or indirectly for food and agriculture, including crops, livestock, forestry and fisheries. It comprises the diversity of genetic resources (varieties, breeds) and species used for food, fodder, fibre, fuel and pharmaceuticals. It also includes the diversity of non-harvested species that support production (soil micro-organisms, predators, pollinators), and those in the wider environment that support agro-ecosystems (agricultural, pastoral, forest and aquatic) as well as the diversity of the agro-ecosystems. (FAO, 1999).

THE ROLE OF AGROBIODIVERSITY

Agro-biodiversity can:

* Increase productivity, food security, and economic returns
* Reduce the pressure of agriculture on fragile areas, forests and endangered species
* Make farming systems more stable, robust, and sustainable
* Contribute to sound pest and disease management
* Conserve soil and increase natural soil fertility and health
* Contribute to sustainable intensification
* Diversify products and income opportunities
* Reduce or spread risks to individuals and nations
* Help maximize effective use of resources and the environment

* Reduce dependency on external inputs
* Improve human nutrition and provide sources of medicines and vitamins, and
* Conserve ecosystem structure and stability of species diversity.

Why it should be preserved?

- Biodiversity is an essential characteristic of most natural ecosystems, and provides many essential ecosystem services.
- Agricultural systems ought to reflect this structure with careful management of agrobiodiversity and preservation of natural ecosystems.
- Genetic diversity increases the potential for plants to be well suited to their environment and adapt to changes.
- Planned biodiversity can be managed by farmers for the benefits of complementarity and ecosystem services.
- Associated biodiversity of soil organisms, pollinators, and Beneficial can serve as built-in regulatory systems.
- Ultimately, breeders' and farmers' choices have interregional and intergenerational consequences when it comes to agrobiodiversity.
- The value of ecological services and environmental benefits of biodiversity, while not reflected in market prices, is a public good worth conserving.

Biodiversity of India:

As per available data, the varieties of species living on the earth are 1753739. Out of the above species, 134781 are residing in India although surface area of India is 2% of the earth's surface. Wild life Institute of India has divided it into ten biogeographical regions and twenty five biotic provinces.

Biogeographical regions are:
(i) Trans Himalayas,
(ii) Gangetic plain,
(iii) Desert,
(iv) Semiarid zone;
(v) Western Ghats;
(vi) Deccan peninsula,
(vii) North eastern zone,
(viii) Coastal lands
(ix) Himalayas,
(x) Islands.

India is one of the twelve mega diversity nations of the world due to the following reasons:

- It has 7.3% of the global fauna and 10.88% of global flora as per the data collected by Ministry of Environment and forest.
- It has 350 different mammals, 1200 species of birds- 453 different reptiles, 182 amphibians and 45,000 plants spices.
- It has 50,000 known species of insects which include 13,000 butterflies and moths.

- It has 10 different biogeographical regions and 25 biotic provinces having varieties of lands and species.
- In addition to geographical distribution, geological events in the land mass provide high level of biological diversity.
- Several crops arose in the country and spread throughout the world.
- There is wide variety of domestic animals like cows, buffaloes, goats, sheep, pigs, horses etc.
- The marine biota includes sea weeds, fishes, crustaceans, molluses, corals, reptiles etc.
- There are a number of hot spots (namely Eastern Ghats, Western Ghats, North Eastern hills etc.).

Importance of Biodiversity:

The living organisms on earth are of great diversity, living in diverse habitats and possessing diverse qualities and are vital to human existence providing food, shelter, clothing's, medicines etc.

The biodiversity has the following importance:

1. Productive values:

Biodiversity produces a number of products harvested from nature and sold in commercial markets. Indirectly it provides economic benefits to people which include water quality soil protection, equalisation of climate, environmental monitoring, scientific research, recreation etc.

2. Consumptive value:

The consumptive value can be assigned to goods such as fuel woods, leaves, forest products etc. which may be consumed locally and do not figure in national and international market.

3. Social value:

The loss of biodiversity directly influences the social life of the country possibly through influencing ecosystem functions (energy flow and biogeochemical cycle). This be easily understood by observing detrimental effects of global warming and acid rain which cause an unfavorable alteration in logical processes.

4. Aesthetic value:

Aesthetic values such as refreshing fragrance of the flowers, taste of berries, softness of mossed, melodious songs of birds, etc. compel the human beings to preserve them. The earth's natural beauty with its colour and hues, thick forest, and graceful beasts has inspired the human beings from their date of birth to take necessary steps for its maintenance. Similarly botanical and zoological gardens are the means of biodiversity conservation and are of aesthetic values.

5. Legal values:

Since earth is homeland of all living organisms, all have equal right to coexist on the surface of earth with all benefits. Unless some legal value is attached to biodiversity, it will not be possible to protect the rapid extinction of species.

6. Ethical value:

Biodiversity must be seen in the light of holding ethical value. Since man is the most intelligent amongst the living organisms, it should be prime responsibility and moral obligation of man to preserve and conserve other organisms which will directly or indirectly favour the existence of the man.

7. Ecological value:

Biodiversity holds great ecological value because it is indispensable to maintain the ecological balance. Any disturbance in the delicately fabricated ecological balance maintained by different organisms, will lead to severe problems, which may threaten the survival of human beings.

8. Economic value:

Biodiversity has great economic value because economic development depends upon efficient and economic management of biotic resources.

In the day to day life, human beings are maintaining their lifestyle at the sacrifice of surrounding species which come from diversity of plants and animals struggling for their existence.

So, it is highly essential for the human beings to take care of their surrounding species and make optimum use of their service, for better economic development. Thus, it is rightly told, survival of the man depends upon the survival of the biosphere.

Uses of Biodiversity:

Biodiversity has the following uses for the development humanity:

- It provides food of all types.
- It provides fibers, sources for the preparation of clothes.
- It provides different types of oil seeds for the preparation of oils.
- It provides new varieties of rice, potato etc. through the process of hybridization.
- It provides different drugs and medicines which are based on different plant products.
- It is very essential for natural pest control, maintenance of population of various species, pollination by insects and birds, nutrient cycling, conservation and purification of water, formation of soil etc. All these services together are valued 16.54 trillion dollars per year.

Threats to Biodiversity:

Biodiversity is considered as a reservoir of resources to be used for the manufacture of food, medicine, industrial products, etc. But with an increased demand of rapid population growth, biodiversity is gradually depleting. A number of plants" and animal species have already become extinct and many are endangered.

The different factors responsible for causing threat to biodiversity are as follows:

1. Habitat destruction:

The primary cause of loss of biodiversity is habitat loss or destruction which is resulted due to the large industrial and commercial activities associated with agriculture, irrigation, construction of dams, mining, fishing etc.

2. Habitat fragmentation:

With increased population, the habitats are fragmented into pieces by roads, fields, canals, power lines, towns etc. The isolated fragment of habitats restricts the potential of species for dispersal and colonization. In addition, the habitat fragmentation also brings about microclimatic changes in light, temperature, wind etc.

3. Pollution:

The most dreaded factor inducing loss of biodiversity is environmental pollution which include air pollution, Water pollution, industrial pollution, pollution due to chemical Pastes, pesticides radioactive materials etc.

4. Over exploitation:

The natural resources are over exploited to meet growing rural poverty, intensive technological growth and globalization of economy. All these factors together may be responsible for the extinction of a number of species.

5. Introduction of exotic species:

The introductions of exotic species are due to:

- Horticulture
- Agriculture.
- European colonisation and
- Accidental transport.

It is seen that some exotic species may kill or eat the native species thereby causing its extinction.

6. Diseases:

Since the animals are more vulnerable to infection, the anthropological activities may increase the incidence of diseases in wild species, leading to their extinction.

7. Shifting or Jhum cultivation:

The shifting or Jhum cultivation by poor tribal people greatly affects the forest structure which is a store house of biodiversity.

8. Poaching of wild life:

A number of wildlife species are becoming extinct due to poaching and hunting.

Table: Endangered and Endemic Species of India

1. Higher plants	15,000	135
2. Mammals	372	69
3. Reptiles and amphibians	580	22
4. Birds	1175	40
5. Fishes	1693	—

Conservation of Biodiversity:

Biodiversity is being depleted by the loss of habitat, fragmentation of habitat, over exploitation of resources, human sponsored ecosystems, climatic changes, pollution invasive exotic spices, diseases, shifting cultivation, poaching of wild life etc.

Since the human beings are enjoying all the benefits from biodiversity, they should take proper care for the preservation of biodiversity in all its form and good health for the future generation i.e., the human being should prevent the degradation and destruction of the habitats thereby maintaining the biodiversity at its optimum level.

Conservation of biodiversity is protection, upliftment and scientific management of biodiversity so as to maintain it at its threshold level and derive sustainable benefits for the present and future generation. In other words, conservation of bio-diversity is the proper management of the biosphere by human beings in such a way that it gives maximum benefits for the present generation and also develops its potential so as to meet the needs of the future generations.

Mainly the conservation of biodiversity has three basic objectives:

(a) To maintain essential ecological processes and life supporting systems.
(b) To preserve the diversity of species.
(c) To make sustainable utilization of species and ecosystems.

Strategies for Conservation of Biodiversity:

The following strategies should be undertaken in order to conserve biodiversity:

(1) All the possible varieties (old or new) of food, forage and timber plants, livestock, agriculture animals and microbes should be conserved.
(2) All the economically important organisms in protected areas should be identified and conserved.
(3) Critical habitats for each species should be identified and safeguarded.
(4) Priority should be given to preserve unique ecosystems.
(5) There should be sustainable utilisation of resources.
(6) International trade in wild life should be highly regulated.
(7) The poaching and hunting of wildlife should be prevented as far as practicable.
(8) Care should be taken for the development of reserves and protected areas.
(9) Efforts should be made to reduce the level of pollutants in the environment.
(10) Public awareness should be created regarding biodiversity and its importance for the living organisms.
(11) Priority should be given in wildlife conservation programme to endangered species over vulnerable species and to vulnerable species over rare species.
(12) The habitats of migratory birds should be protected by bilateral and multilateral agreement.
(13) The over exploitation of useful products of wild life should be prevented.
(14) The useful animals, plants and their wild relatives should be protected both in their natural habitat (in-situ) and in zoological botanical gardens (ex-situ)
(15) Efforts should be made for setting up of National parks and wild life sanctuaries to safeguard the genetic diversity and their continuing evolution.
(16) Environmental laws should be strictly followed.

Conservation Methods:

There are two types of conservation methods namely in-situ and ex-situ conservations.

(a) In situ conservation:

The conservation of species in their natural habitat or natural ecosystem is known as in situ conservation. In the process, the natural surrounding or ecosystem is protected and maintained so that all the constituent species (known or unknown) are conserved and benefited. The factors which are detrimental to the existence of species concerned are eliminated by suitable mechanism.

The different advantages of in situ conservation are as follows:

- It is a cheap and convenient way of conserving biological diversity.
- It offers a way to preserve a large number of organisms simultaneously, known or unknown to us.
- The existence in natural ecosystem provides opportunity to the living organisms to adjust to differed' environmental conditions and to evolve in to a better life form.

- The only disadvantage of in situ conservation is that it requires large space of earth which is often difficult because of growing demand for space. The protection and management of biodiversity through in situ conservation involve certain specific areas known as protected areas which include national parks, Sanctuaries and Biosphere preserves.

Protected areas:

The protected areas are biogeographical areas where biological diversity along with natural and cultural resources are protected, maintained and managed through legal and administrative measures. The demarcation of biodiversity in each area is determined on the basis of climatic and physiological conditions.

In these areas, hunting, firewood collection, timber harvesting etc. are prohibited so that the wild plants and animals can grow and multiply freely without any hindrance. Some protected areas are: Cold desert (Ladakh and Spiti), Hot desert (Thar), Saline Swampy area (Sunderban and Rann of Kutch), Tropical moist deciduous forest (Western Ghats and north East) etc. Protected areas include national parks, sanctuaries and biosphere reserves. There are 37,000 protected areas throughout the world. As per World Conservation Monitoring Centre, India has 581 protected areas, national parks and sanctuaries.

Biosphere reserves:

Biosphere reserves or natural reserves are multipurpose protected areas with boundaries circumscribed by legislation. The main aim of biosphere reserve is to preserve genetic diversity in representative ecosystems by protecting wild animals, traditional life style of inhabitant and domesticated plant/ animal genetic resources. These are scientifically managed allowing only the tourists to visit.

Some importance of biosphere reserves are as follows:

- These help in the restoration of degraded ecosystem.
- The main role of these reserves is to preserve genetic resources, species, ecosystems, and habitats without disturbing the habitants.
- These maintain cultural, social and ecologically sustainable economic developments.
- These support education and research in various ecological aspects,

Some important biosphere reserves are:

Simlipal, (Orissa), Sunderban (West Bengal), Kanha (M.P Kaziranga (Assam) etc. The biosphere reserve net work was introduced by UNESCO 1971.

(b) Ex-situ conservation:

Ex-situ conservation involves maintenance and breeding of endangered plants and animals under partially or wholly controlled conditions in specific areas like zoo, gardens, nurseries etc. That is, the conservation of selected plants and animals in selected areas outside their natural habitat is known as ex-situ conservation.

The stresses on living organisms due to competition for food, water, space etc. can be avoided by ex-situ conservation there by providing conditions necessary for a secure life and breeding.

Some important areas under this conservation are:

(i) Seed gene bank,
(ii) Field gene bank;
(iii) Botanical gardens;
(iv) Zoos.

The strategies for ex-situ conservations are:

- Identification of species to be conserved.
- Adoption of Different ex-situ methods of conservation.
- Long-term captive breeding and propagation for the species which have lost their habitats permanently.
- Short-term propagation and release of the animals in their natural habitat
- Animal translocation
- Animal reintroduction
- Advanced technology in the service of endangered species.

The different advantages of ex-situ conservation are:

(a) It gives longer life time and breeding activity to animals.
(b) Genetic techniques can be utilised in the process.
(c) Captivity breed species can again be reintroduced in the wild.

Some disadvantages of this method are:

(a) The favourable conditions may not be maintained always.
(b) Mew life forms cannot evolve.
(c) This technique involves only few species.

Hot Spots:

Hot spots are the areas with high density of biodiversity or mega diversity which are most threatened at present. There are 16 hot spots in world, out of which two are located in India namely North-East Himalayas and Western Ghats

Different mechanisms involved in the conservation of biodiversity are shown in Fig.

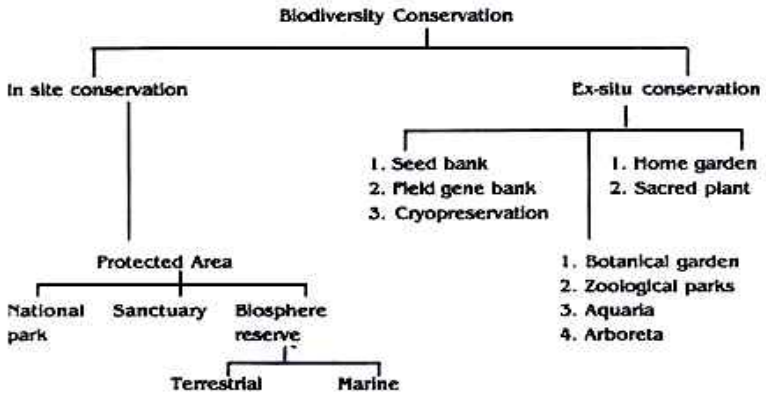


FIG. 5.1 : SCHEME SHOWING BIODIVERSITY CONSERVATION MANAGEMENT SYSTEMS.

Lecture-3

Pre-requisites and basic steps for organic farming; conversion to organic farming-planning and processes in practices- IFS approach-Integration of animal components.

Sustainable agriculture is necessary to attain the goal of sustainable development. The word "sustain," from the Latin ‘sustinere’ (sus-, from below and tenere, to hold), to keep in existence or maintain, implies long-term support or permanence. As it pertains to agriculture, sustainable describes "farming systems that are capable of maintaining productivity and usefulness to society indefinitely. Such systems must be resource-conserving, socially supportive, commercially competitive, and environmentally sound".

According to the Food and Agriculture Organization (FAO), "**sustainable agriculture is the successful management of resources for agriculture to satisfy changing human needs while maintaining or enhancing the quality of environment and conserving natural resources**". All definitions of sustainable agriculture lay great emphasis on maintaining an agriculture growth rate, which can meet the demand for food of all living things without draining the basic resources.

Over time, the International Alliance for Sustainable Agriculture and an increasing number of researchers, farmers, policy-makers and organizations worldwide have developed a definition that unifies many diverse elements into a widely adopted, comprehensive, working definition: **A sustainable agriculture is ecologically sound, economically viable, socially just and humane**. These four goals for sustainability can be applied to all aspects of any agricultural system, from production and marketing to processing and consumption. Rather than dictating what methods can and cannot be used, they establish basic standards by which widely divergent agricultural practices and conditions can be evaluated and modified, if necessary to create sustainable systems. The result is an agriculture designed to last and be passed on to future generations.

Conceived in this sense, sustainable agriculture presents a positive response to the limits and problems of both traditional and modern agriculture. It is neither a return to the past nor an idolatry of the new. Rather, it seeks to take the best aspects of both traditional wisdom and the latest scientific advances. This result in integrated, nature-based agro-ecosystems designed to be self-reliant, resource conserving and productive in both the short and long terms.

Organic farming is one of the several approaches found to meet the objectives of sustainable agriculture.

Organic farming is often associated directly with, *"Sustainable farming."* However, ‘organic farming’ and ‘sustainable farming’, policy and ethics-wise are two different terms. Many techniques used in organic farming like inter-cropping, mulching and integration of crops and livestock are not alien to various agriculture systems including the traditional

agriculture practiced in old countries like India. However, organic farming is based on various laws and certification programmes, which prohibit the use of almost all synthetic inputs, and health of the soil is recognized as the central theme of the method. Organic products are grown under a system of agriculture without the use of chemical fertilizers and pesticides with an environmentally and socially responsible approach. This is a method of farming that works at grass root level preserving the reproductive and regenerative capacity of the soil, good plant nutrition, and sound soil management, produces nutritious food rich in vitality which has resistance to diseases.

Organic farming is one of the widely used methods, which is thought of as the best alternative to avoid the ill effects of chemical farming. The origin of organic farming goes back, in its recent history, to 1940s. During this period, the path breaking literature on the subject published by ***J.I. Rodale*** *in the United States,* ***Lady Balfour*** *in England and* ***Sir Albert Howard*** *in India contributed to the cause of organic farming.*

In simple words organic agriculture is the production system with the optimum utilization of local resources in such a way so that sustainability of production and wellness of the society and environment can be maintained for fairly long time.

Although organic agriculture seems to be just the exclusion of synthetic external inputs but it is the *ideological differences with conventional agriculture* (Sharma, 2001) that makes organic agriculture friendly to society and environment.

Need for Organic farming

The need for organic farming in India arises from the un-sustainability of agriculture production and the damage caused to ecology through the conventional farming practices.

- Science and technology have helped man to increase agricultural production from the natural resources like land. But the realization that this has been achieved at the cost of the nature and environment, which support the human life itself, is becoming clear. It has been fully evident that the present pattern of economic development, which ignores the ecology and environment, cannot sustain the achievement of man without substantial erosion of the factors that support the life system of all living things on the Earth.

- Land, water and air, the most fundamental resources supporting the human life, have degraded into such an extent that they now constitute a threat to the livelihood of millions of people in the country.

- The monoculture of high yielding seeds required external inputs of chemical fertilizers. The fertilizers also destroy soil organisms. The long term effect was reduction of crop yields. The damaged soil was easily eroded by wind and water. The

eroding soil needed use of continuously increasing quantities of fertilizers, much of which was washed / leached into surface and underground water sources.

- The use of machineries in modern agriculture needed to do large scale cultural operations and requires higher amount of fossil fuels besides forcing the farmers to raise the same crops again and again, in order to utilize these machines to their optimum capacities. The result was the reduction of bio-diversity and labour. The high cost of the machines necessitated high profits, which in turn put pressure to raise productivity.

- The crops with high productivity were cultivated which needed increased quantities of fertilizers and pesticides. Increasing use of pesticides resulted in the damage to environment and increased resistance of insects to them. Pesticides harmed useful organisms in the soil.

- The country's farming sector has started showing indications of reversing the rising productivity as against the increasing trend of input use.

- The theme of consumer welfare has become central in the economic activities in the developed countries in the world. Sustainable agriculture based on technologies that combine increased production with improved environmental protection has been accepted as absolutely essential for the maximization of the consumer welfare. Another area to increase the consumer welfare is promotion of the eco-friendly methods in agriculture. No-till, or conservation agriculture, lower input approaches of integrated pest or nutrient management and organic farming are some of them

- The unsustainability of Indian agriculture is caused by the modern farming methods which have badly affected/damaged production resources and the environment.

- The organic agriculture movement in India received inspiration and assistance from IFOAM which has about 600 organizational members from 120 countries. All India Federation of Organic Farming (AIFOF) is a member of IFOAM and consists of a number of NGOs, farmers' organizations, promotional bodies and institutions.

The major advantages of organic farming are as under:

1. Maintains environmental health by reducing pollution;
2. Reduces human and animal hazards by lowering the level of chemical residues in the products;
3. Keeps agricultural production sustainable;
4. Reduces the cost of agricultural production by relying on internal sources for nutrient, pest and disease management;
5. Improves soil health;

6. Ensures the optimum utilization of natural resources;
7. Improves soil physical properties such as granulation, tilth, aeration, and facilitates easy plant root penetration and improves water-holding capacity;
8. Improves supply and retention of soil nutrients, and promotes favourable chemical reactions in the soil.

Conversion Requirements

Organic agriculture means a process of developing a viable and sustainable agro-ecosystem. **The time between the start of organic management and certification of crops and/or animal husbandry is known as the conversion period.** The whole farm, including livestock, should be converted according to the standards over a period of **three years**.

- For a sustainable agro-ecosystem to function optimally, diversity in crop production and animal husbandry must be arranged in such a way that there is interplay of all the elements of the farming management. Conversion may be accomplished over a period of time. A farm may be converted step by step.
- The totality of the crop production and all animal husbandry should be converted to organic management. There should be a clear plan of how to proceed with the conversion. This plan shall be updated if necessary and should cover all aspects relevant to these standards.
- The certification programme should set standards for different farming systems so that they can be clearly separated in production as well as in documentation, and the standards should determine norms to prevent a mix up of input factors and products.
- The standards requirements shall be met during the conversion period. All the standards requirements shall be applied on the relevant aspects from the beginning of the conversion period itself. If the whole farm is not converted, the certification programme shall ensure that the organic and conventional parts of the farm are separate and inspectable.
- Before products from a farm/project can be certified as organic, inspection shall have been carried out during the conversion period. The start of the conversion period may be calculated from the date of application of the certification programme or from the date of last application of unapproved farm inputs provided it can demonstrate that standards requirements have been met from that date of implementation.
- Simultaneous production of conventional, organic, in conversion and/or organic crops or animal products which cannot be clearly distinguished from each other, will not be allowed.
- To ensure a clear separation between organic and conventional production, a buffer zone or a natural barrier should be maintained. The certification programme shall ensure that the requirements are met.
- A full conversion period is not required where full standards requirements have been met for several years and where this can be verified through several means and sources. In such cases inspection shall be carried out with a reasonable time interval before the first harvest.

Duration of Conversion Period

- The establishment of an organic management system and building of soil fertility requires an interim period, the **conversion period**.
- The conversion period may not always be of sufficient duration to improve soil fertility and reestablish the balance of the ecosystem but it is the period in which all the actions required to reach these goals are started.
- The duration of the conversion period must be adapted to:
    - the past use of the land
    - the ecological situation
- Plant products produced can be certified organic when the national standards requirements have been met during a conversion period of at least two years before sowing or in the case of perennial crops other than grassland, at least three years (thirty-six months) before the first harvest of products.
- The accredited inspection and certification agency may decide in certain cases (such as idle use for two years or more) to extend or reduce the conversion period in the light of previous status of the land but the period must equal or exceed twelve months.
- The certification programme may allow plant products to be sold as "produce of organic agriculture in process of conversion" or a similar description during the conversion period of the farm.
- For the calculation of inputs for feeding, the feed produced on the farm unit during the first year of organic management, may be classified as organic. This refers only to feed for animals which are themselves being reared within the farm unit and such feed may not be sold or otherwise marketed as organic.
- Feed produced on the farms in accordance with the national standards is to be preferred over conventionally grown / brought-in feeds.

In general the conversion period for annual crops are 3 years and for perennial crops it is 5 years

Maintenance of Organic Management/ Planing and Processes in practices

Organic certification is based on continuance. The certification programme should only certify production which is to be maintained on a long-term basis. Converted land and animals shall not get switched back and forth between organic and conventional management.

CROP PRODUCTION

Choice of Crops and Varieties

- All seeds and plant material should be certified organic.
- Species and varieties cultivated should be adapted to the soil and climatic conditions and be resistant to pests and diseases.
- In the choice of varieties genetic diversity should be taken into consideration.
- When organic seed and plant materials are available, they shall be used.

- The certification programme shall set time limits for the requirement of certified organic seed and other plant material. When certified organic seed and plant materials are not available, chemically untreated conventional materials shall be used.
- The use of genetically engineered seeds, pollen, transgenic plants or plant material is not allowed.

Diversity in Crop Production

The basis for crop production in gardening, farming and forestry in consideration of the structure and fertility of the soil and surrounding ecosystem and to provide a diversity of species while minimising nutrient losses.

Diversity in crop production is achieved by a combination of:

- a versatile crop rotation with legumes
- an appropriate coverage of the soil during the year of production which diverse plant species

Where appropriate, the certification programme shall require that sufficient diversity is obtained in time or place in a manner that takes into account pressure from insects, weeds, diseases and other pests, while maintaining or increasing soil, organic matter, fertility, microbial activity and general soil health. For non perennial crops, this is normally, but not exclusively, achieved by means of crop rotation.

Fertilization Policy

- Sufficient quantities of biodegradable material of microbial, plant or animal origin should be returned to the soil to increase or at least maintain its fertility and the biological activity within it.
- Biodegradable material of microbial, plant or animal origin produced on organic farms should form the basis of the fertilization programme.
- Accumulation of heavy metals and other pollutants should be prevented.
- Non synthetic mineral fertilizers and brought in fertilizers of biological origin should be regarded as supplementary and not a replacement for nutrient recycling.
- Adequate pH levels should be maintained in the soil.
- Biodegradable material of microbial, plant or animal origin shall form the basis of the fertilization programme.
- The certification programme shall set limitations to the total amount of biodegradable material of microbial, plant or animal origin brought onto the farm unit, taking into account local conditions and the specific nature of the crops.
- The certification programme shall set standards which prevent animal runs from becoming over-manured where there is a risk of pollution.
- Brought-in material (including potting compost) shall be in accordance with standards.
- **Manures containing human excreta (feces and urine) shall not be used.**
- Mineral fertilizers shall only be used in a supplementary role to carbon based materials. Permission for use shall only be given when other fertility management practices have been optimized.

- Mineral fertilizers shall be applied in their natural composition and shall not be rendered more soluble by chemical treatment.

Pest, Disease and Weed Management including Growth Regulators

- Organic farming systems should be carried out in a way which ensures that losses from pests, diseases and weeds are minimized.
- Emphasis is placed on the use of a balanced fertilizing programme, use of crops and varieties well adapted to the environment, fertile soils of high biological activity, adapted rotations, companion planting, green manures, etc. Growth and development should take place in a natural manner.
- Weeds, pests and diseases should be controlled by a number of preventive cultural techniques which limit their development, e.g. suitable rotations, green manures, a balanced fertilizing programme, early and pre-drilling seedbed preparations, mulching, mechanical control and the disturbance of pest development cycles.
- The natural enemies of pests and diseases should be protected and encouraged through proper habitat management of hedges, nesting sites etc. Pest management should be regulated by understanding and disrupting the ecological needs of the pests.
- An ecological equilibrium should be created to bring about a balance in the pest predator cycle.
- Products used for pest, disease and weed management, prepared at the farm from local plants, animals and microorganisms, are allowed.
- Thermic weed control and physical methods for pest, disease and weed management are permitted.
- All equipments from conventional farming systems shall be properly cleaned and free from residues before being used on organically managed areas.
- The use of synthetic herbicides, fungicides, insecticides and other pesticides is prohibited.
- The use of synthetic growth regulators and synthetic dyes is prohibited.
- The use of genetically engineered organisms or products is prohibited.

Contamination Control

- All relevant measures should be taken to minimize contamination from outside and from within the farm.
- Accumulation of heavy metals and other pollutants should be limited.
- For protected structure coverings, plastic mulches, fleeces, insect netting and silage rapping, only products based on polyethylene and polypropylene or other polycarbonates are allowed. These shall be removed from the soil after use and shall not be burnt on the farmland.
- The use of polychloride based products is prohibited.

Soil and Water Conservation

- Soil and water resources should be handled in a sustainable manner.
- Relevant measures should be taken to prevent erosion, salination of soil, excessive and improper use of water and the pollution of ground and surface water.
- The clearing of primary forest is prohibited.
- Relevant measures shall be taken to prevent erosion.
- Excessive exploitation and depletion of water resources shall not be allowed.
- The certification programme shall require appropriate stocking rates which do not lead to land degradation and pollution of ground and surface water.
- Relevant measures shall be taken to prevent salination of soil and water.

IFS approach- Integration of animal components.

Animal Husbandry Management

Management techniques in animal husbandry should be governed by the physiological and ethological needs of the farm animals in question. This includes:

- That animal should be allowed to conduct their basic behavioural needs.
- That all management techniques, including those where production levels and speed of growth should be concerned, for the good health and welfare of the animals.

- For welfare reasons the herd or flock size should not adversely affect the behavioural patterns of the animal.
- The certification programme shall ensure that the management of the animal environment takes into account the behavioural needs of the animals and provides for:
  - Sufficient free movement
  - Sufficient fresh air and natural daylight according to the needs of the animals
  - Protection against excessive sunlight, temperatures, rain and wind according to the needs of the animals
- Enough lying and/or resting area according to the needs of the animal. For all animals requiring bedding, natural materials shall be provided.
  - Ample access to fresh water and feed according to the needs of the animals
  - Adequate facilities for expressing behaviour in accordance with the biological and ethological needs of the species.
- No compounds used for construction materials or production equipment shall be used which might detrimentally affect human or animal health.
- All animals shall have access to open air and/or grazing appropriate to the type of animal and season taking into account their age and condition, to be specified by the certification programme.

The certification programme shall allow exceptions in cases where:

- The specific farm or settlement structure prevents such access provided animal welfare can be guaranteed
- Areas where feeding of animals with carried fresh fodder is a more sustainable way to use land resources than grazing, provided animal welfare is not compromised. Restrictions shall always include a time limit which shall be set for each exception.
  - Poultry and rabbits shall not be kept in cages.
  - Landless animal husbandry systems shall not be allowed.
  - When the natural day length is prolonged by artificial lighting, the certification programme shall prescribe maximum hours respective to species, geographical considerations and general health of animals.
  - Herd animals shall not be kept individually.

The certification programme may allow exceptions, e.g., male animals, smallholdings, sick animals and those about to give birth.

Length of Conversion Period

The establishment of organic animal husbandry requires an interim period, the conversion period. The whole farm, including livestock, should be converted according to the standards set out in this document.

- Conversion may be accomplished over a period of time.
- Replacement poultry should be brought onto the holding at the start of the production cycle.
- Animal products may be sold as "product of organic agriculture" only after the farm or relevant part of it has been under conversion for at least twelve months and provided the organic animal production standards have been met for the appropriate time.
- The certification programme shall specify the length of time by which the animal production standards shall be met.
- With regard to dairy and egg production, this period shall not be less than 30 days.
- Animals present on the farm at the time of conversion may be sold for organic meat if the organic standards have been followed for 12 months.

Brought-in Animals

- All organic animals should be born and raised in the organic holding.
- Organic animal husbandry should not be dependent on conventional raising systems.
- When trading or exchanging livestock, it should preferably take place between organic farms or as part of a long term cooperation between specific farms.
- When organic livestock is not available, the certification programme shall allow brought-in conventional animals according to the following age limits:
  - 2 day old chickens for meat production
  - 18 week old hens for egg production

- 2 week old for any other poultry
- piglets up to six weeks and after weaning
- calves up to 4 weeks old which have received colostrum and are fed a diet consisting mainly of full milk.

- Certification programmes shall set time limits (not exceeding 5 years) for implementation of certified organic animals from conception for each type of animal.
- Breeding stock may be brought in from conventional farms at an annual rate not exceeding 10% of the adult animals of the same species in the organic farm.

Breeds and Breeding

- Breeds should be chosen which are adapted to local conditions.
- Breeding goals should not be at variance with the animal's natural behaviour and should be directed towards good health.
- Breeding shall not include methods which make the farming system dependent on high technological and capital intensive methods.
- Reproduction techniques should be natural.
- The certification programme shall ensure that breeding systems are based on breeds that can both copulate and give birth naturally.
- Artificial insemination is allowed.
- Embryo transfer techniques are not allowed in organic agriculture.
- Hormonal heat treatment and induced birth are not allowed unless applied to individual animals for medical reasons and under veterinary advice.
- The use of genetically engineered species or breeds is not allowed.

Animal Nutrition

General Principles

- The livestock should be fed 100% organically grown feed of good quality. All feed shall come from the farm itself or be produced within the region.
- The diet shall be offered to the animals in a form allowing them to execute their natural feeding behaviour and digestive needs.
- The diet should be balanced according to the nutritional needs of the animals. Products from the organic feed processing industry shall be used.
- Colouring agents shall not be used in organic livestock production.
- The certification programme shall draw up standards for feed and feed ingredients.
- The prevailing part (at least more than 50%) of the feed shall come from the farm unit itself or shall be produced in co-operation with other organic farms in the region.

The following products shall not be included nor added to the feed given to farm animals:

- Synthetic growth promoters or stimulants
- Synthetic appetizers
- Preservatives, except when used as a processing aid
- Artificial colouring agents
- Urea
- Farm animal by-products (e.g. abattoir waste) to ruminants

- Droppings, dung or other manure (all types of excreta) even if technologically processed
- Feed subjected to solvent (e.g. hexane), extraction (soya and rape seed meal) or the addition of other chemical agents
- Pure amino acids
- Genetically engineered organisms or products thereof

This covers both organic and conventional feeding stuffs.

Vitamins, trace elements and supplements shall be used from natural origin when available in appropriate quantity and quality.

The following fodder preservatives shall be used:

Bacteria, fungi and enzymes

By-products of food industry (e.g. molasses)

Plant based products.

Lecture-4

Organic carbon; status and improvement strategies- conservation tillage system

Total organic carbon (TOC) is the carbon (C) stored in soil organic matter (SOM). Organic carbon (OC) enters the soil through the decomposition of plant and animal residues, root exudates, living and dead microorganisms, and soil biota. SOM is the organic fraction of soil exclusive of non-decomposed plant and animal residues. Nevertheless, most analytical methods do not distinguish between decomposed and non-decomposed residues. SOM is a heterogeneous, dynamic substance that varies in particle size, C content, decomposition rate, and turnover time.

Soil Organic Carbon (SOC) is the main source of energy for soil microorganisms. The ease and speed with which SOC becomes available is related to the SOM fraction in which it resides. In this respect, SOC can be partitioned into fractions based on the size and breakdown rates of the SOM in which it is contained (Table 1). The first three fractions listed are part of the active pool of SOM. Carbon sources in the active pool are relatively easy to break down.

SOM contains approximately 58% C; therefore, a factor of 1.72 can be used to convert OC to SOM. There is more inorganic C than TOC in calcareous soils. TOC is expressed as percent C per 100 g of soil.

Table 1. Size and breakdown rates of various soil organic matter fractions.

Soil Organic Matter Fraction	Particle Size (mm)	Turnover Time (years)	Description
Plant residues	≥ 2.0	< 5	Recognizable plant shoots and roots
Particulate organic matter	0.06 – 2.0	< 100	Partially decomposed plant material, hyphae, seeds, etc
Soil microbial biomass	variable	< 3	living pool of soil organic matter, particularly bacteria and fungi
humus	≤ 0.0053	< 100 – 5000	ultimate stage of decomposition, dominated by stable compounds

Why it is important: SOC is one of the most important constituents of the soil due to its capacity to affect plant growth as both a source of energy and a trigger for nutrient availability through mineralization. SOC fractions in the active pool, previously described, are the main source of energy and nutrients for soil microorganisms. Humus participates in aggregate stability, and nutrient and water holding capacity.

OC compounds, such as polysaccharides (sugars) bind mineral particles together into microaggregates. Glomalin, a SOM substance that may account for 20% of soil carbon, glues aggregate together and stabilizes soil structure making soil resistant to erosion, but porous

enough to allow air, water and plant roots to move through the soil. Organic acids (e.g., oxalic acid), commonly released from decomposing organic residues and manures, prevents phosphorus fixation by clay minerals and improve its plant availability, especially in subtropical and tropical soils. An increase in SOM, and therefore total C, leads to greater biological diversity in the soil, thus increasing biological control of plant diseases and pests. Data also reveals that interaction between dissolved OC released from manure with pesticides may increase or decrease pesticide movement through soil into groundwater.

Specific problems that might be caused by poor function: A direct effect of poor SOC is reduced microbial biomass, activity, and nutrient mineralization due to a shortage of energy sources. In non-calcareous soils, aggregate stability, infiltration, drainage, and airflow are reduced. Scarce SOC results in less diversity in soil biota with a risk of the food chain equilibrium being disrupted, which can cause disturbance in the soil environment (e.g., plant pest and disease increase, accumulation of toxic substances).

What is soil organic matter?

Soil is three-dimensional and highly dynamic. It contains abundant plant and animal life. There are four main components of soil: mineral matter, organic matter, air and water

- The ***mineral matter*** is made of sand, silt and clay size particles—the basic texture of the soil.
- The ***soil water*** contains dissolved minerals and is the main source of water and nutrients for plants.
- The ***air*** in the soil is needed for plant roots and soil microorganisms to obtain oxygen.
- ***Organic matter*** includes plant and animal materials in various stages of decomposition. Some soil scientists think of living plant roots and soil microorganisms as part of the soil organic matter.

Dead animal and plant matter begin to decompose as soon as they are added to the soil. Eathworms, beetles, and springtails, the macro (large) to meso (medium) fauna, begin to break large pieces of debris into smaller pieces. At the same time, the microbial population increases rapidly. The microorganisms consume the animal and plant remains and then die, adding themselves to the organic matter. Some organic matter is more readily decomposed than others. The end product of decomposition is **humus**—dark brown or black organic matter that is highly resistant to further decomposition.

Most soil organic matter is present near the soil surface, rather than deeper in the soil. Plant roots become organic matter as their cells slough off or as they die. Aboveground parts of the plant that are not harvested are left on the soil surface when it dies. Earthworms and insects incorporate the surface residues deeper into the soil, but the greatest concentration of organic matter still remains in the top six inches of soil.

Crop cultivation, harvesting, erosion, and natural decomposition gradually reduce the amount of organic matter in soils. However, you can maintain and even increase your soil's current organic matter level through proper management.

The Soil Science Society of America (SSSA) (2008) defines ‘SOM as the total organic fraction of the soil exclusive of un decayed plant and animal residues’.

Since SOM consists of C, H, O, N, P and S, it is difficult to directly measure the SOM content. Most analytical methods of organic matter generally measure only organic compounds or carbon, and estimate SOM through a conversion factor. They are therefore only an approximation of the level of once-living or decomposed matter. SOM is assumed to consist of between 50 to 58 % carbon such that SOM is simply the multiplication of measured soil organic carbon (SOC) with a factor 1.724 to 2. Based on the organic carbon content and the soil specific weight, the organic carbon stock can be derived.

Soil organic matter based on decomposition level

Not all soil organic matter is created equal. Fruit and vegetable wastes are easily degraded because they contain mostly simple carbohydrates (sugars and starches). In contrast, leaves, stems, nutshells, bark and trees decompose more slowly because they contain cellulose, hemicellulose and lignin. The ease with which compounds degrade is determined by the complexity of the carbon compounds and generally follows the order: carbohydrates > hemicellulose > cellulose = chitin > lignin.

In contrast to fresh plant residues, composted organic materials decompose slowly when added to soil because they have already undergone a significant amount of decomposition during the composting process. Well-decomposed organic matters no longer provides as many nutrients for plants and soil microbes as the active pool, but it does play important roles in the soil, such as promoting water and nutrient retention, and prevent soil compaction and crusting.

Soil organic matter is often viewed as the thread that links the biological, chemical and physical properties of a soil. It has been associated with numerous soil functions like nutrient cycling, water retention and drainage, erosion control, disease suppression and pollution remediation.

How soil organic matter influences the soil quality.

1. Stores and supplies plant nutrients (N P K and micronutrients; increases cation exchange capacity.
2. Stabilizes and holds soil particles together as aggregates.
3. Helps soil to resist compaction, promotes water infiltration, and reduces runoff.
4. Aids growth of crops by improving the soil's ability to store and transmit air and water, as measured by improved porosity; water holding capacity, and drought resistance.
5. Makes soil more friable and easier to work so that plant roots can penetrate the soil profile better.
6. Provides a source of carbon and energy for soil microbes which cycle nutrients and fight plant diseases.
7. Reduces the negative environmental effects of pesticides, heavy metals and other pollutants by binding contaminants.

SOM & SOC

Soils are formed over thousands of years as rock is broken downand colonized by plants and soil biota, leading to the formation of soil organic matter (SOM). While SOM is primarily carbon, it also contains nutrients essential for plant growth such as nitrogen,

phosphorus, sulphur and micronutrients**.** Organisms in the soil food web decompose SOM and make these nutrients available. The rate of SOM decomposition and turnover mainly depends upon the interplay between soil biota, temperature, moisture and a soil's chemical and physical composition.

SOC is the main constituent of SOM. SOM is formed by the biological, chemical and physical decay of organic materials that enter the soil system from sources above ground (e.g. leaf fall,crop residues, animal wastes and remains) or below ground (e.g.roots, soil biota). The elemental composition of SOM varies, with values in the order of 50 per cent carbon (Broadbent 1953), 40 per cent oxygen and 3 per cent nitrogen, as well as smaller amounts of phosphorus, potassium, calcium, magnesium and other elements as micronutrients. Soil biota (from microbes to earthworms) contributes living biomass to SOM mixing and breaking down the organic matter through physical and biochemical reactions. These biochemical reactions release carbon and nutrients back to the soil, and greenhouse gases such as carbon dioxide (CO_2), nitrous oxide (N_2O) and methane (CH_4) to the atmosphere

Soil organic carbon in the global carbon cycle

The carbon cycle is essential for life on Earth. While photosynthesis by plants, algae and cyanobacteria is the key mechanism allowing life to capture the sun's energy, carbon dioxide (CO_2) respiration is the main mechanism through which autotrophs and heterotrophs use part of the stored energy to fuel their metabolism. Within the carbon cycle, the soil acts as a major reservoir. Although there is large uncertainty associated with estimating the mean soil carbon (C) content of a biome or soil taxonomic order (with coefficients of variation of the order of 65%), recent estimates for the global soil C reservoir converge (Hiedererand Köchyl, 2012).

It is estimated that global soils contain between 1400 and 1600 petagrams (Pg) of carbon (1 Pg = 1015 g) in the upper meter, and that the next meter of soil contains an additional 500- 1000 Pg C. These estimates imply that the soil organic carbon pool is more than twice the size of the atmospheric carbon pool (ca. 800 Pg) and that it contains about three times the amount of carbon in vegetation (ca. 550 Pg C). The exchange of C between terrestrial and atmospheric reservoirs by photosynthesis and respiration is of the order of 120 Pg C per year in each direction (Houghton, 2003). The overall size of the soil C reservoir is therefore relatively large compared to these gross annual fluxes of C.

Carbon sequestration

Carbon sequestration is defined as the removal of CO_2 from the atmosphere into various longlived chemically bound forms, either on land or in the ocean. Through the process of photosynthesis, CO_2 is sequestered from the atmosphere into plant tissues. Photosynthesis represents the largest transfer of CO_2 in the C cycle, and therefore, is of great importance in understanding how to manage the global C cycle. Carbon sequestration on land (or terrestrial C sequestration) occurs in standing biomass (e.g., trees), long-term harvested products (e.g., lumber), living biomass in soil (e.g., perennial roots and microorganisms), recalcitrant organic matter in surface soil (e.g., humus), and inorganic C in subsoil (e.g., carbonates). Soil organic and inorganic C pools are generally the most long-lived terrestrial C sequestration forms.

Importance of C Sequestration

- Sequestration of soil organic C from plant biomass is a key sequestration pathway in agriculture; offering an offset strategy (i.e., mitigation) for agriculture's other greenhouse gas emissions.
- Soil C sequestration is also important at the farm level to build soil fertility, protect soil from compaction, and nurture soil biodiversity.
- In addition to its vital role of mitigating greenhouse gas emissions, soil C sequestration provides many other significant off-farm benefits to society.
- These benefits include the protection of streams, lakes, and rivers from sediment, nutrient, and pathogen runoff from agricultural fields, as well as enhanced wildlife habitat.
- A full-system cost-to-benefit ratio of soil C sequestration from various conservation agricultural practices has not been adequately addressed, but is needed to more fully appreciate this important pathway.
- Conservation agricultural systems promote soil C sequestration by tipping the balance in favor of C inputs relative to C outputs.

Carbon sequestration can be achieved by maximizing C inputs and minimizing C outputs (Table 2).

Table 2 Strategies to sequester soil organic C

Maximising C input	Minimizing C loss from soil
✓ Plant selection • Species, cultivar, variety • Growth habit (perennial / annual) • Rotation sequence • Biomass energy crops	✓ Reducing soil disturbance • Less intensive tillage • Controlling soil erosion
✓ Tillage • Type • Frequency	✓ Utilizing available soil water • Promotes optimum plant growth • Reduce soil microbial activity
✓ Fertilization • Rate, timing, placement • Organic amendments	✓ Maintaining surface residue cover • Increased plant water use and production • More fungal dominance in soil

Conservation Tillage

- **Conventional tillage** is a system that combines primary and secondary tillage operations to prepare a seedbed. These operations include moldboard or chisel plowing, disking, harrowing, and cultivating. Essentially no residue is left on the surface after conventional tillage is completed.
- **Conservation tillage** is a system that retains protective amounts of crop residue as mulch on the soil surface. After planting, at least 30 percent of the soil surface remains covered by residue. It bears repeating that under a wide range of soil, crop and environmental conditions, conservation tillage can reduce soil loss by 10 to 90 percent over conventional tillage.

These are the three main systems of conservation tillage:

- **No-till** is a method of planting crops that requires no seedbed preparation other than opening the soil so that the seed can be placed at the intended depth. In its simplest form, no-till planting is a one-pass operation accomplished with a multi-component implement that slices through surface residue and the top 3 or 4 inches of soil, drops seeds into the slot, and squeezes the slot back together over and around the seed, leaving little or no visible evidence that the crop has been planted. This system produces the least soil disturbance of the three systems.
- **Ridge-till** is a method of preparing the seedbed and planting on a pre-formed ridge remaining from the previous year's crop. The soil is left undisturbed until this single operation takes place. Ridge tillage differs from no-till planting in that some cultivation is required during the current growing season to form the ridge for next year's crop. This system produces some soil disturbance, but not as much as the strip-till system.
- **Strip-till** is a method of preparing a seedbed on a strip 2 to 12 inches wide and 2 to 8 inches deep. Soil remaining between strips is left undisturbed. Often strip tillage equipment will include a 12-inch subsoil shank to break any tillage pans that may be present. A conventional planter may be used in this system. Of the three systems, this one produces the most soil disturbance.

Benefits of Conservation Tillage

Organic Matter Increases. Switching from conventional tillage to a CT system will generally bring about an increase in organic matter, most noticeably at the surface and within the top 2 inches of the soil. In conventional tillage, plant residues, fertilizers, and other amendments become homogenized and diluted as they are mixed deeply into the soil. But with conservation tillage, the residues and amendments reside in a layer near the soil surface, and plant roots tend to proliferate in the top 2 inches of soil where the nutrients are located.

Biological Activity Increases. As organic matter, in particular soil organic carbon, increases in CT systems, soil biological activity increases. This enhances the diversity of organisms and the ecosystem functions they perform. Chief among these functions is nutrient cycling. Soil microbes decompose organic matter. In doing so, they derive energy for themselves by breaking the long chains of carbon molecules that compose organic matter. As they do this, many plant nutrients attached to these chains are released. These nutrients are absorbed by

the organisms themselves or cycled to plants. When the plants or microbes die, nutrients can be recycled once again. Thus, adding organic matter adds energy to this system and promotes nutrient cycling. The soil microbial biomass serves as both a source and a sink for nutrients.

This biomass includes other organisms that perform additional functions, such as improving soil aggregation and inducing resistance to plant diseases The soil biological community in CT systems differs from that found in conventional tillage systems. In conventional tillage, plowing and other tillage operations bury plant residues deeply into the soil, where they decompose rather quickly, primarily because bacteria are the decomposers.

But in conservation tillage, when the soil is left undisturbed and plant residues remain at the surface, the primary decomposers are fungi. Decomposition is much slower. Slower decomposition results in an increased accumulation of organic materials and a rise in populations of assorted *secondary decomposer* organisms, such as nematodes and arthropods (insects and spiders). Earthworms and other tunneling invertebrates, such as the larvae of some beetles, proliferate in this soil environment. This community of decomposer organisms helps to maintain soil structure and improves infiltration and aeration within the soil. Plant root growth proliferates in the tunnels and channels these organisms provide.

Soil Structure Improves. A primary benefit of soil biological activity is that it builds soil structure, which produces good crop growth and yield. Organic compounds produced during decomposition and other microbial processes act as glues to bind individual soil particles together into aggregates. Soil aggregates are important for maintaining soil *tilth*(tillability) and *friability* (looseness). The stability of these aggregates is also important. That is, soil should maintain its structure while allowing water and nutrients to flow freely.

Tillage destroys aggregate stability and promotes compaction and crusting. CT systems help to build and preserve the aggregate stability of soils. Conservation tillage, in particular no-till, increases water-holding capacity and reduces runoff. Surface residues keep the soil temperature cooler and reduce water evaporation. Plant residues on the soil surface decrease runoff during heavy storms by increasing the infiltration of water into soil. Residues also protect the soil surface from any crusting caused by raindrop impact.

Lecture-5

Sources of organic manures-plant, animal and microbial origin- On-farm resources; FYM, green manures, crop residues, Poultry manure, Sheep and goat manures, bio-gas slurry and vermin-compost

Sources of Plant nutrients

Plant nutrients are essentially supplied through manures and fertilizers. Manures are organic in nature and applied in large quantities. They are also called as organics or organic manures. They are of animal and plant origin and contain more than one nutrient element. Nutrient content in organic manure is low. Fertilizers are inorganic or synthetic and the nutrient content is higher than in manures. Fertilizers are available for a particular nutrient or combination of nutients.

Farmyard manure

Farmyard manure commonly describes a more or less decomposed mixture of livestock dung and urine (mostly from cattle) mixed with straw and litter, which was used as bedding material. It may also contain residues from the fodder fed to the cattle and decomposed household waste.

Farmyard manure is extremely valuable organic manure. Farmyard manure contains large amounts of nutrients. The availability of phosphorus and potassium from farmyard manure is similar to that of chemical fertilizers. Chicken manure is rich in phosphorus. When dung and urine from cattle are mixed, they form a well-balanced source of nutrients for plants.

Many farmers still underestimate the value of animal manure. In many places, it is dried and burned for cooking or just not recognized as a source of nutrients and organic matter. By drying or burning farmyard manure, large quantities of organic matter and nutrients are lost from agricultural systems. Appropriate recycling of nutrients on the farm, especially if it comes from a high-value source, is a principle of organic farming. Therefore, proper handling and use of animal manures are essential to ensure that the nutrients in the manure are preserved and the risks of causing environmental pollution are minimized.

Most farmers do not own animals, and neither do they have access to animal manure. Growing animal feed and integrating livestock into the farm not only provides milk and or meat and other animal products, but also some animal manure In areas with mixed crop-

livestock farming systems manure is likely to be available to most households, although at varying levels.

Improving the value of farm yard manure

Farmyard manure refers to the decomposed mixture of dung and urine of farm animals along with litter and left over material from roughages or fodder fed to the cattle. On an average well decomposed farmyard manure contains 0.5 per cent N, 0.2 per cent $P_2O_5$and .0.5 per cent K_2O.The present method of preparing farmyard manure by the farmers is defective. Urine, which is wasted, contains one per cent nitrogen and 1.35 per cent potassium. Nitrogen present in urine is mostly in the form of urea which is subjected to volatilization losses. Even during storage, nutrients are lost due to leaching and volatilization. However, it is practically impossible to avoid losses altogether, but can be reduced by following improved method of preparation of farmyard manure. Trenches of size 6 m to 7.5 m length, 1.5 m to 2.0 m width and 1.0 m deep are dug.

All available litter and refuse is mixed with soil and spread in the shed so as to absorb urine. The next morning, urine soaked refuse along with dung is collected and placed in the trench. A section of the trench from one end should be taken up for filling with daily collection. When the section is filled up to a height of 45 cm to 60 cm above the ground level, the top of the heap is made into a dome and plastered with cow dung earth slurry. The process is continued and when the first trench is completely filled, second trench is prepared.

The manure becomes ready for use in about four to five months after plastering. If urine is not collected in the bedding, it can be collected along with washings of the cattle shed in a cemented pit from which it is later added to the farmyard manure pit.

Partially rotten farmyard manure has to be applied three to four weeks before sowing while well rotten manure can be applied immediately before sowing. Generally 10 to 20 t/ha is applied, but more than 20 t/ha is applied to fodder grasses and vegetables. In such cases farmyard manure should be applied at least 15 days in advance to avoid immobilization of nitrogen. The existing practice of leaving manure in small heaps scattered in the field for a very long period leads to loss of nutrients. These losses can be reduced by spreading the manure and incorporating by ploughing immediately after application.

Vegetable crops like potato, tomato, sweet-potato, carrot, raddish, onion etc., respond well to the farmyard manure. The other responsive crops are sugarcane, rice, napier grass and orchard crops like oranges, banana, mango and plantation crop like coconut.

The entire amount of nutrients present in farmyard manure is not available immediately. About 30 per cent of nitrogen, 60 to 70 per cent of phosphorus and 70 per cent of potassium are available to the first crop.

Sheep and Goat Manure

The droppings of sheep and goats contain higher nutrients than farmyard manure and compost. On an average, the manure contains 3 per cent N, 1 per cent P_2O_5 and 2 per cent K_2O.It is applied to the field in two ways. The sweeping of sheep or goat sheds are placed in pits for decomposition and it is applied later to the field. The nutrients present in the urine are *wasted* in this method. The second method is sheep penning, wherein sheep and goats are kept overnight in the field and urine and faecal matter added to the soil is incorporated to a shallow depth by working blade harrow or cultivator or cultivator.

Poultry Manure

The excreta of birds ferment very quickly. If left exposed, 50 percent of its nitrogen is lost within 30 days. Poultry manure contains higher nitrogen and phosphorus compared to other bulky organic manures. The average nutrient content is 3.03 per cent N; 2.63 per cent P_2O_5 and 1.4 per cent K_2O.

CROP RESIDUES

Harvest residues (straws, stover and haulms) and process wastes, groundnut shell, rice husks, oil cakes) are the two major components of crop residues. Among these, rice straw and sorghum straw are the popular cattle feeds. Therefore, only small percentage of it will be available for use as organic manure in the fields. This crop residues contribute not only nitrogen but also phosphorous, potassium and minor nutrients (Table) Rice straw, the most readily available material in rice growing areas, is a good source of silica. Nutrients when added as biomass are not in a readily available form for crops and need to be mineralized by microbial activity. It is widely accepted that only a portion of the N applied as biomass to the soil through soil incorporation is recovered by the crop. Thus nutrients from a given quantity of biomass added in a year are available over several years. Also, a given quantity of biomass applied as surface mulch versus as incorporated in soil is likely to have different value and may take different time for degradation. Placed on soil surface the biomass serves as mulch

and potentially helps to prevent the loss of soil moisture, decreases the soil temperature and prevents soil degradation. About 2000 million tonnes (Mt) of straw is produced annually in India.

Nutrient content and potential of crop residues

Crops	Nutrient %			Residue (Mt)	Nutrient potential (Mt)	
	N	P_2O_5	K_2O		Total	Utilizable
Rice	0.61	0.18	1.38	110.50	2.40	0.80
Wheat	0.48	0.16	1.18	82.63	1.50	0.50
Sorghum	0.52	0.23	1.34	12.54	0.26	0.09
Maize	0.58	0.18	1.35	11.97	0.25	0.08
Pearl millet	0.45	0.16	1.14	6.97	0.12	0.04
Barley	0.52	0.18	1.30	2.48	0.05	0.02
Finger millet	1.00	0.20	1.00	5.35	0.12	0.04
Sugar cane	0.45	0.18	1.28	22.74	0.42	0.42
Potato	0.52	0.21	1.06	7.87	0.14	0.14
Ground nut	1.65	0.23	1.37	10.60	0.34	0.34
Total				273.63	5.61	2.47

Water hyacinth, an aquatic weed of 2.92 lakh ha is a nuisance and creates serious problem in normal flow of water, fish transport and human health. Water hyacinth can be used as compost, green manuring, soil mulch, etc. It can contribute about 0.17 million tonnes equivalent of NPK Water hyacinth on dry weight basis can yield 370 l of biogas with average methane content of 69%.

Vermicompost

Among the soil fauna earthworms have attracted more attention than others because of their importance in agriculture. Earthworm gut is the site of production of genuine humic acids which are distinct from the polysaccharide-gum humic acids. About half of the gums secreted by earthworm are in form of mucoproteins that help stabilizing pore space distribution. The earthworm soil casts are richer in available plant nutrients (nitrate nitrogen, exchangeable Ca, Mg, K and P) and organic matter. The earthworms through their casts and

dead tissues supply about 60-90 kg N to the soil. Earthworm eats on fungal mycelia. Earthworms convert farm waste and organic residues into high quality compost.

Nutrient composition of vermicompost
Nutrient Content

1. Organic carbon 9.15 to 17.98 %
2. Total nitrogen 1.5 to 2.10 %
3. Total phosphorus 1.0 to 1.50 %
4. Total potassium 0.60 %
5. Ca and Mg 22.00 to 70.00 m.e / 100 g
6. Available S 128 to 548 ppm
7. Copper 100 ppm
8. Iron 1800 ppm
9. Zinc 50 ppm

Besides the above nutrients the vermicompost also contains Protease, Lipase, Amylase, Cellulase enzymes. Conversion rates: 1000 earth worms may convert 5 kg waste material per day 1000 worms weighs about a kilogram

Advantage of vermicompost

1. Vermicompost is a rich source of nutrients, vitamins, enzymes, antibiotics and growth hormones. So it gives disease resistance to plansts. Nutrient content of vermicompost is higher than traditional composts. It is a valuable soil amendment.
2. Vermicompost harbours certain microbial populations that help in N fixation and P solubilisation. Its application enhances nodulation in legumes and symbiotic mycorrhizal associations with the roots.
3. Superiority of vermicompost over other synthetic growth media is more pronounced in plant nurseries. It can be used as rooting medium and for establishment of saplings in nurseries.
4. It improves taste, lusture and keeping quality of the produce.
5. It has immobilized enzymes like protease, lipase, amylase, cellulase and chitinese which keep on their function of biodegradation of agricultural residues in the soil so that further microbial attack is speeded up.
6. It does not emanate foul odour as is associated with manures and decaying organic wastes

Table5: Nutrient potential of green manures

Green manure	**Biomass(to**	**Naccumulation**
		(Kg/ha)
***Sesbania aculeate*(Dhiancha)**	**22.50**	**145.00**
S.rostrata,	20.06	146.00
***Crotalaria juncea,*(Sunnhemp)**	**18.40**	**113.00**

***Tephrosia perpurea*(self seeding)**	6.80	46.00
Greengram	6.50	60.20
Blackgram	5.12	51.20
Cow pea	7.12	63.30

Nutrientcontentofimportantgreenmanures

Crop	**Nutrientcontent(%ondryweightbasis)**		
	N	P	K
Green manure			
Sesbania aculeate	3.3	0.7	1.3
Crotalaria juncea	2.6	0.6	2.0
Sesbania speciosa	2.7	0.5	2.2
Tephrosia purpurea	2.4	0.3	0.8
Phaseolus trilobus	2.1	0.5	-
Green leaf manure			
Pongamia glabra	3.2	0.3	1.3
Glyricidia maculeata	2.9	0.5	2.8
Azadirachta Indica	2.8	0.3	0.4
Morinda tinctoria			
Calatropis gigantecum	2.1	0.7	3.6

Biogas slurry is a good source of organic manure

The following are the different methods of applying bio-digested slurry as manure:

a) Air dried biogas slurry can be applied by spreading on the land atleast one week before sowing the seeds or transplanting the seedlings.

b) The liquid slurry can be mixed directly with the running water in irrigation canal

c) Biogas slurry can also be coated on the seeds prior to sowing.

d) The digested slurry is filtered in the layer of leaves and the semi-solid slurry can be transported easily and used for topdressing of crops like sugarcane and potato.

e) Biodigested slurry is also being used to fish culture, which acts as a supplementary feed. On an average, 15-25 litre of wet slurry can be applied perday in a 1200 sqpond. Slurry mixed with oil cake or rice-bran in the2-1 ratio increases the fish production remarkably

f) The digested slurry, if mixed with Azospirillum, KMB and PSM @ 200ml each/ acre ensures increase of yield minimum 30% over slurry alone.

In general, organic manures, about 10t/ha in the form of FYM or compost or bio-digested slurry is recommended to be applied once in three years to maintain the organic content of soil, besides providing nitrogen, phosphorus and potassium in the form of organic fertilizers to the crop.

Nutrient content in biogas slurry and FYM

Particulars	Nutrient content						
	Major element (percent)				Trace element(ppm)		
	N	P	K	Fe	Mn	Zn	Cu
Biogas slurry	**1.43**	**1.21**	**1.01**	**4200**	**550**	**150**	**52**
FarmYardmanure	0.94	0.56	0.72	4000	490	100	45

Lecture-6

Off-farm resources; coir pith, press mud, oil cakes, fly ash , bio-compost, minerals, bone meal, bio-fertilizers, traditional preparations

Agro-industries generate residues like husk, hull, shell, peel, testa, skin, fibre, bran, linter, stone, seed, cob, prawn, low grade fish, leather waste, hair, bones, coir dust, saw dust, bamboo dust, etc. which could be recycled or used efficiently through agro-processing centers. In the last three decades, rice and sugarcane residues have increased by 162 and 172 %, respectively. Their disposal problem needs serious rethinking. To some extent these organic residues are used as soil conditioner, animal feed, fuel etc.,

Coir pith compost

The coir dust is a waste product of the coir industry and could be used as organic amendment. The coir waste accumulates in large quantities near the coir industrial units and it can be converted into valuable organic manure by proper composting with the aid of the mushroom fungus ***Pleurotus sojorcaju*.** The composting reduces the volume by 42 per cent besides narrowing the C : N ratio. Once the compost is matured add potash mobilizing bacteria and phosphorus solubilizing bacteria at the rate of 400ml/tonne increase the availability of P_2O_5 and K_2O within 20 days. It has been found to be a good amendment for sodic soils and a moisture conserving material for rainfed crops. The use of coir pith without composting is, however, not advisable. In the fresh form it contains 8 - 12 per cent soluble tannin - related phenolics which can inhibit microbial activity in the soil and cause immobilization of nitrogen. **Coir pith contains more of potassium nutrient.**

Composition of raw and composed coir pith

Properties	**Raw Coir pith**	**Composted coir pith**
Lignin(%)	30.00	4.80
Cellulose(%)	26.52	10.10
Organic carbon(%)	29.00	24.90
N(%)	0.26	**1.06**
P_2O_5(%)	0.01	**0.06**
K_2O (%)	0.78	**1.20**
Ca (%)	0.40	0.50
Mg (%)	0.36	0.48
Fe (%)	0.07	0.09
Mn (ppm)	12.50	25.00

Zn (ppm)	7.50	15.80
Cu (ppm)	3.10	6.20
C:N ratio	**112:1**	**24:1**

Press mud

The major sugar factory wastes are pressmud, filter cakes bagasse and molasses. Pressmud contains about 1.2 percent N, 3.82 per cent P_2O_2, 1.42 per cent K_2O and 11.1 per cent CaO and is a good source of organic matter (4.4 to 35.8%). Because of high calcium content in the pressmud, it is used mainly to reclaim acidic soils substituting gypsum to some extent.

There are two forms of pressmud: pressmud with carbonate process having $CaCO_3$ (66%) and organic matter (4.4%); presumed with sulphitation process having $CaSO_4$ (9.3%) and organic matter (35.8%). The amount equivalent to one kg of 100 percent pure gypsum will be 0.88 kg and10.75 kg of carbonated and sulphated pressmud, respectivley and in terms of sulphur 4.0 and 57.8 kg, respectively. Addition of pressmud improves soil aeration and drainage in heavy soils, whereas in sandy soils it helps in improving the retention of moisture. When added to sugarcane fields it increased the cane yield, improved the juice quality and enhanced the ammonifying power of the soils.

Sugarcane trash

The sugarcane trash removed during cropping to prevent lodging of the canes and the dead leaves collected at the time of cane harvest are generally burnt in the field itself. The valuable nutrients are lost during the process. The trash may be converted into a very useful organic amendment by proper composting and adding mussoorie rock phosphate at the time of composting. As the pressmud itself is rich in Potash and Phosphorus, use of 400ml/tonne of composted material with Potash Mobilizing Bacteria (*Fraturia aurientia)* and Phosphate Solubilizing Bacteria increases the availability of K_2O and P_2O_5 in the wastes. This may be used later like other organic amendments.

Oil cakes

Many oil cakes such as the castor, neem, madhuca, karanja, linseed, rape seed and cotton seed which are non-edible oil cakes may serve as useful organic manure as these contain high amounts of plant nutrients. Most of the non-edible oil cakes are valued much for their alkaloid contents which inhibit the nitrification process in soils. Neem cake contains the alkaloids - nimbin and nimbicidine which effectively inhibit the nitrification process . Similarly, Karanjin (Pongamia pinnata) and (*Madhuca butyracea*) is a potent nitrification inhibitor equal in efficiency to nitropyrin inretarding the nitrification process of ammoniacal nitrogen and increasing the yield, nitrogen uptake and grain protein content of rice . Madhuca cake has been successfully used in coastal saline soils for cultivatiion of rice. soils.

Non edible oil cakes

These are the good source of nitrogenous manures. The amount of nitrogen varies from 2.5% to 7.9%. In addition to nitrogen all oilcakes contain small quantities of phosphorous (0.8 to 2.9%) and potassium (1.1 to 2.2%). Oil cakes are insoluble in water. But their nitrogen becomes quickly available in about a week or ten days after application to crops. All oil cakes have been found to give good result with almost every crop and on all type of soils.

Neem cake as organic manure protects plant roots from nematodes, soil grubs and white ants due to presence of **Limonoid.**

Concentrated organic manures

These are organic manures rich in plant nutrient and low in volume.

1. Oil cakes 2.Blood meal 3. Meat meal 4.Fish meal 5. Horn and hoof meal

Nutrient content of some concentrated organic manures

Manure	Percentage composition		
	N	$P_2 0_5$	$K_2 0$
Caster cake	4.0-4.4	1.9	1-4
Groundnut cake	6.5-7.5	1.3	1.5
Cotton seed cake (decorticated)	6.9	3.1	1.6
Cotton seed cake (undecorticated)	3.6	2.5	1.6
Linseed cake	5.6	1.4	1.3
Coconut cake	3.4	1.9	1.9
Neem cake	**5.2-5.6**	1.1	1.5
Safflower cake (decorticated)	**7,9**	2.2	1.9

Safflower cake(undecorticated)	**4.9**	1.4	1.2
Sesamum cake	4.7-6.2	2.1	1.3
Mahua cake	2.5	0.8	1.9
Niger cake	4,7	1.8	1.3
Pungam cake	4.0	1.0	1.3
Raw bone meal	3.0-4.0	20-25	-
Steamed bone meal	1.0-2.0	25-30	-
Basic slag	4.0	14-18	1.3
Fish slag	4-10	3-9	1.5
Blood meal	10-12	1-2	1.0
Meat meal	9-11	3.5	-
Horn and hoof meal	10-15	1	-
Press mud	1-1.5	4-5	2-7
Guano (Peruvian bird)	11-16	8-12	2-3

Fly ash

Fly ash is a useful organic amendment. At many places acidic mined lands exhibit reduced biological activity. In some strip mine sites with pH ranging from 2.6 to 4.7, the pH was found to range between 5.0 and 7.5 after 10 years of treatment with fly ash and the credit of vegetation on these sites was rated fair to good. With the exception of N, fly ash is rich in P_2O_5, K_2O, Ca and Mg and also contains elements like B, Mo and Zn which are essential for plant growth and are available to the vegetation raised on soils treated with fly ash. Mixing of potash mobilizing bacteria and phosphorus solubilizing bacteria at the rate of 400ml/tonne increase the availability of P2O5 and K2 O with in 20 days

Saw Dust

It is possible to use waste products from timber industry as organic matter. Inoculation with cellulolytic strains of *Bacillus sp*., *Cephalosporium sp*. And *Streptomyces sp*. has been reported to increase the decomposition of saw dust and bark . Mixing of potash mobilizing bacteria and phosphorus solubilizing bacteria at the rate of 400ml/tonne increase the availability of P2O5 and K2 O with in 20 days

The following criteria for judging a good quality compost may be as follows:

1. Dark brown to black.
2. Crumbly structure not packed or lumpy.
3. Largely insoluble in water.
4. C:N ratio from 10 to 20.
5. Dissolves to a large extent in dilute alkali.
6. Beneficial effect upon soil and growing vegetation.
7. Pleasant earthy smell.
8. Will not attract flies.

Leaf composting

Leaf composting, can be achieved by heap or ditch composting or by windrow composting. Windrow are preferred as they allow efficient handling of materials. Provide good aeration, allow sufficient absorption of water and are easy to be formed.

Formation of uniform-shaped windrow from 2.40m – 3.60 m at the base and 2.40m-3.00m high and of any convenient length. Windrow built too high will have excessive compaction at the base resulting in anaerobic conditions. Windrows built too low will not allow sufficient insulation to sustain thermophilic temperature during cold weather. To ensure proper aeration, it is important to break apart tightly compacted leaves.

Biocompost

Biocomposting is a process for the rapid conversion of organic waste into a thoroughly decomposed, stable and humus rich compost for used as a fertilizer and soil conditioner. The process is aerobic and the technology highlights the following activities for large and commercial scale Bio-composting.

Windrowing: The waste materials should be chopped in small size and formed in windrows of 3 meters width and 1.2 meters height, the length according to the availability of the land:

Inoculation: Mixed population of microorganisms are sprinkled over the windrows at the rate of 4kg/tone of waste materials. In case of liquid 2 litres/tone is enough as the population of fungus and bacteria is 100 times more than solid base inoculum.

Aero tilling: The windrow is aero tilled in alternative days by using special type of machine called "Aero tiller" or manual method. It helps for uniform mixing and provides oxygen to the microorganisms.

Application of Spent wash: To maintain the optimum moisture of 50-60% and to maintain temperature between 65-700c for high rate composting, the nutrient rich spent wash is sprayed on the windrows, if it is available. Otherwise, any wash like cow dung wash, vermiwash, kitchen wash, any animal wash etc. can be sprayed.

Bone meal

Long before synthetic commercial fertilizers were available, farmers turned to bone meal to enrich soils in need of amendment. Bone meal is **made from steamed and crushed animal bones, and is rich in phosphorous**, a mineral that plants need for healthy root development and flower growth. It also contains calcium and a little bit of nitrogen, both of which are beneficial to plants. Bone meal is taken up by plants slowly over time, so as long as it is used sparingly, there is little risk of 'burning' plants with too much of this fertilizer.

That makes bone meal an ideal supplement for bulbs and roses in particular, which flourish with an extra boost of phosphorous. A little bit goes a long way, so just one tablespoon is needed for every two square feet. When planting bulbs in the fall, ensure that their spring blooms will be spectacular by adding 1/2 teaspoon per plant to the backfill soil. You can also mix bone meal into the top one to three inches of soil in the spring for other plants.

Blood meal Dried blood meal is a by-product of slaughter house. It contains 10 to 12% highly available nitrogen and 1 to 1.5% phosphorous and 1.0 per cent potassium. It is very quick acting manure and is effective on all crops and all types of soil.

Mineral fertilizer

The mineral fertilizers, which are allowed in organic agriculture, are based on ground natural rock. However, they may only be used as a supplement to organic manures. If they contain easily soluble nutrients, they can disturb soil life and result in unbalanced plant nutrition. In some cases, mineral fertilizers are ecologically questionable as their collection and transport is energy consuming and in some cases natural habitats are being destroyed.

MINERAL FERTILIZERS ALLOWED IN ORGANIC FARMING – A BASIC OVERVIEW

Fertilizer	Origin	Characteristics	Application
Plant ashes	Burned organic material	Mineral composition similar to plants	To compost Around the bade of the plant
Lime	Ground lime stone algae	Buffers low pH Rich in trace elements	Every two to three years when soil pH is low
Stone powder	Pulverized rock	Trace elements The finer the grinding better the adsorbance	To FYM Reduces the volatilization of N and encourages the

			rotting process
Rock phophate	Pulverized rock containing P	Easily adsorbed to soil minerals Weekly adsorbed to organic matter Slow reaction	Rotting process To compost Not to reddish soil

Traditional preparations

Some important formulations for soil enrichment

Preparation of liquid manures Many variants of liquid manures are being used by farmers of different states. Few important and widely used formulations are given below:

Panchgavya – Mix fresh cow dung 5 kg, cow urine 3 lit, cow milk 2 lit, curd 2 lit, cow butter oil 1 kg and ferment for 7 days with twice stirring per day. Dilute 3 lit of Panchgavya in 100 lit water and spray over soil. 20 lit panchgavya is needed per acre for soil application along with irrigation water.

Enriched Panchgavya (or Dashagavya) – Ingredients - cow dung 5 kg, cow urine 3 lit, cow milk 2 lit, curd 2 lit, cow deshi ghee 1 kg, sugarcane juice 3 lit, tender coconut water 3 lit, banana paste of 12 fruits and toddy or grape juice 2 lit. Mix cow dung and ghee in a container and ferment for 3 days with intermittent stirring. Add rest of the ingredients on the fourth day and ferment for 15 days with stirring twice daily. The formulation will be ready in 18 days. Sugarcane juice can be replaced wit 500 g jaggery in 3 lits water. In case of non-availability of toddy or grape juice 100g yeast powder mixed with 100 g jaggery and 2 lit of warm water can also be used. For foliar spray 3-4 lit panchgavya is diluted with 100lit water. For soil application 50 lit panchagavya is sufficient for one ha. It can also be used for seed treatment.

Jivamrut – Mix cow dung 10 kg, cow urine 10 lit, Jaggary 2 kg, any pulse grain flour 2 kg and Live forest soil 1 kg in 200 lit water. Ferment for 5 to 7 days. Stir the solution regularly three times a day. Use in one acre with irrigation water.

Amritpani - Mix 10 kg cow dung with 500 gm honey and mix thoroughly to form a creamy paste. Add 250 gm of cow desi ghee and mix at high speed. Dilute with 200 lit water. Sprinkle this suspension in one acre over soil or with irrigation water. After 30 days apply second dose in between the row of plants or through irrigation water.

Sanjivak – Mix 100 kg cow dung, 100 lit cow urine and 500 gm jaggery in 300 lit of water in a 500-lit closed drum. Ferment for 10 days. Dilute with 20 times water and sprinkle in one acre either as soil spray or along with irrigation water.

BIOFERTILISERS

Atmosphere **contains 78 % nitrogen and 0.03 % carbon dioxide**. Plants are able to assimilate carbon dioxide through photosynthesis even when carbon dioxide content is less, but most of the plants cannot fix atmospheric nitrogen though it is abundant. Some microorganisms are capable of fixing nitrogen, while some can increase the availability of nitrogen and phosphorus.

Rhizobium species suitable for different crops	
Rhizobium sp	Crops
R. leguminosarum	Peas (Pisum) lathyrus, vicia, lentil (Lens)
R. Tripoli	Berseem (Trifolium)
R. phaseoli	Kidney bean (Phaseolus)
R. lupine	Lupinus Ornithopus
R. meliloti	Melilotus, Lucerne (Medicago), Fenugreek (Trigonella)
R. japonicum	Soybean (Glycine)
Cowpea miscellany,	Cowpea, clusterbean, greengram, blackgram, redgram, groundnul, moth bean, dhaincha, sunnhemp, Glyricidia. Acacia. Prosopis. Dalbergia. Albizzia. Indigofera. Tephrosia. Atylosia. Stylo
Separate group	Bengal gram (gram)

Symbiotic Bacteria Bacteria belonging to the genus Rhizobium are capable of fixing atmospheric N2 in association with leguminous crops. Different species of Rhizobium are used for treating the leguminous crops

Rhizobium sp enter the roots of host plants and form nodules on the root surface. The bacteria depend on the host plant for carbohydrates and water while Rhizobium supplies N to the host. Nitrogen fixed by the Rhizobium is translocated through xylem vessels of the host plant mainly in the form **of aspergine and to some extent as glutamine**. Rhizobium species suitable for different crops are multiplied on a peat base in laboratories. This inoculum can be applied in three ways and among them, seed treatment is the best.

Seed Treatment. Depending on the seed rate, the required quantity of jaggary is boiled in water and cooled. Rhizobium inoculum (1.5 kg/ha) is mixed in the jaggary solution and sprinkled over the seeds. Then the seeds are thoroughly mixed to spread the inoculum over the entire surface of the seeds. Seeds are then shade dried.

Soil Treatment. The Rhizobium inoculum is mixed with soil and spread over the field.

Soil Application. If Rhizobium inoculum is not available, 200 kg of surface soil (2 to 10 cm depth) can be collected from the fields where that 'particular leguminous crop is grown

luxuriantly and this soil can be broadcasted over the field where crop is grown for the first time.

Free living organisms

The important free living organisms that can fix atmospheric nitrogen are blue green algae (BGA), Azolla, Azotobacter and Azospirillum. Among them, **BGA and Azolla can survive only in lowland conditions.** Small quantity of inoculum of BGA and AzolIa can be obtained from laboratories and they can be multiplied in the farmers' fields for subsequent application.

Azotobacter and Azospirillum

Azotobacter chroccum is capable of fixing 20 to 30 kg N/ha. It can be applied by seed inoculation, seedling dip or by soil application. The inoculum required is 3 to 5 kg/ha. Application of 5 t/ha of farm yard manure helps in better growth of Azotobacter. **Azotobacter can be used for rice, cotton and sugarcane. Suitable biofertilizer for sugarcane is Acetobacter. Azospirillum inoculum is used for sorghum.**

Phosphobacteria

Inoculated through seed, seedling root dip and soil application methods as in the case of *Azospirillum*. Combined application of bacterial bio-fertilizers, Phosphobacteria can be mixed with *Azospirillum* and *Rhizobium*. The inoculants should be mixed in equal quantities and applied as mentioned above.

Types of bio-fertilizers: There are two types of bio-fertilizers.

1. Symbiotic N-fixation: Rhizobium is the most widely used bio-fertilizers which colonizes the roots of specific legumes to form tumours like growths called root nodules. It is these nodules that act as factories of ammonia production. The rhizobium legume association can fix upto 100-300 kg N/ha in one crop season. The rhizobium cultures of various strains which multiply in roots of suitable legumes and fix nitrogen symbiotically. Almost 50% demands of N are met by these microorganisms in legumes.

2. Asymbiotic N-fixation: This includes Azatobacter, Azospirillium, BGA, Azolla and Mycorrhizae which also fixes atmospheric N in suitable soil medium. They grow on decomposing soil organic matter and produce nitrogen compounds for their own growth and development, besides that they leave behind a significant amount of N in surroundings.

Biofertilizers can be grouped in different ways based on their nature and function.

S. No.	Groups	Examples
N_2 fixing Biofertilizers		
1.	Free-living	*Azotobacter, Clostridium, Anabaena, Nostoc*
2.	Symbiotic	*Rhizobium, Frankia, Anabaena azollae*
3.	Associative Symbiotic	*Azospirillum*
P Solubilizing Biofertilizers		
1.	Bacteria	*Bacillus megaterium* var. *phosphaticum, Bacillus subtilis Bacillus circulans, Pseudomonas striata*
2.	Fungi	*Penicillium sp, Aspergillus awamori*
P Mobilizing Biofertilizers		
1.	Arbuscular mycorrhiza	*Glomus sp. ,Gigaspora sp.*
Biofertilizers for Micro nutrients		
1.	Silicate and Zinc solubilizers	*Bacillus* sp.
Plant Growth Promoting Rhizobacteria		
1.	Pseudomonas	*Pseudomonas fluorescens*

Application of Biofertilizers

1. Seed treatment or seed inoculation
2. Seedling root dip
3. Main field application

Seed treatment

Three packets **(600 grams)** of the inoculant are mixed with 600 ml of rice kanji to make slurry. The seeds required for one hectare are mixed in the slurry so as to have a uniform coating of the inoculant over the seeds and then shade dried for 30 minutes. The shade dried seeds should be sown within 24 hours.

Seedling root dip

This method is used for transplanted crops. Three packets **(600 grams)** of the inoculant are mixed in 60 litres of water. The root portion of the seedlings required for one hectare is dipped in the mixture for 5 to 10 minutes and then transplanted.

Main field application

Ten packets **(2 kg)** of the inoculant is mixed with 50 kgs of dried and powdered farm yard manure and then broadcasted in one hectare of main field just before transplanting.

Liquid biofertilizers

Liquid bio-fertilizers are the microbial preparations containing specific beneficial microorganisms which are capable of fixing or solubilizing or mobilizing plant nutrients by their biological activity.

Blue-green Algae (BGA)

Several species of BGA can fix atmospheric N. The most important species are Anabaena andNostoc. The amount of N fixed by blue-green algae ranges from 15 to 45 kg N/ha. Standing water of 2 to 10 cm in the field is a prerequisite for the growth of BGA. It can grow in a temperature range of 25 to 45°C. Blue green algae inoculum is applied after transplantation of rice crop in the main field. The inoculum required is 10 kg/ha. For higher nitrogen fixation, 3 to 4 t/ha of farmyard manure and 200 kg/ha of superphosphate are applied.

Azolla

Azolla is a free floating fresh water fern. ***Azolla pinnata*** is the most common species occurring in India. It fixes nitrogen due to Anabaena species of blue-green algae present in the lobes of Azolla leaves. A thick mat of Azolla **supplies 30 to 40 kg N/ha**. Unlike blue-green algae, it thrives well at low temperature Azolla is applied to the main field as a green manure crop and as a dual crop. As green manure crop, AzolIa is allowed to grow on the flooded fields for 2 to 3 weeks before transplanting. Later water is drained and Azolla is incorporated by ploughing in. **As a dual crop, 1,000 to 5,000 kg/ha of Azolla is applied to the soil one week after transplanting.** When a thick mat forms, it is incorporated by trampling. The left over Azolla develops again which is trampled in as a second crop. For better growth of Azolla, 25 to 50 kg/ha of superphosphate is applied and standing water of 5 to 10 cm is maintained continuously in the rice fields.

Mycorrhiza and Phospho-micro Organisms

Phosphorus availability and fertilizer phosphorous use efficiency can be increased with mycorrhiza, phosphate solubilizing bacteria and fungi. Mycorrhiza inhabits roots of several crops and solubilises soil phosphates. Inoculation of mycorrhiza increases the pod yield of groundnut.

Some microorganisms like Psuedomonas striate. Aspergillusawaneorii and Bacillus polymyxa are capable of solubilising phosphates. The inoculum of these microorganisms is applied to increase the availability of phosphorus. Saprophytes Microorganisms that are capable of decomposing organic matter at a faster rate can be used as a fertilizer for quick release of nutrients.

Aspergillus, Penicillium, Trichoderma are cellulolytic fungi which break down cellulose of plant material. The natural process of decomposition is accelerated and composting time is reduced by 4 to 6 weeks by the use of inoculants of these organisms.

Lecture-7

Organic waste recycling methods and techniques-composting, vermicomposting, *in situ* composting- System approach.

Compost

A mass of rotted organic matter made from waste is called compost. The compost made from farm waste like sugarcane trash, paddy straw, weeds and other plants and other waste is called farm compost.

The average nutrient content of farm compost **is 0.5 % N, 0.15 % P2O5 and 0.5 % K2O.** The nutrient value of farm compost can be increased by application of superphosphate or rock phosphate at 10 to 15 kg/t of raw material at the initial stage of filling the compost pit.

Farm compost is made by placing farm wastes in trenches of suitable size, say, 4.5 m to 5.0 m long, 1.5 m to 2.0 m wide and 1.0 m to 2.0 m deep. Farm waste is placed in the trenches layer by layer. Each layer is well moistened by sprinkling cow-dung slurry or water. Trenches are filled up to a height of 0.5 m above the ground. The compost is ready for application within five to six months.

Compost prepared by traditional method is usually low in nutrients and there is need to improve its quality. Enrichment of compost using low cost N fixing and phosphate solublizing microbes is one of the possible ways of improving nutrient status of the product. It could be achieved by introducing microbial inoculants, which are more efficient than the native strains associated with substrate materials.

Inoculation with Azotobacter/Azospirillum and phosphate solubilising culture in the presence of 1% rock phosphate is a beneficial input to obtain good quality compost rich in N (1.8%). The humus content was also higher in materials treated with microbial inoculants. Fungi are responsible for the decomposition of many complex plant polymers in soil and compost.

The following basic rules are important for the production of good quality compost:

1. The purpose of composting is to convert organic matter into growth promoting substances, for sustained soil improvement and crop production.

2. The organic matter is partially decomposed and converted by microbes**. The conditions should be a favorable Moisture 50% aeration 50% and C:N ratio 30:1**

3. Soil microorganisms constitute sufficiently to the decomposition of organic matter through their continuous activities. A typical compost earthworm is ***Eisenia foetida.***

4. Certain additives accelerate the conversion and improve the final product. The materials such as lime, earth, gypsum, rock phosphate act as effective additives. The addition of nitrogen (0.1 to 1 %) is important in case of large C:N ratio of the composting material. Addition of lime (0.3 to 0.5 %), if sufficient lime is not present. The preparation of compost takes 2-3 months. The composition of compost varies with in wide limits.

Stages of composting

When organic material is composted in a batch, four stages of the composting process are apparent. Although the same phases occur during continuous composting, they are not as apparent as they are in a batch, and, in fact, they may be occurring concurrently rather than sequentially.

The four phases include:

1) mesophilic phase;

2) thermophilic phase;

3) cooling phase; and

4) curing phase.

In composting process, as the temperature rises above 40°C, the thermophilic organisms are replaced by mesophilic organisms. Immature compost can be harmful to plants. Uncured compost can produce phytotoxins (substances toxic to plants), can rob the soil of oxygen and nitrogen, and can contain high levels of organic acids.

Methods of composting

The process of composting was first initiated in England during the period of **First World War (1914 -1918). The various systems of composting are**

1. **ADCO process (Agricultural Development Company)**
2. **Activated compost process**
3. **Indore process**
4. **Bangalore process**
5. **Coimbatore process**
6. **Rain -water compost**
7. **Vermicompost .** Some are as follows

1. Indore process:

This process is developed in India by **Howard and Ward at the Indian Institute of plant Industry, Indore** Materials needed: a) Straw or organic farm wastes as basic raw materials b) Cattle dung as starter (urine, earth and wood ashes)

Procedure: A compost pit of dimensions of 30 x 14 x 3 feet with sloping sides (narrow at bottom and at wide surface) is prepared and the raw material is spread in layers of 3" thickness. A mixture of urine, earth, and wood ashes is sprinkled and this is followed by 2" layer of dung. The pit is filled up this way until the material occupies a height of 3 feet above the ground level . As air can conveniently penetrate only to a depth of 1.5 to 2.0 feet extra aeration has to be provided, which is done by means of artificial vents (holes) of 4" diameter pipe for every 4 feet length of the pit. The pit is watered twice a day i.e., morning and evening with rose can. The material is turning over 3 times, i.e., First – at the end of the first fort night Second – at the end of the second fort night Third – when the material is two months old in the process of composting.

Observations: I. After 10 days of composting the following things happens A. Synthesis of humus begins i.e., development of fungi and the height of the material is reduced by half B. Check anaerobic decomposition, as indicated by the foul smell and fly breeding C. If there is an anaerobic decomposition ,turn over material for proper aeration D. If insufficient fermentation, hasten by watering the material.

II. At the end of two months A. Fungal activity is over B. Materials become dark C. Now the bacterial aeration takes place D. Stock the material on the ground after 2 months .So 25 % of additional free nitrogen will be fixed from atmosphere. Compost is ready by 3-4 months. One cattle pair produced 50-60 cartloads per year.

2. Bangalore process

This process of composting was developed by Dr.C.N.Acharya in 1949.

1. Basic raw material used: Any organic material 2. Starters or inoculants: FYM or mixture of dung and urine or litter [Undecomposed] 3. Additives:Bone meal or oil cakes, wood ash

Procedure [Pit size: 20 x 4 x 3 feet The basic raw material is spread in a pit of 20 x 4 x 3 feet dimensions to a depth of 6 " layer, moistened with 20-30 gallons of water if the material is dry. Over this FYM or preferably a mixture of dung, urine and litter (undecomposed) from

the cattle shed is placed as a layer of 2" thickness. It is again covered on the top with a layer of earth to a thickness of 6". It is beneficial to mix the earth with bone meal or oil cakes, wood ash etc., to improve manurial value of the compost .The piling of layers is continued till the heap raises above the ground level to a height of 2 feet. Then the heap is kept open for one week to facilitate aerobic decomposition. Later the heap is plastered with a layer of moist clay for anaerobic fermentation to occur. Fissures , or cleavages (cracks) that occur in the clay layer , have to be sealed off periodically .The compost will be ready in 4-5 months period starting from the day of preparation .This process is called as aerobic and anaerobic decomposition of compost. In this process the basic raw material is not so well decomposed as in the other methods .But organic matter and N contents are well conserved. The number of turnings are reduced .The out turn of the compost is relatively greater and cheapest process.

3. The coimbatore process:

The basic raw materials: 1.Raw organic matter 2. Starters : Powdered bone meal and cattle dung and water emulsion prepared by mixing Dung in water at 5-10 kg dung in 5-10 litres water. Procedure [Pit: 12 x 6 x 3 feet]: The basic raw material loosely spread [Pit: 12 x 6 x 3 feet] to a depth of 9" and water is sprinkled till the entire material is moist. Then about one kg of powdered bone meal is broadcasted uniformly above the layer and above this an emulsion of 5-10 kg of fresh cattle dung in 5-10 liters of water is applied. Repeat this process until a heap 2 feet above ground level is formed .Then the entire exposed surface area of heap is plastered with mud to facilitate

In Coimbatore method, there is anaerobic decomposition to start with, following by aerobic fermentation. It is the reverse in Bangalore method. Semi-aerobic fermentation which would takes place for above 4-6 weeks depending upon the nature of the raw material. After 4-6 weeks, the mud plaster is removed to permit aerobic fermentation. If the heap has sunk unevenly which is a sign of defective fermentation, the material is reheaped after forking and moistened .The decomposition is complete in 3-4 months and is fit for application to the field.

Rain watered compost

In dry areas where it is difficult to obtain water for watering, the composting can be done with the aid of rain fall. The compost heap is built up as usual before the rains set in. The

turnings are given during the rainy period at the end of rains the material will be ready for application. About 400 mm rain fall received in 3-4 months is considered sufficient.

Vermicompost

Soil fauna including protozoa to mammals though not considered major is the important source of nutrients. Among the soil fauna earthworms have attracted more attention than others because of their importance in agriculture. Earthworm gut is the site of production of genuine humic acids which are distinct from the polysaccharide-gum humic acids. About half of the gums secreted by earthworm are in form of mucoproteins that help stabilizing pore space distribution. The earthworm soil casts are richer in available plant nutrients (nitrate nitrogen, exchangeable Ca, Mg, K and P) and organic matter. The earthworms through their casts and dead tissues supply about 60-90 kg N to the soil. Earthworm eats on fungal mycelia. Earthworms convert farm waste and organic residues into high quality compost. For this, **Eisenia foetida, Perionyx exacavatus, Eudrillus euginiae and Lumbrius rubellus are important.** These species can be cultured on organic wastes and dung. The technique of culturing them is called vermiculture and using these for decomposing residue to make compost is called vermicomposting.

About 1000 adult earthworms can convert 5 kg waste into compost per day. The earthworm assimilate 5-10% of the substrate and rest passes through the alimentary canal and is excreted as cast. Earthworm cast contains nutrients, vitamins, hormones and antibiotics.

Vermi-compost is a stable fine granular organic matter, when added to clay soil loosens the soil and provides the passage for the entry of air. he multifarious effects of vermicompost influence the growth and yield of crops.

Definition of vermicomposting: Vermicomposting is a method of making compost, with the use of earthworms, which generally live in soil, eat biomass and excrete it in digested form. This compost is generally called vermicompost or Wormicompost.

Definition of Vermiculture: Vermiculture means scientific method of breeding and raising earthworms in controlled conditions.

Types of earthworms in vermicomposting Earthworms belong to phyllum Annelida of Animal Kingdom. They are long and cylindrical in shape and size having a large number of grooves. There are about 3000 species of earthworms in the world which are adapted to a range of environment. More than 300 species have been identified in India. Although,

hermophrodite, two mature earthworms are required to propagate. At the time of egg laying, the clitellum is transformed into hard, girdle like capsule called cocoon. Shedding of cocoon ranges from 1 to 5, only a few of them survive and hatch. The juveniles and again formation of cocoons takes a period of 50-60 days. Normally, the average life span of earthworms varies with species ranging from 1 to 10 years.

Different types of earthworm live in different niches performing different functions. As a result earthworms do not have the same requirements and will not survive under the same environmental conditions.

Epigeic species: Soil surface dwelling or compost preferring species are called epigeic species**.** Examples of such species found include the dung worm, ***Lumbricus rubellus*** **and the tiger worm** ***Eisenia foetida.***

Endogeic species: The top soil dwelling species are called endogeic species. These species usually live in the top 20cm -30 cm depth of soil and feed primarily on soil and associated organic matter (geophages). The most common earthworm species is grey worm ***Aporrectodea caliginosa***

Anecic species: Subsoil dwelling species / deep burrowing species are called anecic species. These species tend to make permanent burrows in the soil, which can be more than three metres deep. Examples: The night crawler *Lumbricus terrestris,* and *Aporrectodea longa.*

Epigeic earthworm like *Eudrilus eugeniae, ,Eisenia foetida* and *Perionyx excavatus* are some of the popularly used worms for vermicomposting . *Eudrilus eugeniae,* a tropical earthworm commonly called African night crawler is large in size grows rapidly breeds fast and is capable of decomposing large quantities of organic material into valuable vermicompost.

Mechanism of vermicomposting Materials consumed by worms undergo physical breakdown in the gizzard resulting in particles < 2 µ, giving thereby an enhanced surface area for microbial processing. This finally ground material is exposed to various enzymes such as protease, lipase, amylase, cellulase and chitinase secreted into lumen by the gut wall and associated microbes. These enzymes breakdown complex biomolecules into simple compounds. Only 5-10% of the ingested material is abosrbed into the tissues of worms for their growth and rest is excreted as cast. Mucus secretions of gut wall add to the structural stability of vermicompost.

Vermiculture industry or vermicompost preparation: 1. Basic raw material: Any organic material generated in the farm like bhusa, leaf fall etc., Horse dung, due to the risk of

Tetanus virus, lethal to human beings is not advisable to be used as feeding material for earthworms. Paddy husk, merigold and pine needles have also advised to be used as feeding materials for earthworms. 2. Starter: Cow dung , Biogas slurry, or urine of cattle 3. Soil animal: Earth worms (Species: Eisenia foetida) 4. Thatched roof/vermished.

Favourable conditions of earth worms in the composting material: A**. pH: Range between 6.5 and 7.5 B. Moisture: 60-70% of the moisture** below and above range mortality of worms taking place C. **Aeration: 50% aeration from the total pore space D. Temperature: Range between 18 ^{0}C to 35 ^{0}C**

Procedure

It is mostly prepared in either pit or heap method. The dimensions either heap or pit are 10 x 4 x 2 feet. The length and width can be increased or decreased depending on the availability of material but not the depth because the earthworms' activity is confined to 2 feet depth only. First of all select a site which is not under any economic use and is shady and there is no water stagnation. The site should also be nearby to water source. **1st layer: bedding material of 1" thick with soft leaves 2nd layer: 9" thick organic residue layer finely chaffed material 3rd layer: Dung + water equal mixture of 2" layer. Co**ntinue the layer up to pile to ground level in the case of pit method and upto 2′ in heap or surface bed method. Protect the worms against natural enemies like ants, lizards, snakes, frogs, toads etc., Maintain proper moisture and temperature by turnings and subsequent staking. At the day of 24th, 4000 worms are introduced in to the pit [1m2 =2000 worms] without disturbing the pit by regular watering the entire raw material will be turned into the vermicompost in the form of worm excreta. The turnover of the compost is 75% [the total material accommodated in the pit is 1000 kg; the out turn will be 750 kg]

Harvesting of the vermicompost from the pit Stop watering before one week of harvest. Sometimes the worms spread across the pit come in close and penetrate each other in the form of ball in 2 or 3 locations. Heap the compost by removing the balls and place them in a bucket. However, under most instances, top layer has to be disturbed manually. Earthworms move downward and compost is separated. After collection of compost from top layers, feed material is again replenished and composting process is rescheduled. The material is sieved in 2 mm sieve, the material passed through the sieve is called as vermicompost which is stored in a polythene bags

Precautions 1. Do not cover vermicompost beds/heaps with plastic sheets because it may trap heat and gases. 2. Do not overload the vermicompost heap to avoid high temperature that adversely affect their population. 3. Dry conditions kill the worms and waterlogging drive them away. Watering should be done daily in summer and every third day in rainy and winter season. 4. Addition of higher quantities of acid rich substances such as tomatoes and citrus wastes should be avoided. 5. Make a drainage channel around the heap to avoid stagnation of water particularly in high rainfall areas in rainy season. 6. Organic materials used for composting should be free from non-degradable materials such as stones, glass pieces, plastics, ceramic tubes/bulbs etc.

Natural enemies and their control The important natual enemies of vermiculture are ants, termites, centipedes, rats, pigs, birds etc. Preventive measures include treating of the site with chlorpyriphos 20 EC at 2 ml/l or 4% neem based insecticide before filling the heap.

Transportation of live worms Live earthworms can be packed with moist feed substrate in a container (card board/plastic) with provision of aeration. Feed substrate quantity should be roughly 0.5-1.5 g/individual for 24 hours of transportation journey. Culture should contain cocoon, juveniles and adults because sometimes adults do not acclimatize to new environment and may even die. Under such circumstances cocoons are helpful for population build up of earthworms.

Conversion rates: 1000 earth worms may convert 5 kg waste material per day 1000 worms weighs about a kilogram

Advantages of composting over direct application: 1. There will be no immobilization in compost because of narrow C:N ratio 2. Application is easy, because the compost is humified and have a structure of crumb and granular. 3. It is hygienic, pathogens and weeds seeds are destroyed.

The rate of application is as Field crops 5-6 t/ha; vegetables 10-12 t/ha; flower plants 100-200 g/sq ft; fruit trees 5-10 kg/tree.

Nutrient content &Advantage of vermicompost

In-situ vermicomposting can be done by direct field application of vermicompost at 5 t/ha followed by application of cow dung (2.5 cm thick layer) and then a layer of available farm waster about 15 cm thick. Irrigation should be done at an interval of 15 days.

Benefits of Vermicompost

i) Vermicompost have higher nutritive value when compared to bio-compost.

ii) Vermicompost have higher number of beneficial organisms like *Azotobacter, Azospirillum* and phosphobacteria.

iii) These beneficial organisms contribute their benefits to the vermicompost.

iv) Vermicompost have growth promoting substances like indole acetic acid and gibberellic acid to influence crop growth.

v) Toxins and other pollutant if any contains in the waste material, it is completely denatured when it passes through earthworm gut.

System approach in recycling of on farm wastes

Farming system is having different components for different situation. However, some basic components exist in any farming system maintained under varied conditions. They are,

1. Crop components and related cropping system in intensive cropped area
2. Crop residue management
3. Allied enterprises. The enterprises may be animal / Alternate land use system / Sericulture / fish farming
4. Recycling of the wastes of allied enterprises.

Lecture-8

Soil and crop management in organic farming; Inter cropping and companion planting, crop rotation, green manures, cover crops and mulching

How to increase the amount of organic matter in the soil

Organic matter permanently undergoes a process of decomposition. In order to maintain or increase the content of soil organic matter, organic material must be applied again and again. The speed of decomposition depends on the climate (in warm and damp conditions, the organic matter is broken down much faster than in cold or dry conditions) and on how green the material is (C/N-ratio).

Activities that increase the level of soil organic matter:

- **Leaving crop residues** on the field, instead of burning or wasting them, as they are the major source of biomass
- **Applying compost:** this is very effective, as part of the organic matter in compost is al-ready stabilised and will remain in the soil for a longer time than fresh plant material • Applying organic manures: as they contain organic material, they help to increase the content of organic matter; at the same time, they can speed up decomposition as they are rich in nitrogen and thus stimulate soil organisms
- **Mulching** with plant materials or agro-wastes: especially applying hardy material (rich in fibres or wood) will increase the organic matter content, as it will remain in the soil for a long time; in addition, it helps to reduce erosion
- **Using green manures or cover crops:** green manures grown on the same field will con-tribute biomass both from the leaves and roots; material grown on another site contributes only the leaves; the younger the plant material is, the faster will it decompose, thus releasing the nutrients faster but adding less to the built up of soil organic matter
- **Suitable crop rotation:** including crops in the rotation which build up soil organic matter (e.g. pastures); especially perennials and crops with a dense root system (e.g. pastures) are very beneficial
- **Reducing soil tillage:** each tillage will speed up the decomposition of organic material, as it aerates the soil and stimulates soil organisms

- **Avoiding soil erosion:** all methods listed before will be in vain unless soils are prevented from erosion; it carries away those parts of the soil which contain most humus and are most fertile

How to produce more biomass on the farm:

• Integrate green fallow periods with green manures in the crop rotation

• Aim at having the soil covered with plants the whole year round, wherever possible

• Integrate fodder cultivation in the farm where possible (grass, fodder hedges)

• Use unproductive space (e.g. along paths, field borders, steep slopes etc.) for planting trees or hedge rows

• Establish agro-forestry systems, where appropriate

• Leave single trees standing in the field (e.g. nitrogen fixing trees), manage them with intense pruning

• Let cattle graze or spend some nights on harvested fields (it can also be the neighbour‘s cattle) in order to profit from their droppings

Introduction Soil cultivation

Includes all mechanical measures to loosen, turn or mix the soil, such as ploughing, tilling, digging, hoeing, harrowing etc. Careful soil cultivation can improve the soil‘s capacity to retain water, its aeration, capacity of infiltration, warming up, evaporation etc. But soil cultivation can also harm the soil fertility as it accelerates erosion and the decomposition of humus. There is not one right way to cultivate the soil, but a range of options. Depending on the cropping system and the soil type, appropriate soil cultivation patterns must be developed.

Creating good growing conditions for plants There are many reasons for cultivating the soil. The most important ones are to:

• Loosen the soil to facilitate the penetration of plant roots

• Improve the aeration (nitrogen and oxygen from the air)

• Encourage the activity of the soil organisms

• Increase infiltration of water

• Reduce evaporation

• Destroy or control weeds and soil pests

• Incorporate crop residues and manures into the soil

• Prepare the site for seeds and seedlings

• Repair soil compaction caused by previous activities

Mulching : is the process of covering the topsoil with plant material such as leaves, grass, twigs, crop residues, straw etc. A mulch cover enhances the activity of soil organisms such as earthworms. They help to create a soil structure with plenty of smaller and larger pores through which rainwater can easily infiltrate into the soil, thus reducing surface runoff. As the mulch material decomposes, it increases the content of organic matter in the soil. Soil organic matter helps to create a good soil with stable crumb structure. Thus the soil particles will not be easily carried away by water. Therefore, mulching plays a crucial role in preventing soil erosion. In some places, materials such as plastic sheets or even stones are used for covering the soil. Here, the term 'mulching' refers only to the use of organic, degradable plant material

What is the use of mulching:

- Protecting the soil from wind and water erosion: soil particles can not be washed or blown away.
- Improving the infiltration of rain and irrigation water by maintaining a good soil structure: no crust is formed, the pores are kept open
- Keeping the soil moist by reducing evaporation: plants need less irrigation or can use the available rain more efficiently in dry areas or seasons
- Feeding and protecting soil organisms: organic mulch material is an excellent food for soil organisms and provides suitable conditions for their growth
- Suppressing weed growth: with a sufficient mulch layer, weeds will find it difficult to grow through it
- Preventing the soil from heating up too much: mulch provides shade to the soil and the retained moisture keeps it cool
- Providing nutrients to the crops: while decomposing, organic mulch material continuously releases its nutrients, thus fertilizing the soil
- Increasing the content of soil organic matter: part of the mulch material will be trans-formed to humus

Selection of mulch materials

- The kind of material used for mulching will greatly influence its effect. Material which easily decomposes will protect the soil only for a rather short time but will provide nutrients to the crops while decomposing. Hardy materials will decompose more slowly and therefore cover the soil for a longer time. If the decomposition of the mulch material should be accelerated, organic manures such as animal dung may be spread on top of the mulch, thus increasing the nitrogen content. Where soil erosion is a problem, slowly decomposing mulch material (low nitrogen content, high C/N) will provide a long-term protection compared to quickly decomposing material. Sources of mulching material can be the following:
  - Weeds or cover crops

- Crop residues
- Grass
- Pruning material from trees
- Cuttings from hedges
- Wastes from agricultural processing or from forestry

Constraints of Mulching

While mulching has a lot of advantages, it can also cause problems in specific situations:

Some organisms can proliferate too much in the moist and protected conditions of the mulch layer. Slugs and snails can multiply very quickly under a mulch layer. Ants or termites which may cause damage to the crops also may find ideal conditions for living.

• When crop residues are used for mulching, in some cases there is an increased risk of sustaining pests and diseases. Damaging organisms such as stem borers may survive in the stalks of crops like cotton, corn or sugar cane. Plant material infected with viral or fungal diseases should not be used if there is a risk that the disease might spread to the next crop. Crop rotation is very important to overcome these risks.

• When carbon rich materials such as straw or stalks are used for mulching, nitrogen from the soil may be used by microorganisms for decomposing the material. Thus, nitrogen may be temporary not available for plant growth (risk of N-immobilisation, see Box below)

• The major constraint for mulching usually is the availability of organic material. Its production or collection usually involves labour and may compete with the production of crops.

Nitrogen Immobilisation

When organic material is applied to the soil, the decomposing microbes multiply quickly. For growth, they need nutrients, especially nitrogen, as plants do. If the applied plant material does not contain sufficient nitrogen, the micro organisms take it from the soil. This process is called nitrogen immobilisation, as the nitrogen is fixed temporarily in the microbes and released only after some time. During this time, the microbes compete with the plants for nitrogen and the crop may suffer from malnutrition.

• Old or rough plant materials should be applied to the soil at least two months before planting or sowing the main crop

• Nitrogen immobilisation can occur when the following materials are applied: straw or grain husks, material containing wood (e.g. twigs, saw dust), half rotten compost

Application of mulch

In any tillage system that maintains at atleast 30 % of the soil surface covered by residue after planting primarily to reduce water erosion.

If the layer of mulch is not too thick, seeds or seedlings can be directly sown or planted in between the mulching material. On vegetable plots it is best to apply mulch only after the young plants have become somewhat hardier, as they may be harmed by the profducts of decomposition from fresh mulch material.

If mulch is applied prior to sowing or planting, the mulch layer should not be too thick in order to allow seedlings to penetrate it. Mulch can also be applied in established crops, best directly after digging the soil. It can be applied between the rows, directly around single plants (especially for tree crops) or evenly spread on the field.

Nutrient Supply by Managing Soil Organic Matter

Plant nutrition in organic farming focuses on sound management of the soil organic matter, which is the main nutrient pool for the plants (beside nitrogen from symbiotic fixation). The organic farmer uses three approaches to ensure a continuous nutrient supply from soil organic matter: • Varying the input of organic material: The amount and the quality of organic matter, which is supplied to the soil, influences the content of organic matter in the soil. A regular supply of organic matter provides the best conditions for a balanced plant nutrition. Estimates say that in humid tropical climates 8.5 tonnes, in subhumid climate 4 tonnes, and in semiarid 2 t of biomass is needed per hectare and per year to maintain soil carbon levels of 2, 1 and 0.5 % respectively.

Crop rotation:

- The crops being grown determine the amount of nutrients the soil needs in order to maintain its fertility. The farmer arranges the rotation in such a way that demand and supply of nutrients (e.g. nitrogen from legumes, nutrients from a green manure crop) fit in the best possible way.
- It invariably includes pulses or green manure crops which leave root nodules in the soil and help in improving residual nitrogen content facilitating to economise nitrogen use. Pulses at plants fixing atmospheric nitrogen find its importance as sequential crop, mixed crop, inter crop, alley crop in cropping system depending upon the need and resource availability of the locality.

Intercropping

It refers to growing of two or more crops simultaneously, on the same piece of land, base crop necessarily in distinct row arrangement. Crop intensification is in both time and space dimensions. There is intercrop competition during all or parts of crop growth

S. rostrata were planted at interval of 1.5m in rice. On 30 days after planting, the green manure is cut at a height of 15 cm from the ground level and incorporated as manure. The ratoon is allowed and the reflesh incorporated while planting second crop of rice

Intensive cropping

Cropping systems based on climate, soil and water availability have to be evolved for realizing the potential production levels through efficient use of available resources.

- The turn around period between one crop and another is minimised through modified land preparation. It is possible when the resources are available in plenty e.g. irrigated land cultivation.
- **Cropping intensity is higher in intensive cropping system.**
- This cropping system should provide enough food for the family, fodder for cattle and generate sufficient cash income for domestic and cultivation expenses.

Companion crop

A **companion crop** is generally defined as a **crop** sown with another **crop** to gain some advantage in yield or **crop** protection from pests. In forage production, a small grain may be seeded with the forage **crop**.

Companion planting, or "intercropping," enhances the growth cycle of plants by releasing chemical attractants or repellents, which provide pollination benefits or protection from insects. **Companion** plants, like onions, release odors that help deter insect infestation.

Cover crops

Groundcovers (GC) is the most widely used conservation agriculture agronomic practice for perennial crops, whereby the soil surface between rows of trees remains protected against erosion. With this technique, at least 30% of the soil not covered by the canopy is protected either by sown cover crops, spontaneous vegetation or inert covers, such as pruning residues or tree leaves. For the establishment of sown cover crops and the spread of inert covers, farmers must use methods in coherence with CA principle of minimum soil disturbance.

Influencing nutrient mobilisation: Soil cultivation improves aeration of the soil and enhances the activity of soil micro-organisms. The farmer can influence the nutrient release from humus by cultivating the soil at the appropriate time, to the appropriate depth, and with the appropriate intensity and frequency. The activity of soil micro-organisms is very important for ensuring a sufficient nutrient supply to the plant. If the micro-organisms find suitable conditions for their growth, they can be very efficient in dissolving nutrients and making them available to plants. Therefore, in organic agriculture it is important to encourage plant health through creating a biologically active soil. Even if soil tests find low rates of available nutrient contents, organically managed soils may still be in a position to provide sufficient nutrients to the plants.

The Main Plant Nutrients and how to Ensure their Supply

Macro- and micro-nutrients Plants require a number of nutrients for healthy growth. The nutrients are generally grouped into macronutrients which are required in considerable amounts (such as nitrogen, phosphorus, potassium, calcium etc.) and micro-nutrients required only in tiny amounts, but which are nevertheless important (such as zinc, manganese, iron etc.). Organic manures usually contain all required nutrients in sufficient amounts and in a balanced composition. Therefore, deficiency of single nutrients can in most cases be avoided by applying compost, animal manure and other organic sources.

Nitrogen One of the most important nutrients limiting plant growth is the element nitrogen (chemical sign: N). Nitrogen is needed to build chlorophyll, which gives the leaves their green colour and enables the plants to gain energy for nutrient uptake and growth. It is also a component of amino acids, a building block of proteins. Nitrogen can be easily lost from the soil through leaching (washed out) or volatilisation (it „evaporates“), if not bound to organic matter.

An important source of nitrogen is the fixation of the element from the air through microbes (rhizobia) associated with certain plant species (especially legumes). Because of their potential to supply nitrogen for other crops, legumes play an important role in organic farming, be it in the form of pulses, cover crops, green manures, hedges or trees. To attain its highest level of nitrogen fixing ability, the legume crop needs good growing conditions.

Phosphorus

Phosphorus plays an essential role in the metabolism of plants in all the processes where transport of energy occurs. Phosphorus improves root growth, and encourages

flowering and ripening of the seeds. It is also essential in livestock nutrition for bone growth and for the metabolism. Deficiency in phosphorus hinders plant growth resulting in poor root growth and delay in flowering and ripening. Plants appear stiff, and their older leaves first take on a dark green colour, and then a reddish one before dying most chemical soils are poor in phosphates. Phosphates available to the plant usually are bound onto soil organic matter or are incorporated into soil micro-organisms, while the soil solution contains only small amounts of phosphorus. Once phosphate is adsorbed onto soil particles, only very small quantities can be dissolved, becoming available for plants. The colonisation of plant roots with mycorrhiza, however, can improve the phosphorus uptake of plants.

Availability of phosphorus be improved:

- The mobility of phosphorus is best at a soil pH of 6.0 to 6.5.
- Rock phosphate is ideally given in addition to elementary sulphur and the bacteria Thiobacillus. It is best mixed into compost or animal manure to avoid being fixated by mineral particles and thus becoming almost unavailable to plants.
- Encourage root growth and thus improve phosphorus uptake. Root growth is enhanced by raising the level of soil organic matter by, for example, covering the soil with mulch (in dry climate). • Grow deep rooting plants
- Humidity in the soil is essential in order to make phosphorus available to plants. • Preferably grow legumes that are adapted to the local conditions.
- Improve the growing conditions for mycorrhiza.

Potassium

Potassium is necessary for the synthesis of amino acids and is involved in the process of photosynthesis and in the plants ability to develop resistance to diseases. Good supply of potassium during growth also improves the storing capacity of what? Plants ideally contain potassium and nitrogen in a 1:1 ratio. Potassium is also essential to animals. It is usually supplied in sufficient amounts by the fodder plants.

The majority of potassium in the soil is incorporated in mineral particles and thus not readily available. Some potassium is adsorbed onto the surface of mineral particles and is more easily available to the plants. Clay and silt soils are rich in potassium.

As potassium is needed most in new tissues and is highly mobile in plants, deficiency results in a premature death of older plant parts first. Soils low in nitrogen and potassium result in stunted plants with small leaves and small and few fruits. In general,

potassium supply can be satisfied through weathering of the mineral underground. The need for potassium is strongly linked to the type of crops being cultivated. Tuber crops are especially sensitive to insufficient supply of potassium.

How can the supply of potassium are improved:

- By ensuring the recycling of crop residues (especially straw) and animal manure which contain potassium.
- By avoiding leaching of the soil through the use of a permanent plant cover and by elevating the level of humus in the soil.
- By covering the soil with mulch.

Lecture-10

Weeds-Ecology- habitat management of weeds- Non-chemical weed management methods; Preventive, Physical, cultural, use of tools and implements and biological measures- good crop husbandry practices

Weed management in organic farming is highly complicated and it would be appropriate to adopt various methods in combination rather than in isolation to manage these weeds from organic source depending upon the nature and intensity of weeds.

- In the world 30,000 species of weeds have been listed.
- Out of which 18,000 cause serious damage to agricultural production.
- 18 weeds are considered as world worst weeds.

Characteristics of weeds

- Weeds are more competitive than the cultivated crops
- They are capable of thriving under stress condition
- They produce enormous seeds
- The weed seeds are easily germinated and
- They undergo pollination very easily.

Non-chemical weed management methods

Weed management methods Classified as

1. Preventive
2. Breeding
3. Cultural / Agronomical
4. Physical/ Mechanical
5. Biological
6. Eradicative practices

I. Preventive methods: Prevent or deny establishment or entry of a new weed species into an area through; Fallow ploughing, Avoid contamination of manure pits, Application of well

decomposed FYM, Use of weed free crop seeds, Prevent movement of weeds with other farm resources, Keep non cropped area clean and Keep vigilance

II. Breeding competitive crops: There is an inverse relationship between the crop growth habit and weed growth. So breeding programme should concentrate on varieties which should emerge faster; grow taller; have wide canopy; prevent light from reaching the weeds and give higher yield.

III. Cultural or agronomical methods

Several cultural practices are employed for creating favourable condition for the crop. These practices if used properly help in controlling weeds. Cultural methods, alone cannot control weeds, but, help in reducing weed population. They should, therefore, be used in combination with other methods. In cultural methods, tillage, fertilizer application and irrigation are important. In addition, aspects like selection of variety, time of sowing, cropping system, cleanliness of the farm *etc*., are also useful in controlling weeds.

1. Field preparation

- The field has to be kept weed free
- Flowering of weeds should not be allowed
- This helps in prevention of buildup of weed seed population.

2. Summer tillage

- The practice of summer tillage or off-season tillage is one of the effective cultural methods to check the growth of perennial weed population in crop cultivation.
- Initial tillage before cropping should encourage clod formation. These clods, which have the weed propagules, upon drying desiccate the same. Subsequent tillage operations should break the clods into small units to further expose the shriveled weeds to the hot sun.

3. Mulching

- Principle is exclusion of sunlight from environment.
- Polythene Sheets, natural materials like paddy husk, ground nut shells, saw dust etc. are used as mulching material.
- The thickness should be enough to cut off light (i.e. 10-15 cm)
- The efficiency of polythene sheet is more (more polythene) if it is applied in continuous sheet rather than in particle farm.
- It is effective against annual weeds and perennial weeds like *Cynodon dactylon* and *Sorghum halapense*.

4. Soil solarisation

- It is also called solar soil heating.
- It is effective against weeds which are produced from seeds.
- Covering the soil with transparent, very thin plastic sheets of 20-25mm polyethylene (PE) film during hottest part of summer months for 2-4 weeks.
- This increases the temperature by 10-12 ^{0} C over the unfilmed control fields. Then weeds seeds are desiccated, which are present at top 5 cm soil depth.
- Eg: *Philaris minor, Avena factua* and the weeds controlled by Solarization. Where as *Melilotus* spp. posses hard seed coat, which is resistant to solarization treatment.

5. Stale seedbed technique

- A stale seedbed is one where initial one or two flushes of weeds are destroyed before planting of a crop.
- This is achieved by soaking a well prepared field with either irrigation or rain and allowing the weeds to germinate.
- At this stage, a shallow tillage or non- residual herbicide like paraquat may be used to destroy the dense flush of young weed seedlings.
- This may be followed immediately by sowing/shallow ploughing. This technique allows the crop to germinate in almost weed-free environment.

6. Blind tillage

- The tillage of the soil, after sowing a crop, before the crop plants emerge is known as blind tillage
- It is extensively employed to minimize weed intensity in drill sowing crops where emergence of crop seedling is hindered by soil crust formed on receipt of rain or irrigation immediately after sowing.

Intercropping

- Having different plant types growing together enhances weed control by increasing shade and increasing crop competition with weeds through closer crop spacing.

Example - Soybean +wheat, Sugarcane + finger millet, Sunflower + groundnut

Crop Rotation

- Weeds tend to infest crops with similar life cycles
- Crop rotations limit the buildup of weed populations and prevent major weed species.
- Altering, narrowly spaced crops with closely spaced crops, shallow rooted with deep rooted crops, cold with warm season crops.

Allelopathy

- Allelopathy crops are especially useful because they have the ability to suppress other plants that attempt to grow around them.
- Rye is one of the most useful allelopathic cover crops which kills weeds like ragweed (by 43%), pigweed (95%), and common purslane (100%).

North Carolina researchers investigated that a combinations of herbicide use and cover crop plantings on weed control and reported Rye and clover showed the highest weed control without herbicides

Use of cover crops for bio -control:

- ✓ *Parthenium* incorporated into soil reduces growth of *Cynodon dactylon*.
- ✓ Leachets of dry plants of cumin control weeds.
- ✓ Velvet bean suppresses purple nut sedge.

2. Use of allelochemicals as natural herbicides:

- ✓ Xanthotoxin inhibits germination and growth of *Lactuca sativa*.
- ✓ AAL toxin (*Alternaria alternate lycopersicii*) e–ffective against dicots at low concentration.
- ✓ The living plant as well as the crop residues with allelopathic potential releases allelochemicals and suppresses weed growth.

Merits of cultural methods

1. Low cost for weed control
2. Easy to adopt
3. No residual problem
4. Technical skill is not involved
5. No damage to crops
6. Effective weed control
7. Crop-weed ecosystem is maintained

Demerits of cultural methods

1. Immediate and quick weed control is not possible
2. Weeds are kept under suppressed condition
3. Perennial and problematic weeds cannot be controlled
4. Practical difficulty in adoption

IV. Physical methods of weed control

Mechanical or physical methods of weed control are being employed ever since man began to grow crops. The mechanical methods include tillage, hoeing, hand weeding, digging, cheeling, sickling, mowing, burning, flooding, mulching *etc*.

1. Tillage

- Tillage removes weeds from the soil, resulting in their death

- It may weaken plants through injury of root and stem pruning, reducing their competitiveness or regenerative capacity
- Tillage also buries weeds
- Tillage operation includes ploughing, discing, harrowing and levelling which is used to promote the germination of weeds through soil turnover and exposure of seeds to sunlight, which can be destroyed effectively later
- In case of perennials, both top and underground growth is injured and destroyed by tillage.

2. Hoeing

- Hoe has been the most appropriate and widely used weeding tool for centuries
- It is however, still a very useful tool to obtain results effectively and cheaply
- It supplements the cultivator in row crops
- Hoeing is particularly more effective on annuals and biennials as weed growth can be completely destroyed
- In case of perennials, it destroyed the top growth with little effect on underground plant parts resulting in re-growth.

3. Hand weeding

- It is done by physical removal or pulling out of weeds by hand or removal by tools called *Khurpi*, which resembles sickle
- It is probably the oldest method of controlling weeds and it is still a practical and efficient method of eliminating weeds in cropped and non-cropped lands
- It is very effective against annuals, biennials and controls only upper portions of perennials.

4. Digging

- Digging is very useful in the case of perennial weeds to remove the underground propagating parts of weeds from the deeper layer of the soil.

5. Cheeling

- It is done by hand using a cheel hoe, similar to a spade with a long handle.
- It cuts and shapes the above ground weed growth.

6. Sickling and mowing

- Sickling is also done by hand with the help of sickle to remove the top growth of weeds to prevent seed production and to starve the underground parts.

- It is popular in sloppy areas where only the tall weed growth is sickled leaving the root system to hold the soil in place to prevent soil erosion.

7. Mowing

- It is a machine-operated practice mostly done on roadsides and in lawns

8. Burning

- Burning or fire is often an economical and practical means of controlling weeds.
- It is used to a. Dispose of vegetation b. Destroy dry tops of weeds that have matured c. Kill green weed growth in situations where cultivations and other common methods are impracticable**.**

9. Flooding

- Flooding is successful against weed species, sensitive to longer periods of submergence in water.
- Flooding kills plants by reducing oxygen availability for plant growth.
- The success of flooding depends upon complete submergence of weeds for longer periods.

10. Mechanical tools

Weeding tools are classified according to the source of power as follows

- Manual weeding tools: Small tools or weeding aids (traditional), Spades or cropping hoes, Pull type hoes, Long handle type weeder, push pull type, heel hoe type weeder
- Animal drawn weeder: Cultivators with shovels, sweeps and duck foot type

Merits of physical methods

- Oldest, effective and economical method
- Large area can be covered in shorter time
- Safe method for environment
- Does not involve any skill
- Weeding is possible in between plants
- Deep rooted weeds can be controlled effectively

Demerits of physical methods

- Labour consuming
- Possibility of damaging crop

- Requires ideal and optimum specific condition

V. Biological weed control:

Involves the deliberate use of living organism like insects, fish, disease causing organisms or competitive plants to limit weed infestation.

Ideal characteristics of a bioagent:

- Host specificity
- Ability to kill weed or prevent its reproduction
- High ability to disperse successfully and to locate its host plant
- Bioagent hardiness (good adaptability)
- Reproductive capacity at a rate sufficient to maintain control of host

Three major strategies are employed in biological control of weeds

1. **Classical approach:** Using insects, fish, competitive plant etc. Constraints of insect bio-control:

Weed	Agent
Lantana (Lantana *camara* L.)	*Crocidosema lantanae, Teleonemia scrupulosa* (Lace bug)
Prickly pear *(Opuntia dillenii)*	*Dactylopius tomentosus* (Cochinealinsect)
Water hyacinth (*Eichhornia crassipes)*	*Neochetina brue hi Neochetina eichorniae*
Salvinia *(Salvinia molesta)*	*Cryptobagus salviniae*

2. **Bio-herbicide approach:** Using microbial plant pathogens for weed control. Fungi are more useful for this and hence the term mycoherbicide is also used to refer bioherbicide.

Commercial Myco-herbicides

Trade Name	Pathogen	Target weed
De-Vine	*Phytophthora palmivora*	Strangler vine (*Morreria odoratd)*
Collego	*Colletotrichum gleosporoides f.sp.aeschynomene*	Northern Joint Vetch *(Aeschynomene virginicd)*
LUBAO II	*C. gleosporoides* f.sp. *Cuscutae*	Dodder (Cuscuta sp.)
Caast	*Alternaria cassiae*	*Cassia obtusifolia*
ABG 5003	*Cercospora rodmanii*	Water hyacinth (*Eichhornia crassipes)*

3.**Naturally occurring herbicides:** Using secondary plant products (allelochemicals) and microbial forms as natural pesticides.

Allelopathy in bio-control programmes:

Use of cover crops for bio-control

- Parthenium- incorporated into soil-Reduction in growth of *Cynodon dactylon*
- Dry plants of Menthi (Cumin)-Leachets controls weeds
- Velvet bean- Suppress purple nut sedge

VI. Eradicative methods:

It is the complete elimination of all the live plant parts including seeds from an area. Deep digging and hand weeding are the most useful methods to remove weeds completely. However, these have to be adopted in a limited area.

INTEGRATED WEED MANAGEMENT

Integrated weed management (IWM) involves the concept of multiple tactics of weed management, maintenance of weed population below economic injury level and conservation of environmental quality. A successful IWM strategy has the principle of enhancing farmers' profitability, environmental protection and responsiveness to consumer preference.

Major components of IWM include:

- Monitoring weeds, shifts in weed flora, appearance of resistant weeds and introduction of new weeds,
- Emphasis on ecological, biological and biotechnological methods of environmental safety, and
- Low cost agronomic strategy for weed management in IWM systems.
- State seedbed,
- Balanced fertilizer use,
- Higher plant population,
- Intercropping / relay cropping, and
- Use of competitive cultivars,
- Supplement herbicide use at minimum possible rate.

Lecture-11

Integrated pest and disease management- bio-control agents, bio-rotational pesticides, minerals, botanicals, soaps trap crops, bird perches and traditional preparations- Sanitation

Integrated Diseases and Pest Management

Organic farming is a production system which avoids inorganic inputs like synthetic chemical fertilizers, chemical pesticides and growth regulators. Many practices in organic farming can contribute to stable pest and disease management. Often avoiding the use of chemical pesticides by itself will play a major role in containing the pest damage. Any pest and disease management strategy in organic farming should include methods for

- Pest and disease avoidance
- Managing the growth environment and
- Direct treatment

I. Avoidance Techniques

To manage pests and diseases effectively, producers need to understand the biology and growth habits of both pest and crop. The type and concentration of pests are often responses to previous crop history, pest life cycles, soil conditions and local weather patterns.

a) Crop rotation

Crop rotation is central to all sustainable farming systems. It is an extremely effective way to minimize most pest problems while maintaining and enhancing soil structure and fertility. Diversity is the key to a successful crop rotation program. It involves: rotating between various crop types, such as annual, perennial, grass and broadleaf crops; each of these plant groups has specific rooting habits, competitive abilities, nutrient and moisture requirements. For example, *S*tem-rot is a common disease in conventionally grown canola on the Prairies, but it can also infect at least a half dozen other field crops. Rotations will not have much effect on pathogens that live indefinitely in the soil, but will shorten the life span of pathogens that can survive only brief periods apart from their hosts.

b) Field sanitation/Crop residue management

Reducing or removing crop residues and alternate host sites can be used to control some insects and many diseases. Incorporating the residue into the soil hastens the destruction of disease pathogens by beneficial fungi and bacteria. Burying diseased plant material in this manner also reduces the movement of spores by wind.

c) Reduced or zero-tillage

Reduced tillage may also reduce the damage by certain pests, as the crop residue creates a micro-climate less preferred by some insects (for example, flea beetles). It is important to maintain a balance between crop sanitation and soil conservation. Lighter soils and those prone to wind and water erosion may require postponing tillage until just before seeding to ensure stubble cover for as long as possible.

d) Seed quality

The use of high-quality seed is especially important in preventing disease. The seed supply should be free of smut, ergot bodies or other *Sclerotia*, and free of kernels showing symptoms of Fusarium head blight infection. Seed analysis by a reputable seed testing laboratory will help determine specific diseases in the seed supply. Relatively few diseases are exclusively seed-borne, and it is more common for pathogens to be transmitted from soil, stubble, or wind, as well as with the seed.

e) Weed management

Although weeds need to be controlled to reduce their impact on crop yield and quality, a field completely free of weeds is not necessarily the best objective. In many cases, weeds provide food and shelter for beneficial insects. Parasitic wasps, for example, are attracted to certain weeds with small flowers. Field experience has shown that the number of predators attacking insects increases and the number of aphids and leafhoppers decreases on certain crops as the diversity of weeds (that act as host plants) increases. A complete history of each field should include any insect or disease infestations, which management methods worked and which did not, and a list of management techniques to try in the future.

II. Managing the growth environment

Soil health management

A healthy soil condition is the first line of defense against pests. A biologically active soil with good drainage supports vigorous crop growth, allowing a higher level of crop competition with weeds. High level of nitrogen results in lush leaf tissue and a dense plant canopy that provides an ideal environment for plant pathogens. Low levels of nitrogen can reduce the incidence of insect outbreaks. A shortage of micronutrients such as zinc or copper can result in disease-like symptoms on crops, while too much of any one micronutrient may be toxic. The addition of composted livestock manure improves soil quality, including

increasing the population of soil micro-organisms that compete with soil-borne plant pathogens.

Crop and variety selection

Farmer's awareness of insects and diseases in the proximity of the farm is very important and should influence the crop and the variety of crop to be grown. These choices must also fit in with the crop rotation plan that has been developed. Some insect pests are specific to certain crops, such as wheat stem sawfly, while others, such as grasshoppers, will attack numerous crops. The situation is similar with diseases. The selection of insect- and disease-resistant cultivars can be a useful tool, but under no circumstances can genetically modified varieties be used in organic systems. Wheat varieties with solid stems are more resistant to wheat stem sawfly than hollow-stemmed varieties.

Intercropping

The practice of intercropping (where two crops are grown at the same time) can reduce pest problems by making it more difficult for the pests to find a host crop. This technique also provides habitat for beneficial organisms. Strip-cropping row crops with perennial legumes often leads to better pest control. In particular, alfalfa attracts many beneficial organisms that can destroy insect pests in neighboring crops.

Seeding date

Planting should be scheduled so that the most susceptible time of plant growth does not correspond to the peak in pest cycles. Early seeding reduces crop damage caused by grasshoppers, aphids in cereal crops, wheat midge in spring wheat, barley yellow dwarf virus in barley and wheat, powdery mildew in peas and pasmo in flax.

Seeding rate

Using a higher seeding rate to affect insect or disease infestations may have different results. More plants in a field may reduce the impact of a given aphid population on individual plants, but they may create a more favourable habitat for insects that prefer a dense canopy, such as true armyworm. A dense leaf canopy can also create a moist soil surface and elevated humidity within the crop, conditions favourable to certain leaf disease pathogens.

Depth and timing of seeding

Optimum seeding depth is also important. Deep seeding in cold soils may result in seedling blights and damping-off, especially in pulses and small seeded crops. Seeding depth should generally be no deeper than required for quick germination and even emergence.

Trap strips

Seeding trap strips around the edge of a cropped field or along a fence row helps lure insect pests to a specific area where they can be managed more easily. For example, planting brome grass near a wheat field attracts wheat stem sawflies and their native parasites away from the wheat crop. Similarly, a trap strip of potatoes planted much earlier than the main crop would attract Colorado potato beetles to the area.

Tillage

Tillage can be properly timed before seeding, after harvesting and during summer fallow to reduce populations of insect pests such as cutworms and grasshoppers that spend part of their life cycles in the soil or stubble. Tillage can help starve insects in the spring or during fallow, prevent adults from laying eggs in the soil and expose overwintering insects to predators and inclement weather.

Roguing

Roguing refers to the labour-intensive practice of walking the fields to remove diseased or insect-infested plants. Roguing may not be practical for large fields, but could be suitable for seed plots or crops having highly infectious and destructive diseases (for example, bacterial blackleg in potatoes and certain viruses in other crops).

iii. Direct treatment

At times, the organic producer will find that, despite all the best efforts, an insect or disease pest will grow to levels that cause substantial crop damage. At this point, direct treatment may be necessary.

Monitoring

Insect monitoring traps are useful in determining which insect pests are present in a field and whether they are at economically important levels. It is imperative that the producer has a positive identification of the insect or disease causing damage before choosing a method of treatment.

Traps

Certain types of insect hormones called pheromone may be used as attractants to monitor population levels of insects such as bertha armyworm, diamondback moth, cabbage

looper and European corn borer, or to simply attract insects into a trap. Yellow board traps are also used to trap sucking pest.

Biological control

In a healthy, balanced ecosystem, biological control by natural predators is constantly occurring. The more diverse a cropping system becomes, the greater the spectrum of insect species and micro-organisms within it. This leads to the development of more natural predators within the ecosystem. Ladybugs, ambush bugs, hoverfly larvae, lacewings, spiders, birds, frogs, toads and a host of other insects are predators of aphids, bertha armyworm larvae, sunflower beetles, beet webworms, and both grasshopper eggs and adults. The destructive wheat midge may also be partially controlled by a parasitic wasp, but crop damage may still occur. Various types of fungi are insect parasites and can either kill their insect hosts or reduce their ability to reproduce. Very few biological controls are available to reduce the effects of plant diseases, as most commercial products do not perform well if the disease is already established in the crop. Mycoparasitism is a form of bio-control where one fungus parasitizes another. Although this process occurs with many fungi under laboratory conditions, it hasn't been successful under field conditions.

Natural insecticides

Organic certification standards prohibit the use of synthetic pesticides. Permitted disease-management products include copper (fixed copper and copper sulphate), lime-sulphur mixes, elemental sulphur, vinegar, soap and silica. Bordeaux mixture is considered a restricted substance, and farmers should contact their certifying body before using it. Although these products are allowed, it may not be cost-effective or feasible to apply them to field crops. Scientific evidence on product efficacy should be researched before using them.

Natural products for pest control

Using various naturally occurring substances and products for pest and disease control. Eg. Neem, pongamia, herbal extracts, NPV, Trichograma, Trichoderma etc.

The four main mechanisms involved in the biocontrol are

- Antagonist may parasite the other organism (**Parasitism**)
- Antagonist may secrete metabolites (antibiotics) harmful to the pathogens **(Antibiosis**)
- Antagonist may compete with the pathogens for nutrients or space **(Competition)**
- Antagonist may cause death of the parasite by producing enzymes **(Lysis).**

Biological methods

Biological control basically means, "The utilization of any living organism for the control of insect-pest, diseases and weeds". This means use of any biotic agents for minimizing the pest population either directly or indirectly. Biological control of insect-pests is gaining recognition as an essential component of successful IPM. Classical biological control involves deliberate and natural establishment of natural enemies in areas where they did not previously occur. In addition to deliberate introduction of bio-control agents, proper attention needs to be given for conservation and augmentation of natural enemies that already exist in an area. This should be treated as important as many insect predators are much more susceptible to insecticides than the pests they attack. Biological control agents for insect pests are available in nature abundantly and work against crop pests naturally called as Natural Control. The pest management programme where natural enemies of the crop pests form the core component is designated as Bio Intensive Pest Management (BIPM).

The most commonly used bio-agents in BIPM are broadly classified into 3 categories

- Parasitoid
- Predators
- Microbes causes diseases in insects.

Parasitoid:

These are the insects, either equal or lesser than the size of the host insect (pest) and always require passing at least one stage of their life cycle inside the host system. Due to their high multiplication rates they are of vital importance in the biological control of insect-pests.

Insect parasitoids (parasites thriving on insects): insect parasitoids include

Trichogramma Egg parasite against eggs of gram pod borer, rice stem borer, castor semilooper, cabbage diamond back moth etc. BraconhebetorAgainst insect-pest of coconut and sugarcane Brachymeria Against the pupae of several pests of plantation crops. ElasmusEriborTrichospilusGonlozusTetrastichusChelonus

Predators:

These are those insects, which are generally bigger than the host and feed on several of the pests by predating upon them externally. They will be consuming several of the insect-pests during their life cycle and hold a key role in minimizing pest population in field conditions. Commonly used predators are: Chilizonus, Cryptolaemus, Scymnus, Meoichlus and Pharoscymnus.

These predators feed on mealy bugs, coccids, scales and mites on citrus, grapevine and guava. Mirid predator Crytorhinus attack brown plant hopper of rice. Insect predator Chrysopa is effective on aphids, mealy bugs and young caterpillars. Chrysoperlasp prey upon several of the soft bodied insects such as aphids, leaf hoppers etc; ladybird beetle against aphids and mealy bugs; spiders against a varied number and types of insects especially in rice ecosystem.

Microbial organisms:

These are those micro-organims which are capable of causing diseases in insects as a result they lose their appetite, subjected to several physiological disturbances, leading to the ultimate death of the insect. The microorganisms exploited in biological control of insect-pests are broadly classified into a) Insect viruses b) bacteria c) Entomo Pathogenic Fungi d) Entomo Pathogenic Nematodes and other organisms like protozoans and Rickettesia etc. while several antagonistic fungi and bacteria are being successfully used in minimizing the plant disease incidence. Nematode pest management by using biotic agents is also one of the most promising areas and gaining much desired importance in the current scenario of organic farming.

Pathogens causing diseases in insects and destroying them:

- These include Viruses Nuclear polyhedroses viruses (NPV): They cause typical "tree top disease" in insects. The infected insect loose appetite becomes restless and reaches the apical portion of the plant due to O_2 depletion inside the insect system. Later the infected insect will be dying by hanging itself in an inverted 'V' shape on the apical portion of the plant. Effective against lepidopteran insects in different crops. Ha NPV is used for the management of *Helicoverpa armigera* while Sl NPV is meant for *Spodoptera litura*. Similarly, castor semilooper is managed by Ach NPV and red hairy caterpillar by Am NPV.

Types of Microbial Bio-pesticides

i) Bacterial bio-pesticides

ii) Fungal bio-pesticides

iii) Viral bio-pesticides

S.N	Category	Products	Target pest	Major crops
1.	Bacteria	*Bacillus thurungiensis* *Bacillus sphaiicus, Bacillus subtilis* , *Pseudomonas fluorescens*	Lepidoptera Mosquitoes, Flies Fungal pathogens Fungal pathogens	Maize, vegetables, soybean, groundnut, wheat, peas, oilseeds, Rice
2.	Fungi	*Trichoderma viride* *Trichoderma harsonum* *Trichoderma hamatum*	Fungal pathogens	Wheat, rice, pulses, vegetables, and spices
		Beauveria bassiana *Verticillium focanii* *Metorhizium anisopiie* *Paecilomyces lilacimis* *Nomuraea rileyi*	insect pests such as boll worms, white flies, root grubs, tea mosquito bugs.	Pulses oilseeds, spice, and vegetables
3.	Virus	Nuclear polyhedrosis virus (NPV) of *He!icoverpa armigera, Spodoptera* sp. and *Chilo infescatellus*	American Boll worm, tobacco caterpillar and shoot borer	Sunflower, Vegetables

Method of application of bio-pesticides

a) Seed Treatment

1. Prepare 5% jaggery solution by boiling 500 gm of jaggery in 10 litres of water for 15-20 minutes. Depending on the quantity of seeds to be treated sufficient solution has to be prepared.
2. The solution have to be cooled.
3. Open the pack of bio-pesticide and mix the contents in the above solution. The general recommendation is l0 gm of bio-pesticide/kg of seed.
4. Heap the seeds to be treated on a polythene sheet and the bio-pesticide solution is poured over the seeds and mixed thoroughly.
5. Dry the seeds in shade and sown immediately.

b. Nursery bed

a. One kilogram of bio-pesticide is mixed with 100 kilograms of good quality soil. Forest soil can also be used. In areas where forest soil is not available, 30 kg of well rotten dung can be mixed with 70 kg of soil to prepare a good soil mixture.

b. The soil mixture can be used as a nursery soil or it can be mixed with the nursery bed of size 10 square meters.

c. Soil drenching

1. Prepare a solution by adding 10 gram of bio-pesticide in a litre of water and stir the solution well.
2. Using a rose can to drench the soil with the solution.

d. Seedling dip

1. Prepare a solution by adding 10 grams of bio-pesticide in a litre of water.
2. Dip the seedlings in this solution for 30 minutes.

e. Soil application

1. A mixture is prepared by adding 5 kilograms of bio-pesticide and 100 kilograms of well rotten cattle dung or farm yard manure (FYM).
2. Broadcast this mixture on soil immediately before ploughing. On a standing crop, after the application of this mixture, light irrigation has to be given.

f. Foliar spray

1. One kilogram of bio-pesticide is mixed in 150 litres of water and sprayed on the crop. The solution is sufficient for once.

Bio-pesticides:

Natural occurrence of diseases caused by micro-organisms is common in both insects and weeds and is a major natural mortality factor in most situations. Use of micro-organisms for pest control involves their culture in artificial media and later introduction of larger amounts of inoculums in to the field at appropriate time. Many fungi and bacteria can be handled in this way but insect viruses have the limitation that they have to be raised in living insects. As the biocontrol agents (microbial pathogens) are applied on the targeted pests in much the same way as chemical pesticides, they are often termed as bio-pesticides or natural pesticides.

Bacillus thuringensis, a bacterial pathogen infesting a wide range of insect pests, is the most common microbial insecticide in use today. It is used against caterpillars that attack a wide range of crop. Unlike most other chemical insecticides, it can be used on edible products up to the time of harvest. It is selective in action and does not harm parasites, predators or pests. The bacteria come in several commericla formulations such as Dipel, Delfin, Halt, Spicturin, Biolep, BioAsp etc. Another bacterium B popilaleis also commonly available against white grub Popillae japonica and Hototricha sp. Amongst insect pathogenic

fungi, commercial preparations of Verticilliumlecanii are available for the control of aphids, thrips and white fly under glass house conditions.

Botanicals

Some weeds like *Lantana* sp., notchi, tulasi, adathoda etc., act as natural repellant to many pests. Trees like pungam, wood apple, anona and their byproducts have excellent insecticidal value in controlling diamond back moth, heliothis, white flies, leafhoppers and aphid infestation. Most commonly used botanicals are neem (*Azadirachta indica*), pungamia (*Pungamia glabra*) and mahua (*Madhuca indica*). Neem seed kernel extract (2 to 5%) has been found effective against several pests including rice cutworm, diamond back moth, rice BPH, rice GLH, tobacco caterpillar, aphids and mites. The pesticidal ingredients of neem formulations belonging to general class of natural products called triterpenes, more specifically, limonides. They act as repellents and also disrupt growth and reproduction in insects. Commonly known limonoids are azadirachtin, meliantriol, salannin, nimbin and nimbidin. The efficiency of vegetable oils in preventing infestation of stored product pests such as bruchids, rice and maize weevils has been well documented. Root extracts of asparagus work as a nematicide for plant parasitic nematodes. Similarly leaf extracts of many plants can inhibit a number of fungal pathogens.

Trap cropping

Trap cropping is the planting of a trap crop to protect the main cash crop from a certain pest or several pests (See Appendix I). The trap crop can be from the same or different family group, than that of the main crop, as long as it is more attractive to the pest. There are two types of planting the trap crops; perimeter trap cropping and row intercropping. Perimeter trap cropping (border trap cropping) is the planting of trap crop completely surrounding the main cash crop. It prevents a pest attack that comes from all sides of the field. It works best on pests that are found near the borderline of the farm. Row intercropping is the planting of the trap crop in alternating rows within the main crop.

Advantages of trap cropping

1. Lessens the use of pesticide
2. Lowers the pesticide cost
3. Preserves the indigenous natural enemies
4. Improves the crop's quality
5. Helps conserve the soil and the environment

Tips for successful trap cropping

• Make a farm plan. This will guide you on where the trap crops are to be sown or planted.

• Learn to know and identify the pests.

• Select a trap crop that is more attractive to the pest than the main crop. Ask for assistance from your local agriculturist.

• Monitor your plants regularly.

• Immediately control the pests that are found in the trap crop. Prune or remove the trap crops once the pest population is high, otherwise they will serve as the breeding ground and the pests will attack the rest of your farm.

• Be ready to sacrifice your trap crop as an early crop and destroy them once pest infestation is high.

• Always keep farm records.

Bird perches

• Bird perches are resting places for predatory birds to rest and to look for preys; such as insect pests of cotton, peanuts, and cowpeas. Predatory birds prefer to look for prey in field crops where they have places to rest.

• To make bird perches, use bamboo or wooden poles or tree branches. Erect either of these at regular intervals in the field.

• To have live bird perches within the field, plant *Setaria s*pecies (foxtail cultivars). These plants are found to be attractive to predatory birds. The birds feed on their seeds. In cotton field, plant *Setaria i*n every 9th or 10th row of cotton. Once the birds are on the field, they prey on cotton bollworms and other insects.

Pheromone

A pheromone is a chemical or mixture of chemicals released by an organism to the outside (in the environment) that cause a specific reaction in a receiving organism of the same species. Pheromones are species specific and their safety to beneficial organisms makes them ideal components of integrated pest management systems. Of the various pheromones, the sex and alarm pheromones are used for pest management in India. Sex pheromones help in mass trapping and mating disruption of insects. 'Gossyplure' is a sex pheromone commercially available for the control of cotton pink boll worm *Pectinophora gossypiella*. Large scale trials in mating disruption in India on the pink boll worm was demonstrated to increase 35% yield (Sharma and Singh, 1982) in Gujarat area using 50-75 g of Gossyplure pheromone per hectare with 7 applications at 8-22 days interval. The feasibility of mass trapping *Spodoptera litura* and *Helicoverpa armigera* with 1000 traps over 175 ha of cotton

showed a reduction in insecticidal application. Pheromones are also available for *Earias Vitella, Oryctes rhinoceros, Rhyncophonrus ferugenius, Cosmoplitius sordidus* and *Scirpophaga incertulas.*

Lecture-12

Indigenous technical knowledge (ITK) in organic agriculture – rationale and principles- General , indigenous practices for soil, nutrient, weed, water, pest and disease management in farming- ITK's in farmers practices

Indigenous Knowledge Systems:

Definition and Concepts

The term 'indigenous' is often interchangeably used with terms such as 'traditional' or local' (Wang, 1988).

Fisher (1989) defined the term 'indigenous' as "systems that are generated by internal initiative within a local community itself."

Indigenous knowledge (IK) is the systematic body of knowledge acquired by local people through the accumulation of experiences, informal experiments, and intimate understanding of the environment in a given culture. IK is local knowledge that is unique to a given culture or society (Warren, 1987).

IK is the information base for a society which facilitates communication and decision-making (Warren, 1990). IK is the actual knowledge of a farming population which reflects their experiences based on traditions as well as recent experiences with modern technologies (Haverkort, 1991). This knowledge is far more than technical methods and cultivation practices of farmers. It entails many kinds of insights, wisdom, perceptions, and practices related to people's resources and environments. It is not static. Experimentation, screening, and integration of knowledge represent activities of farming, as much as tilling of the soil.

Indigenous Knowledge on Food Production

1. In Cauvery delta of Tamil Nadu, black gram {*Phaseolus mungo*) and green gram (Phaseolusaureus) are sown as relay crops in rice fields after the long duration rice crop. Farmers sow the gram seeds in a standing rice crop just one week before the harvest of the long duration rice crop. The moisture level and soil condition of the standing rice crop is optimum for the germination and establishment of the gram seedlings. After the harvest of the paddy crop, the seeds get the required sunshine and yield is good. The entire crop is grown under zero cultivation.

2. In certain pockets of the Tanjore Delta of Tamil Nadu, peanuts are sown in late December, immediately after the harvest of the rice crops. The peanuts are grown without irrigation. The water requirements are met from the soil moisture stored from the previous rice crop. The farmers weed the peanuts by hand at the 20th and 40th days after sowing. During hand weeding at the 20th day, the soil clods are broken and fine tilth is produced to act as a soil mulch. This soil mulch maintains the soil temperature and prevents soil moisture evaporation.

3. The available micro-climatic conditions in the bunds of the paddy fields are utilized for growing crops such as pulses, and vegetables.

Indigenous Soil Classification System

First, the criteria used by technical scientists for classifying the soil may not match the criteria considered by farmers—the latter being concerned with the soil's usefulness for cropping.

Secondly, sampling and analysis required to classify a soil using conventional technical methods are often too laborious, timeconsuming, and costly for applied Con-farm') researchers and extensionists to carry out.

Thirdly, researchers and extensionists may lack the skills and training required to identify and analyze a soil in the field. Fourthly, interpretation of conventional soil classification manuals may be beyond the technical ability of some researchers and extensionists.

In Malabar (Kerala state, India), farmers classify soils into three categories based on the productivity:

(1) Pasheemah Koor - rich clay soil,

(2) RasheePasheemahKoor - moderately clay, and

(3) Rashee Koor - loose lime soil (Dharampal, 1983).

Soils of the arid zone in India are mainly sandy in nature. Hence, farmers collect clayey clods from the nearby ponds during summer, transport it to their fields by bullock cart and incorporate it at the time of land preparation (Gupta, 1991).

Tamil Nadu farmers believe that sandy soil is more suitable for groundnuts because during the flowering stage, the pegs could easily penetrate into the loosely structured sandy soil and the pod formation is easier when compared to the clayey soils (Gnanadeepa, 1991).

Indigenous Soil and Water Conservation Methods

Expensive mechanized conservation methods, high labor requirements to carry out maintenance for which farmers do not have enough time, and top-down approaches resulted in the poor adoption of soil and water conservation technologies (Reij, 1991). On the other hand, indigenous methods for conserving soil and moisture are cost-effective (Sanghi and Kerr, 1991).

Farmers in Kerala and Kamataka states of India have used vetiver grass for more than a century for protecting bunds from erosion (Warren, 1991c). Trees planted above and below crop fields decrease the intensity of soil erosion in Claveria, Philippines (Fujisaka, 1986).

Farmers in the Indo-Gangetic plain break the soil crust by hoeing or plowing to renew gaseous exchange and thereby preventing wilting of crops (Randhawa, 1983).

In Auroville, Pondicherry, India, new crops are planted before harvesting the existing crop (Jhunjunwala and Deshingkar, 1984). Aurovilleans mulch trees with leguminous crops and use certain legumes which can survive for six months in the dryland so that some biological activity always takes place in the soil instead of leaving it dry. Farmers in eastern Uttar Pradesh state of India reclaim alkaline soils by applying large quantities of farmyard manure and water (Balasubramanian, 1987).

Summer or fallow plowing is a common practice followed by farmers in north as well as south India. Dharampal (1983, p.247) provided the rationale behind this practice:

The farmers however know from experience that the soil at the surface, and which has been well heated by exposure to the sun, is that which yields the best returns. It is not uncommon to see them before the hot season plough their more valuable lands, so as to expose as much of the soil to the reviving influence of the sun

The problems of soil erosion in cultivated fields is minimized by certain indigenous soil conservation practices in dry regions of Andhra Pradesh (Sanghi and Kerr, 1991):

1. Field bunds without waste weirs (mainly in areas with low rainfall and deep soils);
2. Field drains and conservation drains with waste weirs (mainly in higher rainfall areas in black as well as red soils). Usually the height of the waste-weirs is gradually increased so that siltation is increased and the fields are converted into terraces over the years;
3. Perennial grasses on field boundary bunds;
4. Construction of small stone checks across rills within the field (in shallow red soils with low rainfall); and
5. Sub-division of land holdings into smaller fields.

Some farmers utilize moisture conservation practices:

1. Deep ploughing during the summer (in black soils with low rainfall and in red soils with high rainfall);
2. Sowing seeds across the major slope; and
3. Furrowing as a part of seeding and intercultural operation (for maize in red soil areas).

The occurrence of manjanathi trees (Morindatinctoria) indicates high moisture content in the soil according to Tamil Nadu farmers (Selevanayagam, 1986). Intercropping, relay cropping and sequential cropping protect the soil from the heat of the tropical sun and from the physical damage of rainfall, wind, and soil erosion.

Indigenous Cropping Systems

Crop rotation is a traditional agricultural practice that, in addition to its agronomic value, is important in managing soil borne plant pathogens (Thurston, 1992). According to Palti (1981), farmers have followed crop rotations for hundreds of years because of the following advantages: (1) more efficient use of nutrients, (2) improved soil texture, (3) water conservation, (4) weed control, and (5) management of soil borne pathogens. Crop rotational practices that involve a rotational pattern of cereals, legumes, oilseeds, and vegetables result in the availability of a wide variety of food for marginal farmers (Rajasekaran, Warren and Babu, 1991).

Growing black gram, green gram, and cow pea as intercrops in groundnut fields helps Tamil Nadu farmers to meet subsistence as well as market needs in India (Rajasekaran, Warren and Babu 1991). Intercropping with a native species of wild lupin (*Lupinus mutabilis*) helps farmers to meet the nitrogenous requirements of potatoes and cereals in the highlands of Bolivia (Wilson and Peter, 1988). Intercropping has been practiced not only for its nutritional value but also for the ability to control weeds and insect pests. Moreover, intercropping generally reduces the yield variability because the different species are not equally affected by adverse environmental conditions.

Rah is a traditional system of seed bed cultivation practiced in the past in Thana district of Maharashtra state with the use of wood twigs, leaves, grass, and cow dung with earth on top (Mahapatra, 1983). This system was appropriate in the past, given the existing ecology and the level of development of technology. Even now a well-regulated utilization of forest resources for rab may be sound from the viewpoint of preservation of soil fertility and a pollution-free environment. Indigenous cropping systems such as crop rotation, mixed cropping, intercropping, and swidden cultivation exhibit a high degree of stability. These systems provide farmers with opportunities for harvesting diverse crops from the same land; increasing total land productivity; maintaining and improving soil fertility through the use of

legumes; and above all, reducing or avoiding risks of crop failures due to weather (Hoque, 1984).

Indigenous Practices on Seeds and Sowing

The extensive use of seed beds and the subsequent transplanting of carefully selected healthy seedlings are examples. If seeds are continually used season after season, they lose their yield potential and may contain admixtures like weed seeds and other crop seeds. Hence, Tamil Nadu farmers normally change their seeds at least once in three years (Gnanadeepa, 1991). Early formed earheads of rice possess higher germination capacity, hence farmers in Tamil Nadu prefer to collect seeds from them (Gnanadeepa, 1991). Closer spacing for the short duration rice crop and wider spacing for the medium duration rice crop is practiced by farmers to ensure optimum plant population and higher yield (Gnanadeepa, 1991). However, the farmers adopt an excess seed rate and almost similar spacing during both the seasons.

Indigenous Crop Nutrient Management Practices

Farmers of eastern Tamil Nadu strongly believe that the gingelly (sesamum) crop depletes soil fertility. The rationale behind this belief is that gingelly is a soil exhaustive crop, and its duration is only 80-85 days. Within this short period, the crop has to convert the nutrients into fats and oils. This belief is also highlighted by a well known Tamil proverb, '*Ellupotta vayalil kollu kooda vilayathu" meaning "Even horsegram does not perform well in the field where gingelly was grown as a previous crop." Though Tamil Nadu farmers are aware of the value of green manure in increasing the soil fertility, they are unable to adopt this practice since they seldom find any gap to include green manure crops in their cropping pattern (Gnanadeepa, 1991). Dryland farmers of Tamil Nadu state believe that the growth of the weed aduthinnapalai indicates low fertility of soil (Selvanayagam, 1986). According to him, farmers also found that weeds shoot up in fertile soils immediately after rains.

Indigenous Water Management Systems

Farmers who depend on wells, tubewells, and canals for irrigation adopt a wide variety of indigenous water management practices. These practices aim at conserving the energy to be used for irrigation, assuring an economical usage of irrigation water, and irrigating at correct stages of the crops. Tamil Nadu farmers avoid irrigating the rice nursery on the fourth day after sowing (Gnanadeepa, 1991). The rationale behind this practice is that after three days of sowing, the plumules of the rice seeds start emerging. At that time, if the water stands on the field, the plumule is affected. Gnanadeepa (1991) also found that farmers in Tamil Nadu adopt alternate wetting and drying of rice main fields to improve soil aeration. This process enhances microbial activity though many farmers are not aware of the scientific rationality behind this practice.

Based on the rainfall and soil type, the fanners in Andhra Pradesh state evolved the following major types of water harvesting systems (Kerr and Sanghi, 1992):

1. Individual farm ponds for supplemental irrigation or percolation: These are observed in areas where rainfall is high (more than 750 mm per annum) and where the existing crops are highly sensitive to moisture stress at critical stages.

2. Community tanks for regular irrigation or percolation: These are mainly found in red soil areas under a wide range of rainfall conditions (500-1200 mm per annum). Tanks for percolation purposes are used primarily in areas with red soil with low to medium rainfall.

3. Khadins (earthem embankments across the gullies) for harvesting moisture in the root zone: The khadins are observed in areas with very low rainfall (less than 500 mm per annum) but deep soil. This system has evolved essentially to recharge the root zone during the khariff season for raising a post rainy season crop under residual moisture as observed in Jaisalmer and Barmer districts of Rajasthan state.

Indigenous Plant Protection Strategies

Indigenous plant protection strategies comprise a wide variety of non-chemical methods adopted by local farmers to minimize the pest and disease incidence in cereal and legume plants (Edwards et al., 1990; Thurston, 1992). These methods include effective utilization of resources, performing religious rituals and practicing mechanicalmethods of pest control (Upawansa, 1989). Moreover, these measures do not aim at complete eradication of pests but rather minimizing the pest population. In addition, these measures prevent soil and air pollution, and they are friendly to earth worms and other living creatures of the environment. Reddy (1988), a practicing farmer of Kamataka state, enumerated indigenous plant protection strategies that were adopted by him:

1. Immerse the banana suckers in boiling water for a few minutes to check root knot nematodes;

2. Companion planting of onion, carrot, rose, garlics, banana and coconut keeps pests under control;

3. If marigold flower plants are grown 15 feet apart, both the aroma of the plant and the flowers check about 40 percent of insect multiplication in the cabbage family crops;

4. The root excretion of the marigold plants attracts and kills the root knot nematodes in Solanaceous crops;

5. Spraying neem leaf or oil cake solution checks pests to a certain extent because of its bitter taste and strong repellent odor;

6. Finely pulverized chili powder mixed with neem oil is found to be effective against the pests of cabbage and cauliflower;

7. Maida and fine salt powder mixed well and dusted over cabbage results in the death of larvae due to thirst; and

8. Natural predators like birds, frogs, snakes, and snails should be protected as they check the pest population considerably. In short, Reddy abstained from the application of chemical pesticides and thus, successfully moved towards low input sustainable agriculture. Some farmers in Mangrol village of Gujarat state follow the practice of growing sorghum and pearl millet as a border crop in groundnut fields (Mane, 1989). This practice is believed to protect the groundnut crop from salty air in the coastal area. Mane (1989) also provided certain plant protection strategies that have been adopted by farmers in the semi-arid region of Gujarat state:

1. Application of fresh cowdung should be avoided to keep the potato tubers free from pest attack; 2. Keep the sweet potato field wet to prevent the attack of mangra, an insect pest; and 3. Irrigate the sugarcane crop to check the termite population.

In Thanjavur delta of Tamil Nadu state, the early paddy crop sown in June-July will be ready for harvest in September-October. If cloudy weather is prevalent during this period, the micro-climatic conditions at the root zone will be conducive for the multiplication of brown plant hoppers, so the farmers will deliberately disturb the microclimatic conditions at the root zone by draining the water and drying the fields (Balasubramanian, 1987).

In rice nurseries, the whole nursery is irrigated in order to submerge the plants for some time before the water is drained to wash away the insects. This practice helps the young seedlings to recover from the attack of thrips, an epidemic pest of rice nurseries (Upawansa, 1989).

Pulichai (Hibiscus cannabinus) seeds are sown inter-mixed with rice in upland rice fields to control termite attacks (Balasubramanian, 1989).

Intercropping onion and turmeric also prevents termite attacks on turmeric rhizomes. Planting castor as a border crop around chillies not only prevents dropping of flowers but also acts as a trap crop for Prodenia sp. The symptoms of the disease khaira are not found if moong plant is plowed in situ after the harvest of pods (Gupta and Saha, 1989). Farmers in Gujarat state grow 2-3 lines of okra (Abelmoschusesculentus) plants surrounding the cotton fields (Gupta, 1991). Farmers believe that cotton pests prefer okra when compared to the cotton plant and attack the okra first. Farmers uproot and eradicate the okra plants after the pest attack is completed. A mixture of groundnut cake seeds and flour of the kidamari (Aristolochiabracteata) when put near the field burrows kills rats to a certain extent (Gupta, 1991). Drums operated with flowing water scare rats (Upawansa, 1989). They also disturb the communication between male and female insects and reduce the mating of insects.

Upawansa (1989, p.l7) recorded several indigenous mechanical methods of pest control: 1. A rough large broom made of bamboo tops or strong twigs without leaves is used to brush the standing rice crop; 2. A gummed rope is drawn across the rice fields. Sometimes, the back of the winnowing fan, called kulla, is gummed and the crop is winnowed. The insects get struck to the ropes or fans; 3. A recent practice for brown plant hopper is lighting powerful fire crackers near infested spots. According to farmers, this practice gives very good results; and 4. Discarded robes of Buddhist monks are ignited in pest infested areas. Groundnut farmers in Tamil Nadu trap and kill the moths of the red hairy caterpillar by hanging a broad-mouthed vessel filled with water and a little kerosene near electric lights (Sashi and D'Silva, 1989). Selected herbal treatments are also adopted by Sri Lankan farmers (Upawansa, 1989, p. 18):

1. Daluk (Euphorbia antiauorum) chips with milk are placed at the point of impounding irrigation water to control thrips;

2. Creepers called kaluweV (Derris scanders) are placed in a similar way to control hoppers;

3. The plant called Mahapatta is crushed and spread in affected areas to control hoppers;

4. Areca nut flowers and young coconut leaves are hung in several places to demarcate affected areas;

5. Fresh Gliricidia (Gliricidiapurpurea) leaves are applied as a mulch to control the vector of virus mosaic disease; and

6. A solution prepared from Mimosa pudica. and an extract of cattle urine, margosa leaves, and asafoetida is used as a general purpose insecticide.

Ashes were recommended for disease control in many parts of ancient India (Raychaudhuri, 1964). Summer plowing helps to conserve the soil moisture and manages the pest attack and weeds by exposing the soil to the sunlight (Gnanadeepa, 1991). This indigenous practice was highlighted by the Tamil proverb, "Chithiraimathapuzhuthi, pattharaimatthuthangam" meaning the fine soil tilth received due to summer plowing can be compared to pure gold (Kandaswami, 1978). In addition, Vijayalakshmi (1991) also recorded a number of Tamil proverbs highlighting indigenous plant protection practices. In summary, proverbs represent the wisdom of our ages.

Indigenous Dryland Management Practices

Sequential as well as mixed cropping helps the resource-poor dryland farmers in managing risk situations and also in meeting subsistent food needs. Red gram and groundnuts are grown as intercrops in dryland regions of Tamil Nadu, Gujarat, and Andhra Pradesh states. This practice is well established due to their value as legumes, oilseeds, and fodder for livestock (Venkataratnam, 1990). Farmers in arid regions of northern India sow hajra (pearl millet) utilizing the pre-monsoonal showers (Gupta, 1987). Grasia tribes of Gujarat state practice mixed cropping and strip cropping to enable the different sizes of root length to reach the varied levels of ground water (Shankaran, 1988).

Dharampal (1983, p.236) provided the rationale behind the practice of mixed cropping by dryland farmers:

It has been found by experience that mixed crops not only thrive in the same field; Thejowar serves as a shelter to the sugar cane, from the violent heat of the sun, during the most scorching season of the year.

Dryland farmers in Tamil Nadu state watch the weather pattern using a farming calendar which is usually referred to as panchangam. This calendar divides the season into 'karthis' definite periods enjoining the farmer to manage his/ her resources of land (Venkatratnam, 1990). Suitable timings for various dryland management practices such as timely preparation of land, sowing, weeding, and harvesting are also provided in the calendar.

Indigenous Natural Resource Management Systems

It is now apparent that many indigenous agricultural and natural resource management systems evolved in ways that did not override the carrying capacity of the environment (Warren, 1991a). The key to successful natural resource management for sustainable agriculture lies in partnerships between researchers, extensionists, and farmers (Pretty and Sandbrook, 1991, p.ll). They further explained the need for such partnerships:

1. Integrated approaches such as agroforestry, IPM, and integrated nutrient conservation require a greater range of scientific knowledge and understanding if all the apparently conflicting goals are to be met;

2. Each of these integrated approaches requires detailed local ecological and socio-economic knowledge on livelihood systems—who better to provide this than the rural people themselves; and

3. These partnerships help researchers, extension workers, and policy makers to understand some of the complexities and achievements of rural people's livelihoods.

Ecologically stable and economically viable indigenous natural-resource management systems provide the basis for a major part of small-scale agriculture.

Incorporating Indigenous Knowledge Systems into Agricultural Research and Extension Organizations

The capacity of farmers using their indigenous knowledge systems to classify and evaluate the technological innovations in the local environment can complement station-based development of agricultural technologies (Raman and Balaguru, 1990). The endeavor to build market-oriented agriculture on the energy intensive model must accept ecological and economic strengths of indigenous practices which could be secured with a desired blend of tradition and modernity (Sankaram, 1991). The introduction of high energy technologies

through the application of chemical fertilizers, agrochemicals, machinery, and modern methods of irrigation in developing countries was a complete departure from traditional agriculture and has led to pollution and land degradation (Ezaza, 1989). Lack of relevance to small farm conditions was found to be one of several constraints in the station research technologies. Sanghi and Kerr (1991) provided a specific example to support the above statement. The conventional graded bunding system is not an appropriate soil and moisture conservation technology under small-scale dryland farming conditions due to the following reasons:

1. Continuous bunds leave corners in some fields thus creating the risk of losing the piece of land to the neighboring farmer;

2. Contour farming causes inconvenience in field operations (particularly where multi-row implements are used) and reduces the efficiency of operations (where the desi plough is used) due to repeated cultivation in the same direction;

3. Systems based on a central water course provide benefit to some farmers at the cost of others with regard to disposal of excess runoff; and

4. The overall system emphasizes only long-term gains, hence creating an impression that short-term gains are not possible through such measures (Sanghi and Kerr, 1991, p.2).

On-Farm Research

Successful adaptive research trials create working relationships between experimenting cultivators, agricultural researchers, social scientists, and generate interfaces between the different networks (Box, 1987). Ashby (1987) conducted on-farm varietal trials by involving farmers successfully. The stages of on-farm trials include: types of trials and stages of on-farm research, establishing varietal trials with farmer participation, evaluating varietal trials with farmer participation, participatory research with groups, and group evaluation of trial results. The experience of the on-farm research trials shows that this activity can provide breeding programs with important information to streamline the selection of new varietal materials for specific farming systems (Ashby, 1986; Ashby, 1987).

On-farm participatory research is a means by which two bodies of knowledge can be brought together and can interact so that the solution of small-scale farming problems can take place over a shorter period of time than in conventional research and with greater confidence that the results will be adopted (Fernandez and Salvatierra, 1989). In farmer participatory research, questions to be investigated are determined by farmers rather than scientists (Waters-Bayer, 1987). Biggs (1988) distinguished four modes of relationships between scientists and farmers in on-farm research: (1) Contract, (2) Consultative, (3) Collaborative, and (4) Collegiate. The first three types fall under a transfer-of-technology paradigm whereas the collegiate mode possesses the essence of the farmer-first approach. In the collegiate mode, the formal research system strengthens informal research at the farmer and community level, and enhances farmers' capacity to make demands on the formal system.

Indigenous Organizations

In addition to ignoring local knowledge and skills, many development efforts have ignored existing formal and informal local institutions (Pretty and Sandbrook, 1990). Indigenous organizations are crucial for sustainable resource use and development because they can act as institutions for resource management and control. They enforce rules, provide incentives, and apply penalties for eliciting behavior conducive to rational and effective use of local resources. Local associations are embedded in local social structures and characterized by voluntary, personalistic, face-to-face transactions; hence, they tend to be

highly participatory and reflect well with their members' interests (Cook and Grut, 1989). Leaders of local organizations have a comprehensive understanding of existing strengths and weaknesses in their own organizations and are exceptionally open to trying new management and planning mechanisms for development (Warren, 1992d).

The working methods of farmer organizations and nongovemmental grassroots support organizations are highly participatory, greatly reduce the researoh/ extension linkage problem and are built upon local knowledge (Bebbington, 1989). These organizations often have the capacity to distribute technology and train indigenous agricultural promoters. Warren (1992d) developed a systematic approach for the identification of indigenous organizations, analyses of their structures and functions, and existing capacity for development.

For further reading:

Indigenous Technical Knowledge to Enhance the Productivity of Crops

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Indigenous Knowledge refers to the unique, traditional, local knowledge existing within and developed around the specific conditions of women and men indigenous to a particular geographic area. Indigenous knowledge is the local knowledge - that is unique to a given culture or society. It is the basis for local-level decision in agriculture, health care, food preparation, education, natural resource management, and a host of their activities in rural communities. Indigenous knowledge is the information base for a society, which facilitates communication and decision-making. The term indigenous technical knowledge is often camouflaged with the belief that isassociated with forthcoming happenings and the innovations made by the farmers to solvespecific problems. Some of the related terms are

- **Indigenous Technical Knowledge (ITK):** is specifically concerned with actual application of the thinking of the local people in various operations of agriculture and allied areas.
- **Belief:** change in behavior of insects, animals and vegetation indicating a forthcoming event without any scientific rational but could be true in happening.
- **Innovation:** outside the arena of ITK, but scientifically based development of practices

  using the locally available resources to solve specific problems.

Diversity of Indigenous Knowledge

- Adaptive skills of local people usually derived from many years of experience, that have often been communicated through oral traditions and learned through family members over generations.
- Time-tested agricultural and natural resource management practices, which pave the way for sustainable agriculture.
- Strategies and techniques developed by local people to cope with the changes in the socio-cultural and environmental conditions.
- Practices that are accumulated by farmers due to constant experimentation and innovation.
- Trial-and-error problem-solving approaches by groups of people with an objective to meet the challenges they face in their local environments.
- Decision-making skills of local people that draw upon the resources they have at hand.

General indigenous knowledge based agricultural practices

a. Weather forecasting

Observations of the sky and the movements of the clouds

- Movement of clouds in one direction is a sign of rainy weather while the opposite direction is meant dry weather.
- A red sky at sunrise and sunset considered a warning for rainless days ahead.
- Cold nights with mist and dew considered signs of impending dry weather while hot and warm feelings during the day signaled rainy evenings.
- Thunder in summer and lightening inrainy season brings heavy rain.
- Drizzling occurring in '**Aani'** (June-July), assures rain in **'Avani'** (Aug-Sep).
- Rainfall during 'Marghali' (Dec-Jan) does not benefit the crop but affects the grains.
- If the rainbow appears in the east in the evening or west in the morning, it will rain if the sky becomes dark near the horizon, there will be instant rain.
- If the north wind blows, it will surely rain.

Animal's behavior

- Ants raising mounds is an indication of rains
- If ants move to higher ground with eggs, it will rain

- If termites fly in the evening, it will be rainfall
- Dragonflies flying low indicate the rainfall
- Flight of butterflies from north to south and death of more insects indicates the arrival of rains
- Rain follows the frogs crocking
- It will rain if the cattle look at the sky frequently

b. Soil and Water Management

Soil and water are the basic resources and these must be conserved as carefully as possible. The pressure of increasing population neutralizes all efforts to raise the standard of living, while loss of fertility in the soil itself nullifies the value of any improvements made. It is well known to every farmer that it is the top soil layer, which sustains agricultural production. Once this layer is lost or eroded, nothing can be done to replace it within a short period of time. In a predominantly agricultural country like India, where droughts and floods cause chronic food scarcity, adequate soil conservation programme, not only increases crop yield, it also prevents further deterioration of land. Methods to control surface runoff and soil associated erosion have been practiced in India from times immemorial.

A list of some documented ITKs on soil and water conservation measures under different categories.

S. No.	Categories	Name of ITK
1	Agronomic Measures	• Intercropping • Cultivation and sowing across the slope • Wider row spacing and deep interculturing • Mixed cropping • Cover cropping • Criss –cross ploughing • Hoeing with local hoes • Set furrow cultivation • Application of manure (FYM) • Strip cropping • Green capping

		• Green manuring
2	Tillage	• Conservation furrows with traditional plough • Deep ploughing • Summer ploughing/ Off-season tillage • Repeated tillage during monsoon season
3	Bunding & Terracing (Mech. & Vegetative barrier)	• Vegetative barrier • Stone bunding • Nala check with soil filled in cement bags • Compartmental bunding • Peripheral bunding/ Field bunding • Ipomea as vegetative barrier • Cosnervationbench terrace • Loose stone surplus • Stabilization of field boundary bund with *Vitexnegundo* • Strengthening bunds by growing grasses • Bund farming of pulse crops inkharif under rainfed situation • Earthen bunds • Stone-cum-earthen bunding • Live bunding by raising Cactus • Grass Plantation on field boundaries (filter strip) • Growing of *Saccharummunja* as vegetative barrier on field boundaries
4	Land Configuration	• Use of indigenous plough for formation of broad bed &furrows • Furrow opening in standing crops local implement hoe (Dawara) for moisture conservation • Levelling the plots by local leveler • Opening up set furrow • Conservation furrow

5	Soil Amendment / Mulching	• Application of tank silt • Application of ground nut shells • Sand mulching • Gravel sand mulching • Retention of pebbles on the soil surface • Retention of sunflower stalks • Mulching of Sal leaf in turmeric • Crop residue application in the field
6	Erosion Control & Runoff Diversion Structures	• Sand bags as gully check • Loose boulder checks • Stone waste weir • Waste weir (stone / sorghum stubbles) at the outlet of the field • Brushwood structure across the bund • Grassed waterways • Spur structure • Nala plugging
7	Water Harvesting, Seepage Control & Ground Water Recharge	• Seepage control by lining farm ponds with white soil • Harvesting of seepage water • Wells as runoff storage structures • Rain water management using indigenous rain gauge • Farm pond • Percolation pond / tank • Ground water recharging through ditches and percolation pits • Well recharging through runoff collection pits • Dug wells • Haveli / Bharel system • Bandh system of cultivation • Earthen check dams

		• Field water harvesting • Nadi farming system • Collection of sub-surface runoff water and recycling in Diara land • Rain water harvesting from roof top and road surfaces • Rain water harvesting in Kund / Tanka

- For soil improvement in 'theri' lands of Tuticorin district, 200 tonnes to tank silt are applied per acre followed by 50 tonnes per year for the next few years (Farmers of Erode and Tuticorin District in Tamil Nadu)
- Crop residues and tree branches are burnt on the soil surface at the end of summer to improve the structure of clayey soil and to make ploughing easy (Farmers of Konkan Region in Maharashtra)
- Grassed water ways are adopted or vettiver (*Vetiveriazizanoides*) is planted along the bunds in red soils to check the soil erosion (Farmers in Karnataka, India)
- Leaves of *Buteamonsperma* are spread in the field and burnt along with the dried wheat stalks to improve the soil fertility (Farmers in Gujarat)

b. Rainfall

- Rain water received during the 'Magha' constellation is stored and used as a growth promoter on the standing crop in future (Farmers in Gujarat)
- Large number of fireflies seen at night on the forest trees is a sign that the monsoon will start early (Farmers in Maharashtra)
- If there is rain, accompanied with lightning and mild thunder on the second day of Jayastha month (May – June), there will be no rain for the next 72 days (Farmers of Gujarat)

C. Preparatory cultivation

- During the summer season itself, the soil is prepared well and applied powdered FYM (Farmers in Kerala)
- A piece of raw / fermented areca nut with a betel leaf and a pinch of lime are chewed together and spit on the ploughing spot. If the soil becomes blackish, it indicates that

the soil is well tilled. Or if the soil turns reddish, then the test is negative, indicating the need for more tillage. (Farmers in Assam)

d. Seeds and sowing

- Seeds soaked in 'Kumkum' (Vermilon) water or milk for a day before sowing germinate faster and grow in to healthy plants. (Farmers in Karnataka)
- For increasing the productivity of seeds, it is to be soaked in the paste of ankola fruit (*Alangiumsalvifolium*) or in its oil or in the oil of 'Shelshmantak' fruit (*Cordialatifolia*) (Farmers in Maharashtra)

e. Manures and manuring

- In 'theri' lands of Tuticorin district, FYM is applied by all the farmers to improve the structure and fertility of soils (Farmers in Tamil Nadu)
- If leaves of *Agaves pp.* are cut and embedded in the soil for 45 days, the soil is enriched (Farmers in Karnataka)
- About 10 kg. ofneem cake is soaked in 10 lit. of cow urine along with ½ kg. of waste asafoetida and leaf over night. In the next day, it is sprayed for 1 ac. after dilution as liquid manure (Farmers in Tamil Nadu)
- About 20 – 25 kg. of common salt is mixed with 10 cartloads of natural manure and applied during May to increase crop yields (Farmers in Gujarat)
- To make the soil soft and fertilized, gingelly plants in full bloom may be cut in to pieces and mixed with the soil (Farmers in Maharashtra)

f. Weed Management

- To control *Cyanodondactylon*, harvested dried stalks of cumin crop are spread in the field. As the stalks decompose and mix with the soil, the weed is destroyed (Farmers in Gujarat)
- Farmers change the variety of paddy crop in each season to control the weeds like 'dhakura' (Farmers in Uttaranchal)
- Manure made with mango leaves is applied in the fields to control *Cyperusrotundus* (Farmers in Tamil Nadu)
- Use of multivariate seeds (MVS) for mixed sowing Navathaniyam as intercrops

g. Pest and Disease Management

- To control pod borer (*Leucinodesorbonalis*) in vegetables, *Anethumsowa* plants are sown in rows along with the vegetables. The strong odour of the plant repels the insects (Farmers in Karnataka)
- Mint and marigold are planted in the farm to repel insects (Farmers in Philippines)
- To project the plants from the white ants, *Vetiveriazizanoides, Euphorbia tirucalli*and *Calotropisgigantea* are planted at intervals in the field (Farmers in Maharashtra)
- A mixture is made with 1 lit. ofneem oil, 3 kg of fine sand and 3 kg. of cow dung and heaped in shade covering with a moist sack for 3 days. On the fourth day, the mixture is dissolved in 150 lit. of water and sprayed to control all sucking pests (Farmers in Tamil Nadu)
- About 10 kg of dried cow dung is ground in to fine powder and mixed with ash (obtained from brick kiln) and dusted in the early morning, to control pests and diseases (Farmers in Tamil Nadu)
- Garlic acts on a wide spectrum of organisms in unrelated crop plants singly or in combination with neem products, chilli, asafoetida etc. Besides garlic is effective against bacteria, fungi and nematodes (Farmers in Tamil Nadu)

h. Storage Pest Management

- Lime juice is mixed with grains and then sun dried before storage to prevent insect pests (Farmers in Nigeria)
- Seeds are safely stored inearthern pots after mixing with the leaves of neem and *Vitexnegundo*(Farmers in Gujarat, India).

General ITKs to enhance crop productivity

Tillage system

Tillage systems alter the soil seed bank dynamics and depth of burial of weed seeds. Studies have found that almost 75% of the seedbank was concentrated in the upper 5 cm of soil in no-till fields. In the moldboard plough system however, the seedbank is more uniformly distributed over depth. Other conservation tillage systems are intermediate to these two systems.

Pitcher pot irrigation

Clay pitcher irrigation is a simple, low-cost solution for dryland farming. Unglazed clay pots with a capacity of seven to ten liters, a conical body and narrow neck, are buried approximately one meter apart with the mouth of the pot exposed. Crops are planted around each pot which is filled with water every four to seven days.

Also pots are placed one foot away from fruit trees trunk (mango, sapota*etc*.) a pit with size of 1 feet depth and 2 feet width is dug. Small size hole is made at the bottom of the pot. Pot is buried in the pit, in such a way that it's rim is slightly above ground level. Water is poured into the buried pot and the pot is closed with mud lid, one pot of water (10lit) is enough to irrigate a tree for about 4-5 days. In this method water droplet is applied directly to the root zone of the crop. Water wastage will be avoided and water loss due to evaporation will be minimized. So, water can be effectively saved through this method of irrigation.

Crop residues as mulches

Mulches controlling effect of the evaporation of soil moisture and hence play an important role in agriculture. Locally available materials like paddy straw, sugarcane dried leaves and coir pith can be very well used as mulch material. Crop yield increase has been noticed in the mulch field due to better water use and reduction of soil loss by runoff. Use of straw mulch minimized evaporation. Sugar dried leaves and paddy straw mulch act as barrier between the incident radiation and soil surface's.

Cotton + redgram + castor- intercropping

In a cotton field redgram and castor crops are cultivated as an intercrop, during the month of July-August. Cotton seeds are sown at the same time. Redgram seeds are sown in line 12 feet spacing of 12 feet interval. Castor seeds are sown as a border crop around the cotton field. Seed rate required for red gram is 2 kg/ac. Castor seeds ½ kg/ac. Cotton is harvested with in a period of 5 months. Redgram is harvested in the 6 months. Duration of the castor crop is 10 month.

Legumes

We have known for centuries that legume plants enrich the soil. Certain groups of soil bacteria are able to convert nitrogen into nitrates. The symbiotic bacteria group called ***Rhizobia*** form nodules on the roots of legumes (clovers, peas, alfalfa, soy, etc.). Plant roots chemically "invite" the bacteria to come on in. The nodules formed are filled with ammonium, which supplies the plant with nitrogen, and fertilizes the soil when the plant dies. Inside the nodules, the rhizobia receive food (carbohydrates & proteins), oxygen, and protection from predators such as amoebas.

Eg.Naripayaru, cowpea, sunhemp, daincha (greenmanure)

Seed treatment using Cow's urine: Dilute 500 ml of cow's urine in 2½ litres of water. Tie the seeds (paddy) to be sown in small bags and soak them in cow's urine extract for half an hour. Shade dry the seeds before sowing.

Cowpea along with cotton: Sowing of cowpea along with cotton in clay soil facilitates germination of cotton by early germination and root penetration (it makes soil porus on root portion).

Sheep penning

During non cropped season or after harvest, sheep penning is done with 100 sheep's in the field for about 4-5 days. Every day morning the palce of penning can be changed. So that the entire field will be covered. At the time of penning, sheep manure enriches the soil fertility. Existence: 75 years

Green leaf manuring

It is the application of land of green leaf, gathered from shrubs and trees growing in waste lands. Green leaf is gathered from all sources by farmers during the rice planting season. Different kinds of shrubs growing on tank bunds, waste lands, field bunds etc., are used. In addition, lopping from trees on the road side and forest area also gathered for use as green leaf manures. Common shrubs and trees useful for this purpose are,neem, teak, pungam and erukku leaves.

- Addition of organic manure
- Availability of nutrients
- Improvement of soil structure.

Using Neem Leaves as Manure in nursery

In the paddy nursery site (before sowing of paddy seeds) neem leaves are incorporated into the soil and puddle and allow decomposing for two weeks. Paddy seeds sown after this treatment are believed to become resistant to pest and disease.

In main field:

10% Amirthakaraisal should be sprayed before transplanting. Otherwise for 1 acre of land, 300-500 litres of Amirthakaraisal should be mixed with water and irrigated.

Plant growth regulator

Panchagavya

Panchagavya, an organic product has the potential to play the role of promoting growth and providing immunity in plant system. Panchagavya consists of nine products viz. cow dung,

cow urine, milk, curd, jaggery, ghee, banana, Tender coconut and water. When suitably mixed and used, these have miraculous effects.

- Cow dung - 7 kg
- Cow ghee - 1 kg

Mix the above two ingredients thoroughly both in morning and evening hours and keep it for 3 days

- Cow Urine - 10 liters
- Water - 10 liters

After 3 days mix cow urine and water and keep it for 15 days with regular mixing both in morning and evening hours. After 15 days mix the following and panchagavya will be ready after 30 days.

- Cow milk - 3 liters
- Cow curd - 2 liters
- Tender coconut water - 3 liters
- Jaggery - 3 kg
- Well ripened poovan banana – 12 nos

Preparation

All the above items can be added to a wide mouthed mud pot, concrete tank or plastic can as per the above order. The container should be kept open under shade. The content is to be stirred twice a day both in morning and evening. The Panchagavya stock solution will be ready after 30 days (care should be taken not to mix buffalo products. The products of local breeds of cow is said to have potency than exotic breeds). It should be kept in the shade and covered with a wire mesh or plastic mosquito net to prevent houseflies from laying eggs and the formation of maggots in the solution. If sugarcane juice is not available add 500 g of jaggery dissolved in 3 liter of water.

Generally panchagavya is recommended for all the crops as foliar spray at 30 % level (3 litre panchagavya in 100 litres of water.

Amirthakaraisal

In paddy field 10% Amirthakaraisal should be sprayed before transplanting. Otherwise for 1 acre of land, 300-500 litres of Amirthakaraisal should be mixed with water and irrigated. Fresh cow dung - 10 kg, Cow's urine - 10 litres, Jaggery - 1 kg and Water - 100 litres. The mentioned ingredients should be added to a cement tank or suitable container and mixed well. The extract would be ready for use the next day. It improves the soil fertility and gives good yield.

Weed Management

- If weeding is not done, it will reduce three fourths of crop yield.
- Weeding is not required under dry land condition. If weeding is not done the weed growth is controlled naturally and it also helps to conserve moisture.
- Repeated ploughing will reduce weed population.
- Cultivating rice once in three years in garden lands to control 'Arugu' *(Cyanodondactylon)*weeds.
- 'Raising and ploughing the green manure crops like daincha, *(Sesbania sp.),* kolingi*(Tephrosiapurpurea)* in the field before their flowering to reduce weed population.
- Raising *Calotropisgigantea* as a green manure to check the growth of Aarai (Mars/tea*quadrifolia)* weed.
- Growing horse gram to control nut grass *(Cyperusrotundus).*
- Growing cowpea as a green manure to control nut grass.
- Allowing swine in the fields to eradicate nut grass.
- Frequently ploughing the fields by wooden plough made up of neem trees and frequent application of neem cake in the soil to control nut grass.
- Dissolving 1 kg of salt and 100g. ofsarvodaya soap in 10 lit. of water and spraying this solution to control all the weeds except nut grass.
- To control nut grass in the field 50kg of neem cake is applied both at the time of ploughing and sowing.

- Dissolving 200 g. of salt in 1 lit. of water and spraying to eradicate congress weed (P*artheniumhysterophorus).*
- Continuous submergence of field for some time controls the weeds.
- Keeping the irrigation channels free from weeds.

Pest and Disease Management

- Uninterrupted drizzling of rain leads to the occurrence of more number of pests and diseases.
- Winds blowing at the end of December bring lot of pests.
- Crops grown in alkaline soils are more prone to disease attack.
- Small lamps are placed on either side of the house entrance and light from the lamps acts as a light trap and the farmers are able to identify the pest outbreak.
- To prevent the attack of aphids and mite flies, sorghum or pearl millet is grown very closely in 4 rows around the fields to act as a shelter so that these pests cannot enter the fields.
- Growing Thangarali' (*Tecoma stands)* and 'Sevvarali' *(Nerium oleander)* as border crops, which act as trap crops and control the insect pest.
- Thiruneeru' (Sacred ash) is dusted on the crops to reduce pest attack.
- Kitchen ash is applied to control aphids.
- To control the sucking pests, 5 kg of tobacco powder is soaked in a mixture of 10 lit. of cow urine and 5 lit. of water for 5 days. Then it is filtered and diluted with 80 it of water and sprayed.
- Mixing cow urine, neem oil and tobacco decoction together and spraying on crops controls sucking pests.
- Leaves of *Calotropisgigantea* and *Strychnos nux-vomica* and neem cake are soaked in water in a mud pot and fixed in the field. Moths are attracted towards the smell, fall inside and die.
- Leaves of notchi (*Vitexnegundo)* and pungam *(Pongamiapinnata) are* also used to control moths.

- About 5 kg of Calotropis leaves is soaked in a mixture of 10 lit. of cow urine and 5 lit. of water for 5 days. Then it is filtered and diluted with 80 li. of water and sprayed to control defoliation.
- Spraying sarvodaya soap solution to control mealy bugs.
- To control nematodes, pungam *(Pongamiapinnata)* or iluppai *(Bassialatifolia)* cakes are applied.
- Cow dung, Cow urine, calotropis leaves and neem cake are put in a pit near the irrigation channel. After decomposing, it is mixed with irrigation water.
- Grinding the leaves of *Calotropisgigantea* with the fruits of *Datura metal,* soaking in water for 15 days, filtering and spraying to control the entire pest.
- Two handsful each of leaves of thumb (*Leucasaspera),* kuppaimeni *(Acalypha indica),* thulasi*(Ocimumcanum). Datura metal,* neem, nochi (V/tex *negundo), 5* fruits of *Datura* mefa/and handful each of neem cake and lluppai *(Bassialatifolia)* cake are pounded together and soaked in water in earthen pot for 10 days. Then it is filtered, diluted (100ml./lit.) to which 100ml. of sarvodaya soap solution and 100 ml. of neem oil added and sprayed to control all insect pests.
- Leaves of *Ocimumcanum,* seeds of *Nerium oleander and* fruits *Datura metal* are taken in equal quantities, powdered and soaked in cow urine for 10 days. Then it is filtered and diluted 100ml./lit.) to which 100 ml. of neem oil is */added* and sprayed to control all insect pests.
- Neem oil and neem seed kernel extract are the general organic pesticides used to control many pests.
- During the night time on full moon day of Tamil month 'Karthigai' (Nov.-Dec.) 'Chokkapanai' (Community firing) is performed as a part of celebrations in a common place in the village by which the pests get attracted and killed. Ash from this fire is dusted on the crops to control ' sucking pests.
- Crop wastes are burnt and its ash is dusted on the fields to control diseases.
- If neem cakes are applied as basal fertilizer there will not be any incidence of diseases.

- One kg of leaves of seemaikaruvel *(Prosopisjuliflora)* is pounded and diluted with water and sprayed to control yellow mosaic virus.
- Spraying cow urine to control many pests.
- A mixture of extracts of garlic and neem cake is *sprayed* to control aphids.
- Planting 'Pirandai' *(Cisssusquadrangularis)* vines around fields to protect against termites.
- Grow castor on the fields to control pest as trap crop.
- Spread neem leaves over the nursery to control termite damage.
- Putting neem cake inside a gunny bag and placing it in the irrigation channel controls mites.
- Termites destroy the seedlings in nursery grown in dry land condition. To control these termites, apart from putting the neem leaves, sheep wool and human hairs are also put. Termites eating these hairs die.
- Pouring decoction of finger millet roots on the root zone of crops to control termites.
- Before planting tree seedlings, dried leaves and trashes are burnt in the pits to protect the seedlings against termite attack.
- Dusting ash in the pits before planting tree seedlings also helps to prevent termites.
- `Sprinkling 5% common salt solution to reduce termite attack on the trees.
- After the harvest of tobacco leaves, their stems and roots are ploughed in-situ to control the termites
- Tobacco soaked water is poured on the ant mounds to control them.
- Any spraying is to be done in the early morning.
- Take 30gm. of grounded nirium seed in 10 lit. of water for 1 hour and mix with khadi soap and spary to control thrips, aphids, whiteflies and leaf eating caterpillars.
- A mixture is made with 1 lit of neem oil, 3kg of fine sand and 3 kg of cow dung and heaped in shade covering with a moist sack for 3 days. On the fourth day, the mixture is dissolved in 150 lit. of water and sprayed to control all sucking pests.

- About 10kg of dried cow dung is ground into fine powder and mixed with ash (obtained from brick kiln) dusted in the early morning, to control pests and diseases.
- Garlic acts on a wide spectrum of organism in unrelated crop plants singly or in combination with neem products, chilli, asafetida etc. Besides garlic is effective against bacteria, fungi and nematodes.
- Calotropis leaf extract is applied at the place of termite attack

Preparation of Neem Leaf Solution against locust

The solution was prepared by soaking 1 kg of neem leaves in 5 litres of water. The solution was left over night and then filtered through a cloth. The solution was taken in a backpack sprayer. The solution was added two days prior to the arrival of locusts.

- Azadirachtin content of neen leaves in known scientifically pest control

Effect on Locusts

For the first 6 days, the crops were consistently attacked by migrating locusts. Initially they were controlled using burning car tyres but when the infestation got severe neem solution was sprayed over the crops and directly on the locusts. After the spray, there was no damage caused to the plants by the locusts.

Cow dung Spray Controls BLB:

Case study: In a pot culture experiment conducted at TRRI, Aduthurai, Foliar spraying of cowdung extract twice and cow's urine twice at 10 days interval from 24 hours after inoculation of Bacterial Leaf blight culture into the plants reduced the disease intensity when compared to untreated check. It was similar to chemical spray constituting nickel nitrate and streptocyline(0.06%). In the cowdung treated pots, yield obtained was maximum i.e. 5050 kg/ha and in control 4120 kg/ ha was recorded. Cow's urine pots recorded an yield of 4900 kg/ha. Hence foliar spray of cowdung extract (20%) starting from initial disease appearance and 10 days later could check BLB and prevent yield loss (www.tnau.ac.in)

Plant based bio-pesticide

- Take 1 kg each of the following leaves - tobacco, mango, neem and adathoda. Take fresh turmeric ¼ kg and garlic 1 kg. Pound the leaves with water and make it into paste. Pound turmeric and garlic separately and mix all the ingredients together.
- Take the above mixture in a mud-pot filled with 3 litres of water and boil it for 15 minutes. Filter the extract and spray it on the crops.

- This effectively controls sucking pests, termites, root grubs etc. For 1 acre, 2 litres of extract is required for spraying.

Chilli garlic extract for bollworms:

Take 3 kg of green chillies and remove the pedicel. Grind this thoroughly into paste and soak it in 10 litres of water overnight. Take ½ kg. Garlic, grind thoroughly soak it over night in 10 litres of water. Prepare 2 extracts separately and add khadi soap solution at the rate of 5 ml/litre. Filter the solution and add 80 litres of water and spray immediately. This controls all the bollworms and the army worms.

- Pungent taste of spray after fermentation of garlic and green chillies may not like by larval stage. Same condition may deteriorate the adult settling behavior (AnandPrakash*et al.*, 2008).

Neem kernel powder for sucking pests:

Take 5kg of Neem Kernel powder and put in a cloth pouch. Soak this pouch in 3-4 litres of water overnight. Next day, sqeeze the pouch and filter the extract. To this add 100 litres of water and 100 ml of Khadi soap solution. Spray this solution during morning or evening hours. This controls all the bollworms, leaf folders, defoliators and sucking pests.

- Azadirachtin content of neen leaves is known scientifically pest control

Cow dung extract to control grasshopper in paddy

2 kgs of cow dung, 1 litre of cow's urine and 1 kg of neem cake was taken in a large vessel; To this 20 litres of water was added. The mouth of the vessel was tied securely with a cloth and left as such. After 3 days, the cloth was removed and the extract was stirred well. Then, it was allowed to settle and the supernatent was used. To every litre of the extract, 100 ml of khadi soap solution and 9 litres of water was added and sprayed with a sprayer.

Rat control

- Rats do not live in the fields where sheep penning is being practiced
- Pieces of palmyrs (*Borassusflabellifer)* leaves are tied on the pole and fixed on the field. The sound produced by the leave scares away the rats
- Providing owl stands near the rat holes will help in reducing the rat damage
- Reduce rate population, rat holes are dug and rats are killed after each harvest
- To catch the rats, a trap made up of wire loops on bamboo pegs is being used
- Use of soaked rice as bait attracts more rats

- **Extract of *Ipomoea fistulosa:*** Leaves of *Ipomoea are boiled in water and* filtered. Sorghum grains are boiled in this extract and placed near the rat burrows. It is believed that rats die after eating it.

Others:

- Pots are filled with water and buried in the field.
- By leaving space (nearly 1 ft) in all four sides of field
- Cow dung balls are immersed in kerosene and kept in 8-10 places of the field randomly
- Papaya- cut in to pieces and spread around the farm (3 fruits/ acre)

Bird's scaring

- Stones are thrown with the help of 'Kavan' made by tying a long pieces of gunny thread to both ends of small piece of leather.
- Tying a black cloth to a long pole and placing it in the centre of fields to drive away the crows.
- Waste magnetic tapes taken from audio-cassettes are tied in the fields in a criss-cross manner, which the birds will think as a net.
- Pieces of palmyrs (*Borassusflabellifer)* leaves are tied on the pole and fixed on the field. The sound produced by the leave scares away the rats
- Beating drums and using crackers

Storage Pest control

- Drying the seeds and grains of all the crops on the new moon day before storing to avoid pestattack.
- Grains, trees etc. harvested on full moon day are more prone to storage pest attack.
- Mixing the dried leaves of neem with seeds while storing them.
- While storing the seeds of food crops, they are filled in container to its $3/4^{th}$ height, covered with arough cloth on which leaves of neem, pungam and nochi (*Vitexnegundo)*are placed to the remaining volume and finally covered with sand up to its mouth.Generally seeds stored with their outer coat/shell escape from storage pests.

- Finger Euphorbia *(Euphorbia tirucalli)* provides protection against a wide range of grain pests. Branches of the plant are burnt to obtain its ash. One tea cup full of ash is mixed with 20 litres of grains. This provides protection against grain pests.
- Coconut oil, peanut oil and sesame oil can be used for protection from bruchids. 5 ml of the oil is mixed with 1 kg of grain legumes. This provides protection up to six months from bruchids.
- Custard Apple *(Annonareticulata)* leaves have insecticidal and antifeedant properties. Fresh or dried leaves mixed with the produce or placed in layers between the produce gives effective control against rice moth 3-4 months.

Conclusion

- ITK is Cheaper and Safer, harmless and local resources
- Though ITK's are sustainable, using them alone will not meet out the food requirement of exploding population So, to keep in phase with the present level of food production and to sustain the higher levels of productivity ITK's are to be integrated with organic and modern technologies without adding any chemical form of inputs.
- Keeping ITK as a basis during the process of developing technologies results in a basket of sustainable options rather than fixed packages.
- Validating farmers' experiments creates an environment of respect for local people and village level extension workers thus leading to their increased participation and empowerment.
- Many ITKs are not adopted everywhere throughout India because of constraints in adoption and

unawareness of the effectiveness of such practices.

Thus validation of the ITKs is a logical step to qualify and quantify the effectiveness of these practices. Suitable modifications of the traditional practices through on-farm research would help in developing appropriate and acceptable technologies for different local environments.

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Lecture-13

Benefits and problems in organic farming

The growth of organic agriculture in India has three dimensions and is being adopted by farmers for different reasons. First category of organic farmers are those which are situated in no-input or low-input use zones, for them organic is a way of life and they are doing it as a tradition (may be under compulsion in the absence of resources needed for conventional high input intensive agriculture). Second category of farmers are those which have recently adopted the organic in the wake of ill effects of conventional agriculture, may be in the form of reduced soil fertility, food toxicity or increasing cost and diminishing returns. The third category comprised of farmers and enterprises which have systematically adopted the commercial organic agriculture to capture emerging market opportunities and premium prices. While majority of farmers in first category are traditional (or by default) organic they are not certified, second category farmers comprised of both certified and un-certified but majority of third category farmers are certified. These are the third category commercial farmers which are attracting most attention. The entire data available on organic agriculture today, relates to these commercial organic farmers.

International standards

The International Federation of Organic Agriculture Movements (IFOAM) has produced a set of international organic standards, laid down by people from many countries. These give guidelines about what organic farming is and how it should be practiced on the farm.

International standards are also used to help countries set their own standards, which take into account different farming systems. Many countries have an organic standards authority which lays down national standards and awards a symbol to farms which have followed the standards. This symbol then allows farmers to market certified organic produce. This is important, as it ensures that people know that the food which they buy is organic.

The main principles of organic farming were laid down by IFOAM in 1992.

- To produce food of high nutritional quality in sufficient quantity.
- To interact in a constructive and life enhancing way with all natural systems and cycles.
- To encourage and enhance biological cycles within the farming system, involving micro-organisms, soil flora and fauna, plants and animals.
- To maintain and increase long term fertility of soils.
- To use, as far as possible, renewable resources in locally organized agricultural systems.
- To work, as far as possible, within a closed system with regard to organic matter and nutrient elements. This aims to reduce external inputs.
- To work, as far as possible, with materials and substances which can be reused or recycled, either on the farm or elsewhere.
- To give all livestock living conditions which will allow them to perform the basic aspects of their innate behaviour.
- To minimise all forms of pollution that may result from agricultural practices.

- To maintain the genetic diversity of the agricultural system and its surroundings, including the protection of plant and wildlife habitats.
- To allow agricultural producers a living according to the UN human rights; tocover their basic needs and obtain an adequate return and satisfaction from their work, including a safe working environment.
- To consider the wider social and ecological impact of the farming system.

Scientific Constraints of Organic Farming

- Very high volume of organic materials required to meet the nutrient demand. [Eg.Average application of 234 kg N/ha/year for a system, this requires either 46 t of rawFYM (0.5 % N) or 15.5 t vermicompost (1.5 % N)]. Finding the right combination ofsources are essential to meet the demand.
- Efficiency of organic manures is higher than inorganic? Hence, standardization offrequency and quantity of application of manures are essential to develop packageof practices in a system mode.
- Mismatch between time of nutrient release from organic materials and crop nutrientdemand (Mineralization of N from VC is high in first 30 days) in most of the casesaffects the yield. Hence, split application of enriched manures for various crops andsystems needs to be standardized.
- Higher incidence of weeds under organic conditions is reported by organic growers.There is a need to identify crop specific non-chemical weed management strategies.
- Estimation of heavy metals and other residues are essential when the different kindof organic inputs (including municipal waste) are used.
- Development of climate friendly smart farms and carbon farming techniques areessential to promote organic farming.
- While a great deal of research has been conducted on the impact of non-organicsystems on the environment, there are very little data available on thepositive impacts of organic farming.
- Great need for research on the human health benefits of consuming organic foodscompared to non-organic food diets. The conversation needs to be expanded beyondthe argument over "is organic more nutritious or not" and encompass full analysis ofdifferent agriculture and food systems, their environmental impacts, and their impactson public health.

Constraints expressed by organic growers (total number of organic farmers surveyed: 98)

- Lack of marketing facilities (43.7 %),
- Non availability of premium prices (39.5 %),
- Difficulty in control of weed, pest and diseases (35.4 %),
- Limited availability of organic manures (31.3 %)

Organic Farming: Advantages

a) **Nutrition**

The basic purpose of farming is to provide food for human beings and animals. If the farming is done organically, it dramatically improves the mineral content of the food produced. The modern conventional methods use plenty of fertilizers. On the otherhand organically nourished plants provide a better quality food to people and animals alike.

b) **Poison-free**

One of the important benefits of organic food is that it is free of contamination with health harming chemicals. Because of the toxic chemicals in industrialized societies there has been a profound upward trend in the incidence of diseases.

c) **Food Tastes Better**

Animals and people have the sense of taste which allows them to discern the quality of the food they eat. No surprise that organically grown food tastes better than that conventionally grown. The tastiness of a grown fruit and vegetables is directly related to its sugar content, and which in turn is a function of the quality of nutrition that the plant itself has enjoyed.

d) **Food Keeps Longer**

Organically grown plants are nourished naturally. And therefore the metabolic integrity of their cellular structure is superior to conventionally grown plants. And it directly affects its storage. Organically grown foods can be stored longer and the latter's susceptibility to rapid mould and rotting is not seen in organically grown foods.

e) **Grower Benefits**

Most diseases and insect pests are resisted by a healthy plant grown organically in properly balanced soil. US doctor and soil nutrition pioneer Dr. Northern who conducted many experiments to test the hypothesis during the 1930's proved it.

f) **Lower Input Costs**

Expensive agrichemicals are not used in organic farming. Additionally, the greater resistance of their organically grown crops to pests and the diseases save farmers significantly in expensive insecticides, fungicides and other pesticides.

g) **Drought Resistance**

Drought tolerance is an important benefit of organically grown plants. As such plants are grown organically and naturally, they can manage easily in sudden changes of nature such as drought and floods.

h) **Added Value**

There is a continuous increase of consumer interest in the organically grown food. They are even ready to pay premium prices for it as they recognize the greater food value of the organic grown food. Organic food attracts price premiums of up to 30%.

- Organic farming may produce profits for farmers from 'carbon credit' trading systems being developed around the world.

- Organic agriculture can help feed the hungry by allowing them to produce their own food and by reducing the need to import subsidized food.

- It can provide commercial opportunities where poor farmers can produce a diverse range of certified organic surpluses to be exported at premium profit.

Because organic methods exclude the use of chemical inputs, poor farmers have less capital outlay on fertilizers and pesticides, and thus have less dependency on multinational seed and chemical companies.

i) Organic farming protects the environment

- Organic farming does not pollute air, water and land with chemical fertilizers and pesticides.
- Organic farming helps reduce global warmingbecause its methods use animal manure and cover crops like clover and legumes to enrich the soil which helps sequester CO2 from the atmosphere.
- Organic farming helps reduce global warming because it uses less energy than conventional farming.

J) Organic farming's benefits to biodiversity

Gabriel *et al* (2010) show that, compared to paired conventional farms, the organic farms studied

had:

- an overall 12% increase in biodiversity,
- more plant diversity,
- greater floral diversity,
- more earthworms,
- more insects,
- more butterflies,
- increased numbers of some types of birds , Also:
- The biodiversity benefits in areas with a high proportion of organic farming were higher than in those with a low proportion

K) Greater plant biodiversity

- Greater floral species richness and abundance within the crop, crop margins and non-farmed areas on organic farms (Gabriel et al. 2006).
- Up to six times more species within the crop on organic farmscompared to conventional farms (Shepherd et al. 2003).
- More frequent occurrence of rare arable species on organic farms (Hole et al. 2005).

Organic Farming: Disadvantages

a) Productivity

Superior productivity is the chief purpose of farming. By massive inputs of chemicals and machinery, in a very short term the yield is possible. However, over the longer time frame, productivity advantages vanish away. In organic farming soil health is given importance. Whereas industrialized agriculture thrashes the land.

b) Cultivation

Organic farmers usually use the method of cultivation to remove weeds prior to sowing. While their counterparts have the benefit of sowing by direct drilling of seed into soil which is herbicide treated.

c) GM Crops

Organic growers do not use genetically modified food crops. Such GM crops are engineered to tolerate herbicides and even resist pest. Directorate-General for Agriculture of the European Commission declared a report which claims that productivity gains attributed to GM crops are usually negligible. The main advantage is convenience only. Additionally, GM crops may be associated with harm to both human health and the environment.

d) Time

No doubt, in organic farming a greater interaction between a farmer and his crop is required for observation. In addition, timely intervention and weed control are also important elements. And therefore it is more labour intensive and time consuming method than chemical/mechanical agriculture. Naturally a farmer can produce more crop using industrial methods than by solely organic methods.

e) Skill

To farm organically it requires more skill. An organic farmer does not have a convenient chemical fix on the shelf for every problem he encounters. He has to engage careful observation and greater understanding the problem rather than simply putting a plaster over its effect.

Future of Organic Farming

The social and economic viability and sustainability of organic farming just keeps getting better and better. A steep increase in the area of organic land has been observed in most parts of the world. Even consumers are increasingly demanding organic food. All the above factors make organic farming the future of farming.

Lecture-14

Organic farming : Promotional activities; role of government and NGO's- action plan-policy considerations.

PRODUCTION

- India ranks 9th in terms of cultivable land under organic certification.
- The certified area includes 10% cultivable area with 1.78million hectare and rest 90% (3.99 million hectare) is forest and wild area for collection of minor forest produces.
- The total area under organic certification is **5.77 million hectare** (2018-19)
- India produced around **1.70 million MT** of certified organic products
- Sugarcane, Cotton, Basmati rice, Pulses, Tea, Spices, Coffee, Oil Seeds, Fruits and their value added products.The production is not limited to the edible sector but also produces organic cotton fiber, functional food products etc.
- **Madhya Pradesh** has covered largest area under organic certification followed by Rajasthan, Maharastra and Uttar Pradesh.

EXPORTS

- India exported **132 products** last year **(2018-19)**.
- The organic agricultural products export realization was around **641 million Euros** from **4.58 lakh** tonnes of produces
- Organic products are exported to EU, US, Switzerland, Canada, South East Asian countries and South Africa.
- **Oil seeds (48%)** lead among the products exported followed by cereals & millets (10.%), Tea and coffee (9%), Dry fruits (9%) Spices and condiments(8%),

Ways to improve marketing:

Government officials involved in promotion of organic agriculture, agricultural scientists, extension officers and NGO's have revealed that multi-pronged initiatives at different levels would contribute towards enhancing greening of Indian agriculture.

From Policy perspective:

Experience elsewhere shows that government has to play a key role in developing organic agricultural production and enhancing marketing opportunities. Towards this there is a need to have policy framework to address greening agriculture in India. Policy change in favour of organic agriculture can make positive difference for changing the market condition in terms of encouraging production of bio-inputs, which in turn can propel changes in cropping pattern in favour of organic practices. Currently the attention given both in terms of policy framework and institutional dynamics towards organic agriculture is only marginal. Though Ministry of Commerce registers farmers wishing to convert to organic operations but

farmers are asked to seek technical assistance from the Ministry of Agriculture that is unavailable (Scialabba Nadia, 2000).

Involvement of government not just in standardization and accreditation procedures but also through proactive "support to inspection and certification and market-oriented services are necessary to provide equal opportunities. Otherwise, the export of certified organic products risks becoming that only large farmers, or highly organized groups of small holders, can afford" (Scialabba Nadia, 2000).

From Farmers Perspective:

- Existing subsidy structure enhances the use of chemical fertilizers and pesticides as agricultural inputs. Gradual phasing out subsidy on synthetic fertilizers / pesticides and initiating promotion of bio-inputs would contribute towards enhancing organic agriculture production in the country.
- Providing grant assistance in the form of financial assistance for converting the traditional farm in to organic farm and intending support to meet certification fees especially for the small and marginal farmers etc.
- Enlarging the scope of work of the agricultural extension services by focusing on organic agriculture through collaborative engagement with NGOs, who are actively working in the rural areas (Scialabba, Nadia, 2000).
- Special insurance scheme for organic farm that reduces the risk of farmer in case of failure of crop.
- Promote contract farming based on organic agriculture as it has been done in Madhya Pradesh.

From NGO's perspective:

Non-Government Organizations have being playing crucial role in promoting organic agricultural practices in the country. Change of agricultural practices is equivalent to changing the culture and mindset of farmers, which can only be achieved by a long drawn interface. NGOs have demonstrated capabilities to this effect. An illustrative example of public private partnership is the successful story of Spice Board's involvement of NGOs to enhance organic production of spices in Kerala, Tamilnadu, Andhra Pradesh and North Eastern states (Shenoy, 2003)

From Traders Perspective:

- Ministry of Commerce & Trade should have a dedicated cell for providing organic agricultural export market potentials in different parts of the world and the price premium that different products attract.
- Establishment of Special Organic Agriculture Trade Zone (OATZ) for domestic as well export market can help the traders in easily accessing the farm products as well as customers for organic products like agro- based food processors and retailers.

- In order to develop organic export networking potentials there is need for inventorizing the early exporters by providing special tax relaxation for traders/exporters those who have organic trading/export.
- In order to enhance the domestic organic agricultural market government need to make efforts towards promotion of organic food processing industries by providing subsidies and financial assistance as well as facilitates to market development. As the shelf life of some of the organic produce is shorter establishment of dedicated cold storage facilities could help in enhancing the prospects of exports.

From Green Input Producers Perspective:

- Popularization of existing scheme for promotion of bio-fertilizers and extension of these schemes for other bio-inputs.
- Assistance in marketing of bio-inputs thorough govt. network and also involving the network of co-operative societies at village level for distribution of these bio-inputs.
- Improving infrastructure facilities like roads, transportations, storage facilities etc would enhance forward and backward linkages in the organic products supply chain.
- Promotion of corporate research for organic agronomic practices, bio- control of diseases and pest, bio-fertilizers etc.

From Institutional Perspective:

There is urgent need for giving strategic attention to organic farming efforts. This requires inter ministerial coordination among various ministries at the Centre. Towards this forming a Steering Committee consisting of various Ministries at Central Government level would be a move in the right direction. Such as..

- Ministry of Agriculture,
- Ministry of Commerce & Trade,
- Ministry of Environment,
- Ministry of Science & Technology,
- Ministry of Finance etc.

Such committee would helps on increasing effectiveness of policies &programmes evolved towards promotion of organic agriculture.

At the state level some of the important institutions that require coordinated action plan include: Agriculture universities, State agriculture department, Private business organizations and NGOs

Each of these institutions can enhance their contribution towards greening agriculture by developing a detailed program of action. In the following an attempt has been made to briefly dwell action programs for some of these institutions. For instance:

Agriculture Universities can have action programs by reorienting their current 1) Educational activities 2) Research Agenda and 3) Extension services Programmes as given in the matrix below:

Education	**Research**	**AgricultureExtension**
From: • Traditional subjects and practices which are mainly focused on high productivity	***From:*** • Research that is focused on increasing productivity through high use of chemical fertilizer & pesticides. • Research focused on development of chemical based methods for controlling plant diseases and pests.	***From:*** • Information dissemination that is focused on conventional land use and cropping practices.
To: Incorporation of new subjects and syllabus with focus on sustainable and organic agriculture. Introduction to environmental impact of agriculture practices.	***To:*** Research that give adequate attention to alternative patterns of agriculture with emphasis on environmental benign and sustainable agriculture.	***To:*** • Global and local market opportunities for organic agriculture produces and price premium they carry. • Organic agronomic practices, organic control of diseases and pests. • Encourage use of Bio-fertilizers and Bio-pesticides.

To build green agricultural input market in India it is not sufficient to incentivize production but there is a need to focus efforts in generating market demand. Collaborative and concerted efforts between input producers, farmers, agriculture scientific community, government officials and traders at different level i.e. central government, state government, district and village level could give the required strategic attention to organic agriculture. There are examples from elsewhere which showcase these types of collaboration for organic products market development Such collaboration would be create avenues for marketing organic produce internally and expedite the necessary linkages for enhancing our presence in the international organic products market.

Some of the domestic markets:

- 24 lettered mantra
- Fab india
- Organic India
- Food world
- Dobdensfreen.newdelhi
- Parakhvitaran
- Shrishti
- Future foods
- ISKCON
- Green world

International markets:

- Basic super markets,germany
- Coop,Switzerland
- Migros,Switzerland
- Sinsbury.UK
- Spencers,UK

Lecture-15

Economic evaluation of organic production systems - cost - benefit analysis and comparison with conventional systems.

Introduction

Organic farming systems have attracted increasing attention over the last one decade because they are perceived to offer some solutions to the problems currently besetting the agricultural sector. Organic farming has the potential to provide benefits in terms of environmental protection, conservation of non-renewable resources and improved food quality. India is bestowed with lot of potential to produce all varieties of organic products due to its diverse agro-climatic regions. In several parts of the country, the inherited tradition of organic farming is an added advantage. This holds promise for the organic producers to tap the market which is growing steadily in the domestic market related to the export market. In India, the land under certification is around 1.5 million ha. But, there is considerable latent interest among farmers in conversion to organic farming. However, some farmers are reluctant to convert because of the perceived high costs and risks involved in organic farming. Despite the attention which has been paid to organic farming over the last few years, very little accessible information actually exists on the costs and returns of organic farming in India. The empirical evidences of efficiency analysis of organic and conventional farming systems are scarce or even absent.

So, the present paper focuses mainly on the issues like economics and efficiency of organic farming vs conventional farming in India. Four states namely Gujarat, Maharashtra, Punjab and U.P were purposively selected for the present study. Similarly, four major crops i.e., cotton, sugarcane, paddy and wheat were chosen for comparison. A model based non-parametric Data Envelopment Analysis (DEA) was used for analyzing the efficiency of the farming systems. The crop economics results showed a mixed response. Overall, it is concluded that the unit cost of production is lower in organic farming in case of cotton and sugarcane crops where as the same is lower in conventional farming for paddy and wheat crops. The DEA efficiency analysis conducted on different crops indicated that the efficiency levels are lower in organic farming when compared to conventional farming, relative to their production frontiers. The results conclude that there is ample scope for increasing the efficiency under organic farms.

Economics and Efficiency of organic farming vs conventional farming in India

Organic agriculture is developing rapidly; its share in agricultural land and farms continues to grow in many countries. At present almost 57.8 million ha are managed organically by more than 2.7 million producers (based on 2018 consolidated data). Australia holds 22.69 m ha 39.3 per cent of the world's organic land, followed by Argentina 3.07 m ha (5.3 per cent) and USA 2.03 m ha (3.5 per cent). Global demand for organic products remains robust, with sales increasing by over five billion US dollar per year.

India is bestowed with lot of potential to produce all varieties of organic products due to its agro-climatic regions. In several parts of the country, the inherited tradition of organic farming is an added advantage. This holds promise for the organic producers to tap

the market which is growing steadily in the domestic market related to the export market. Currently, India ranks 33rd in terms of total land under organic cultivation and 88th position for agriculture land under organic crops to total farming area in the World. The cultivated land under certification is around 2.8 million ha.

Market opportunities

The total volume of export during 2017-18 was **4.58 lakh MT.** The organic food export realization was around **INR 3453.48 crores** (**515.44 million USD).** Organic products are exported to USA, European Union, Canada, Switzerland, Australia, Israel, South Korea, Vietnam, New Zealand, Japan etc. In terms of export value realization **Oilseeds (47.6%)** lead among the products followed by Cereals and millets (10.4%), Plantation crop products such as Tea and Coffee (8.96%), Dry fruits (8.88%), Spices and condiments (7.76%) and others.

Tamil Nadu has 12,675 hectares of land certified under organic farming. At present, there are 5,000 farmers who practice certified organic farming. Many of the organic farmers of the state are, however, functioning without certification.

Economics of paddy (basmati) cultivation in Punjab

The per acre economics of paddy cultivation in Punjab state both under organic and conventional farming is presented in table 1. Most of the sample organic farmers in this region are following the concept of 'Natural farming' or 'Zero- budgeting'.

Table 1 Economics of Paddy cultivation in Punjab (Rs per acre)

	OF	CF
Land preparation	1265	1307
Seed cost	320	279
Sowing cost	1790	1815
Fertilizer cost	1955	1760
Inter cultivation/Weeding	1245	471
Plant protection cost	310	928
Irrigation cost	310	72
Harvesting cost	1180	771
Threshing cost	510	300
Marketing cost	440	115
Total cost of cultivation	9325	7818
Yield (Kg)	1335	1836
Price (Rs)	19.5	15
Fodder (Qtl)	11.2	12.5
Price (Rs)	100	94
Total revenue	27153	28715
Net returns	17828	20897
Cost of production (per Qtl)	701	427

The cost of production (variable) per quintal of paddy was Rs.701 under organic farming (OF) where as Rs.427 in conventional farming (CF). It is almost 64 per cent higher

in OF than CF. The average cost of cultivation of paddy in OF was Rs.9325 per acre while the same in CF was Rs.7818 per acre. The cost of cultivation was nearly 19 per cent higher in OF when compared to CF. Average yield per acre of paddy was 13.35 and 18.36 quintals respectively in OF and CF. The absolute difference between the yield levels was 5.01 quintal per acre. But, the unit price of paddy was higher (30 percent) in OF relative to CF. There was no significant unit price differences in fodder prices. The average net returns per acre of paddy cultivation were Rs. 17828 and Rs.20897 respectively in OF and CF. However, the differences between the gross returns per acre of these farming were marginal (Rs.95 only). Among different cost break-ups, the actual costs on weeding and harvesting operations were significantly higher in OF when compared to CF. It clearly indicates the more labor incentive nature of OF than CF. The relative costs on fertilizer application was higher in OF while the same on plant protection was higher in CF. The costs on the remaining cost items were more or less equal in both types. Since, many organic farmers are started practicing organic methods from last two or three years, it takes some more time to stabilize or increase the yields further under organic farming. The premium prices for paddy helping the organic farmers in Punjab to cover their higher costs to some extent.

Economics of wheat cultivation in Punjab

The comparison of crop economics of wheat cultivation between organic and conventional farming systems is presented in table 2. Most of sample organic farmers in the state were cultivating 'Bansi' (local) variety of Wheat. The cost of production per quintal was Rs.644 under OF. But, the same in case of CF was Rs.315. The cost of production per quintal of wheat was more than double in OF.

Table 2 Economics of Wheat cultivation in Punjab (Rs per acre)

	OF	CF
Land preparation	1050	1010
Seed cost	1240	1285
Sowing cost	275	261
Fertilizer cost	1163	1520
Inter cultivation/Weeding	1350	495
Plant protection cost	92	435
Irrigation cost	142	130
Harvesting cost	1300	840
Threshing cost	710	330
Marketing cost	217	130
Total cost of cultivation	7539	6436
Yield (Kg)	1170	2042
Price (Rs)	22.3	10.3
Fodder (Qtl)	11.4	16.4
Price (Rs)	233	227
Total revenue	28747	24755
Net returns	21208	18319
Cost of production (per Qtl)	644	315

It was due the lower (nearly half) yields under organic farming. But, the overall cost of cultivation per acre was slightly higher (17 per cent) in OF when compared to CF. The market price realization of per kg wheat was significantly higher in OF (117 percent). However, the gross returns per acre of wheat cultivation in Punjab were Rs.28747 and Rs.24755 respectively for OF and CF. The data indicates almost 16 per cent higher gross returns per acre of wheat under OF over CF. However, per acre net returns difference between OF and CF was Rs.2889. It clearly shows the high profitability of wheat cultivation under organic farming in Punjab. As the organic farmers gains more experience under OF, higher yields can be expected on par with CF. Among different crop operations, the higher costs under organic farming were observed in weeding, harvesting and threshing. Most of sample organic farmers are following manual harvesting and threshing practices for good quality of wheat grains and straw. Due to that the costs on labor per acre was higher under OF. The costs on fertilizers and plant protection chemicals were significantly higher under conventional farming. Overall, there is huge potential for domestic as well as export market for organic wheat from Northern states.

Economics of cotton cultivation in Punjab

The details of economics of organic cotton farming vis-à-vis conventional farming are summarized in table 3. Many of the sample organic farmers are cultivating desi variety of cotton where as conventional farmers are growing Bt cotton varieties. The cost of production per quintal of cotton under OF was Rs.662 while the same in case of CF was Rs.1112. The cost of production in OF was almost 40 per cent lower than CF.

Table 3 Economics of Cotton cultivation in Punjab (Rs per acre)

	OF	CF
Land preparation	967	850
Seed cost	125	1250
Sowing cost	150	125
Fertilizer cost	333	2250
Inter cultivation/Weeding	1332	650
Plant protection cost	33	4550
Irrigation cost	380	150
Harvesting cost	1967	2500
Marketing cost	140	130
Total cost of cultivation	5427	12455
Yield (Kg)	825	1125
Price (Rs)	28	28.5
Total revenue	23100	32063
Net returns	17673	19608
Cost of production (per Qtl)	662	1112

The average cost of cultivation per acre of cotton were Rs.5427 and Rs.12455 respectively under organic and conventional farming. There is a huge difference of Rs.7028 (66 %) between these farming types. The mean yield per acre of OF was 73 per cent of conventional farming. The unit price realization of cotton was almost same under both

production systems. Total gross returns per acre of organic farming were 72 per cent of conventional farming. But, in case of net returns per acre, the share has increased up to 90 per cent. The mean differences between the OF and CF net returns per acre was Rs.1935. It clearly demonstrates the high efficiency of organic cotton farming when compared to conventional farming in Punjab.

Among various cost components, inter cultivation /weeding and irrigation costs were higher in organic farming. But, the costs on seeds, fertilizers and plant protection chemicals were significantly higher in conventional farming. Actually, the major problem for organic cotton farming was lack of premium prices. Establishment of organic cotton export channels either by government or private organization would really enhance the incomes of the farmers in Punjab. The results clearly reveal that organic farmers can safely earn almost equal amount of net margins per acre as conventional farmers.

Economics of paddy cultivation in Uttar Pradesh

The costs and returns of paddy (basmati) cultivation both under organic and conventional farming types are presented in table 4. Most of the sample organic farmers are practicing the method of 'Natural farming' or Zero-budgeting concept in their farms. The most common basmati varieties growing in this region are Pusa – 1 and Pusa -1121.

Table 4 Economics of paddy cultivation in Uttar Pradesh (Rs per acre)

	OF	CF
Land preparation	3482	3444
Seed cost	501	511
Sowing cost	1136	1400
Fertilizer cost	1082	930
Inter cultivation/Weeding	622	375
Plant protection cost	350	521
Irrigation cost	2281	3300
Harvesting cost	2082	2214
Threshing cost	1555	1671
Marketing cost	140	80
Total cost of cultivation	13231	14446
Yield (Kg)	1518	1807
Price (Rs)	15.8	16.9
Fodder (Qtl)	10.5	11.8
Price (Rs)	70	93
Total revenue	24719	31636
Net returns	11488	17190
Cost of production (per Qtl)	870	803

The average cost of production per quintal of paddy (basmati) under organic farming was Rs.870 while the same in conventional farming was Rs.803. The cost of production per quintal under OF was 8 per cent higher than CF. The mean yield per acre in OF accounted for 84 per cent of the conventional farming yield. The average gross returns per acre of

conventional farming were nearly 28 per cent higher than organic farming. The average net returns per acre of paddy cultivation were Rs.11488 and Rs.17190 respectively for OF and CF. No premium prices were available for organic paddy in Uttar Pradesh. The yield levels under organic farming were lower (16%) than conventional farming. Among different cost items, weeding cost was significantly higher in organic farming. The costs on plant protection chemicals and irrigation were significantly higher in conventional farming. It clearly indicates that organic farming increased water- use-efficiency of the farm. Lack of premium prices as well as absence of export market channels limits the expansion of organic farming in the state.

Economics of sugarcane cultivation in Uttar Pradesh

The detailed break-up of the cost of cultivation of sugarcane in Uttar Pradesh state is presented in table 5. Most of the sample organic farmers were growing CoS 88230 variety of sugarcane while majority of conventional growers were using CoS 88230 or CoS 767 varieties.

Table 5 Economics of sugarcane cultivation in Uttar Pradesh (Rs per acre)

	OF	CF
Land preparation	2892	3533
Seed cost	4090	5065
Sowing cost	1514	1313
Fertilizer cost	1935	1904
Inter cultivation/Weeding	3113	3217
Plant protection cost	420	687
Irrigation cost	2750	2687
Harvesting cost	3495	2847
Marketing cost	2190	1846
Total cost of cultivation	22399	23099
Yield (Kg)	27364	24333
Price (Rs)	1.95	2.02
Total revenue	53360	49153
Net returns	30961	26054
Cost of production (per ton)	820	951

The cost of production of sugarcane per ton was Rs.820 under organic farming. But, the cost of production per ton was 16 per cent higher under conventional farming. The mean yield per acre was 12 per cent higher under organic farming.

The average cost of cultivation per acre of organic farming accounted for 97 per cent of the conventional farming cost. The gross returns per acre of OF was 9 per cent higher than CF. However in case of the net returns per acre, this value gone up to 19 per cent. The results conclude that the cultivation of sugarcane was more profitable under organic farming than conventional farming. Premium prices did not exist for organic sugarcane production in U.P. Creation or addition of premium prices would further increase the profitability of organic sugarcane production. Among different cost components, the costs were more or

less equal in both farming systems. One of the major benefits under organic sugarcane cultivation was the crop can thrive for more than three years without any yield loss. So, organic farmers can significantly reduce their seeds and sowing costs and reap more benefits. Production of organic jaggary or any other value addition measures would further boost organic sugarcane production in the state.

Economics of wheat cultivation in Uttar Pradesh

The economics of wheat cultivation under organic farming vis-à-vis conventional farming is summarized in table 6. Most of sample organic farmers were cultivating Bansi or 292 varieties of wheat. But, many conventional farmers were growing PBW-343 or WL-711 varieties. The cost of production of wheat per quintal was Rs.620 under organic farming. The same under conventional farming was slightly lower at Rs.609 per quintal. But, the average cost of cultivation per acre was lower in organic farming (8 per cent) when compared to conventional farming. The average yield levels were 1519 and 1682 kg respectively under OF and CF. However, the gross returns per acre was higher (15 per cent) in organic farming than conventional farming. This share has further gone up to 39 per cent in case of net returns per acre. The unit price realization was 28 per cent higher in OF. These results clearly demonstrate that the cultivation of wheat under organic farming is more profitable than conventional farming method. Between different cost components, the costs on weeding and inter culture was higher in organic farming. But, the costs on irrigation were higher under conventional farming. Further expansion in green or organic export market channels will yield higher net incomes per acre to farmers in U.P state.

Table 6 Economics of wheat cultivation in Uttar Pradesh (Rs per acre)

	OF	CF
Land preparation	2298	2571
Seed cost	1281	1034
Sowing cost	663	674
Fertilizer cost	981	1054
Inter cultivation/Weeding	656	432
Plant protection cost	85	214
Irrigation cost	994	1532
Harvesting cost	1510	1674
Threshing cost	844	879
Marketing cost	106	159
Total cost of cultivation	9418	10223
Yield (Kg)	1519	1682
Price (Rs)	13.4	10.5
Fodder (Qtl)	14	13.8
Price (Rs)	222	193
Total revenue	23463	20324
Net returns	14045	10101
Cost of production (per Qtl)	620	609

Economics of sugarcane cultivation in Maharashtra

The cost of cultivation of sugarcane per acre in Maharashtra between organic and convention farming is compared in table 7. Most of sample organic farmers are practicing the method of 'Natural Farming' or Zero-budgeting concept. The most popular varieties under organic and conventional farming systems are Co 86032 and CoC 671/ Co 8014 respectively. The cost of production of sugarcane per ton was Rs.589 in case of organic farming where as the same under conventional farming was Rs.745. The COP under OF accounted for 79 per cent of the same in CF. The mean cost of cultivation per acre was lower (20 per cent) under organic farming compared to conventional farming. The average yields were almost equal under both the farming systems. The gross returns per acre was slightly higher (8 per cent) under OF than CF. But, the difference has increased to 35 per cent in case of net returns per acre. The results clearly lend support to organic farming in Maharashtra than conventional farming. Most of the sample organic farmers are also adding value through organic jaggery production and syrup preparation. Among different break-up costs, the costs on sowing and irrigation were slightly higher under organic farming than conventional farming. But, the costs on fertilizer application and plant protection chemicals were significantly higher under conventional farming. Overall, development of organic output market channels will create lot of value addition to organic jaggery in Maharashtra.

Table 7 Economics of sugarcane cultivation in Maharashtra (Rs per acre)

	OF	CF
Land preparation	3675	4100
Seed cost	4825	5300
Sowing cost	1313	1120
Fertilizer cost	2344	5450
Inter cultivation/Weeding	3313	4300
Plant protection cost	275	1550
Irrigation cost	3588	3040
Harvesting cost	2375	2700
Marketing cost	838	760
Total cost of cultivation	22546	28320
Yield (Kg)	38375	38000
Price (Rs)	1.6	1.5
Total revenue	61400	57000
Net returns	38854	28680
Cost of production (per ton)	589	745

Economics of cotton cultivation in Gujarat

The detailed break-up of cost of cultivation of cotton in Gujarat is presented in table 8. Most of sample organic farmers were growing devraj variety while many of the conventional farmers cultivating Bt cotton or V-797 variety of cotton. Agrocel Industrial Limited at Rapar office is providing the technical service, inputs and buyback arrangements for organic farmers. The cost of production of cotton per quintal was Rs.784 in organic

farming. The cost of production was almost 8 per cent higher under conventional farming. The mean average yield per acre of organic farm accounted for 90 per cent of the same in conventional farm. The average costs of cultivation per acre were Rs.9906 and Rs.12088 respectively under OF and CF. The COC per acre was almost 22 per cent higher under conventional farming. The unit price realization under organic farming was 25 per cent higher when compared to conventional farming. The gross returns per acre were 13 per cent higher under organic farming than the conventional farming. But, in case of net returns per acre this gap has become wider (Rs.7187). Overall, the results conclude that the cultivation of cotton under organic farming is more profitable than conventional farming. Among different cost components, the costs on fertilizer and plant protection chemicals were significantly lower under organic farming than conventional farming. The organic farmers in the study region are enjoying the benefits of Agrocel Industries in form of quality inputs and zero marketing costs.

Table 8 Economics of cotton cultivation in Gujarat (Rs per acre)

	OF	CF
Land preparation	939	1600
Seed cost	206	281
Sowing cost	443	375
Fertilizer cost	1586	2675
Inter cultivation/Weeding	1946	1800
Plant protection cost	110	478
Irrigation cost	1161	1291
Harvesting cost	3515	3525
Threshing cost	0	0
Marketing cost	0	63
Other costs	0	0
Total cost of cultivation	9906	12088
Yield (Kg)	1263	1400
Price (Rs)	35	28
Total revenue	44205	39200
Net returns	34299	27112
Cost of production (per Qtl)	784	863

Economics of Chilli cultivation in Karnataka

The per acre average yield of chilli on organic farm (4.10 tonnes) was comparatively lower than that of inorganic farm (4.86 tonnes).The average per tonne market price of organic chilli (9830) was found to be higher than that of inorganic chilli (6300). The average transportation cost of organic chilli was 265 per tonne as against 285 per tonne of inorganic chilli. The commission charges paid during marketing of organically produced chilli was 644 per tonne, as against 432 per tonne for inorganically produced chilli because of more cost charged by the middlemen since there was high premium price for organic chilli since the organic farmers send their produce to the distant markets like Belgaum, Pune, Solhapur and Bangalore as they unable to get the premium price in the local market. The organically

produced chilli could fetch premium price in the market. The total marketing cost was 3726 for organic chilli and 3485 for inorganic chilli.

The return structure in chilli clearly revealed that the gross returns per acre were higher (40303) on organic farms compared to that of inorganic farms (30618) with a positive net return on both the categories of the farms. The net return on organic farm was 25484 and was 15044 on inorganic farms. Though the yield levels on organic farms were lower compared to inorganic farms, the net returns were higher because of the premium price received and lower cost of cultivation. The B:C was also higher on organic farms (2.72) compared to inorganic farms (1.97).

Table 9 Economics of chilli cultivation in Karnataka (Rs per acre)

Particulars	Organic farms	Inorganic farms
Seeds	372.00	491.00
Farm yard manure	1822.00	2261.00
Green manuring seeds	444.00	-
Vermicompost	1437.00	-
Biofertilizers	162.00	-
Biopesticide	745.00	-
Chemical fertilizers	-	1887.00
Plant protection chemicals	-	1110.00
Human labour	5114.00	5690.00
Bullock labour	2849.00	2081.00
Machine labour	905.00	1035.00
Interest on working capital	969.00	1019.00
Cost of cultivation (per acre)	14819.00	15574.00
Yield (tonne per acre)	4.10	4.86
Market price (per tonne)	9830.00	6300.00
Gross returns (per acre)	40303.00	30618.00
Net returns (per acre)	25484.00	15044.00
B:C	2.72	1.97

Economics of tomato cultivation in Maringá, Brazil

In organic tomato cultivation, it is possible to achieve 80% of the total productivity in terms of volume (kg) obtained in the conventional system and, over time, with increased soil fertility and improvements in management, the organic system could obtain a significant improvement in production.

Table 10 Economics of tomato cultivation in Maringá, Brazil ((US$/1,000m²)

Particulars	Organic farm	Conventional farm
Cost of cultivation	2,058.45	2,817.33
Production (kg/1,000m²)	5,515.30	9423
Cost per kg	1.44	0.78
Net revenues (US$/1,000m²)	7,954.37	7,374.41
Net Return	5,895.92	4,557.08

The difference in productivity between the two cultivation systems is minimized when it is considered that, in the organic system, there is a lower inflow of external inputs and the

selling price of the final products is higher.

Economics of brinjal cultivation in Maharashtra

It is reveled from the table that the productivity of inorganic brinjal was highest (179.05 q/ha) than that of organic brinjal of pargaon unit (118.20 q /ha) and Agricultural College, Pune unit (34.20 q/ha). However, the organic brinjal of pargaon organic unit was profitable with B: C ratio of 1.79 than that of inorganic brinjal with B: C ratio of 1.60. The productivity of inorganic brinjal was comparatively more than that of organic brinjal. However the organic farms were profitable than that of inorganic farms due to higher prices realized for output.

Table 11 Economics of brinjal cultivation in Maharashtra

Particulars	Organic farm	Conventional farm
Cost of cultivation	198078.72	200281.66
Yield (q/ha)	118.20	179.25
Rate (Rs. /q)	3000.00	1783.00
Grass return (Rs. /ha)	354600.00	319602.75
Net Return	156521.28	119321.09
B: C	1.79	1.60

Conclusion

Overall, the above findings summarizes that relatively organic farming is a production system which has little lower productivity, needs more labour and low energy inputs, and has a changing net income level relating with unit product selling prices. Though organic farming gives relatively lower yields in the initial years, its continuous practice will help to build up the soil fertility, there by to get increased yield in the later years. The difference in cost of cultivation between organic farming and inorganic farming is marginal as per the study. The organic inputs are mostly produced on the farms by the farmers themselves. Proper practicing of it will lead to higher net returns to the farmers because of the premium price the organic produce fetch. Hence farmers should be convinced by the extension workers about its economics to achieve its larger scale adoption.

The result from the different crops indicates that, the efficiency levels are lower in organic farming when compared to conventional farming, relative to their production, marketing and obtaining premium price frontiers. The results conclude that there is ample scope for increasing the efficiency under organic farms.

Lecture-16

Organic certification – procedures – certification agencies in India – labeling and accreditation processes

Certification:

Certification is a procedure by which a third party gives written assurance that a product, process or services is in conformity with certain standards.

Organic certification:

It means that the farm and the farmer's methods inspected by an organic certifying group to ensure that they comply with guidelines on organic farming. Each certifying group has a code of standards, which is available to interested people.Certification is the key to the National Organic Programme. The certification process focuses on the methods and materials used in production.

A third party- an organic certifying agent evaluates producers, processors and handlers to determine whether they conform to an established set of operating guidelines called organic standards. Those who conform are certified by the agent and allowed to use a logo, product statement or certificate to document their product as certified organic.

Purpose of certification

Organic certification addresses a growing worldwide demand for organic food. It is intended to assure quality andprevent fraud. For organic producers, certification identifies suppliers of products approved for use in certifiedoperations. For consumers, "certified organic" serves as a product assurance, similar to "low fat", "100% wholewheat", or "no artificial preservatives".Certification is essentially aimed at regulating and facilitating the sale of organic products to consumers. Individualcertification bodies have their own service marks, which can act as branding to consumers.Most certification bodiesoperate organic standards that meet the National government's minimum requirements.Some certificationbodiescertify to higher standards.

Types of certification

1. Foreign certification:

The inspection and certification of export orientedorganic projects in developing countries are certified by the certification bodies based in importing countries.The main disadvantage is that costs of certification are highdue to frequent plane trips and Western salaries.

2. Co-certification:

Local branch officers of Western Certification programmesconduct inspection along with local inspection staff who speak local languages andfamiliar with local conditions. However, inspection work is supervised by the HeadOffice. This reduces number of plane trips.

3. Indigenous certification:

Indigenous certification bodies can usually offercheaper inspection fees as less traveling is required and only local salaries have to becovered.They support the development of domestic markets. For export to the European Union (EU) market, a certificate from acertification body, which is accepted by EU competentauthorities is essential. Toregulate the export of certified organic products, the Director General of Foreign Trade, Government of India has issued a public notice according to which no certifiedorganic products may be exported unless they are certified by an inspection andcertifying agency duly accredited by one of the accreditation agencies designated by Government of India.

Certification process

To certify a farm, the farmer is typically required to engage in a number of new activities, in addition to normalfarming operations:

Study the organic standards, which cover in specific detail what is and is not allowed for every aspect offarming, including storage, transport and sale.

Compliance — farm facilities and production methods must comply with the standards, which mayinvolve modifying facilities, sourcing and changing suppliers, etc.

Documentation — extensive paperwork is required, detailing farm history and current set-up, and usuallyincluding results of soil and water tests.

Planning — a written annual production plan must be submitted, detailing everything from seed to sale:seed sources, field and crop locations, fertilization and pest control activities, harvest methods, storagelocations, etc.

Inspection — annual on-farm inspections are required, with a physical tour, examination of records, and anoral interview.

Fee — an annual inspection/certification fee (currently starting at $400–$2,000/year, in the US and Canada,depending on the agency and the size of the operation).

Record-keeping — written, day-to-day farming and marketing records, covering all activities, must beavailable for inspection at any time.

In addition, short-notice or surprise inspections can be made, and specific tests (e.g. soil, water, plant tissue) may berequested.

For first-time farm certification, the soil must meet basic requirements of being free from use of prohibitedsubstances (synthetic chemicals, etc) for a number of years. A conventional farm must adhere to organic standardsfor this period, often, two to three years. This is known as being in *transition*. Transitional crops are not consideredfully organic.

Certification for operations other than farms is similar. The focus is on ingredients and other inputs, and processingand handling conditions. A transport company would be required to detail the use and maintenance of its vehicles,storage facilities, containers, and so forth. A restaurant would have its premises inspected and its suppliers verifiedas certified organic

Steps in certification

1. **Identification of a suitable certifier:** The producer or farmer makes contactwith certifying agency. Certifying agency gives information on standards fee application, inspection, certification and appeal procedures.
2. **Submission of an application:** Submit application along with field history,farm map, record keeping system, etc.
3. **Review of application**
4. **On-farm inspection**
5. **Final review:** The outcome of the review may be a) approval for organic certification, b)request for additional information, c) Notification of non-compliance, d) Denial ofcertification

If certification is granted, the producer can begin marketing his or her productsas organic. The producer is free to use the seal of certifier.

Examples of non-compliance may include

1. In adequate records of manure application, equipment clearing on-farms whereconventional production is also done and compost preparation.
2. A farm that has had chemicals used on it and is in its 1st or 2nd year oftransition to organic production cannot be granted certification because theland must be free from prohibited pesticides and fertilizer for a minimum ofthree years
3. The contract indicating scope, obligation, inspection and certification, sanctionand appeals, duration, fee structure is executed. The costs of certification depends onsize of farm, type of

production system, group of farmers, location of unit, travel timeto reach the inspection site and costs for travel, food and accommodation duringinspection.

4. The farm has to undergo inspection at least once a year. Inspectors verify thatorganic practices such as long-term soil management, buffering between organicfarms and neighbouring conventional farms and record keeping are being followed. Processing inspections include review of the facilities of clearing and pest controlmethods, ingredient transportation and storage and record keeping and audit control. The inspector evaluates the performance of the farm activities with the help offarmers' statements and records and by viewing the fields, animals and farm buildings. He can take samples for laboratory testing and may conduct unannounced inspections. The inspector transmits his/her findings to the certification body as a written report.

5. Final review and certification The certification body compares the results of the inspection with therequirements of the organic standards. A certification committee decides whether certification may be granted or not and then the agency issues approval or denial ofcertificate. The certificate is given for current years harvest only and hence annualcertification is required. The operator can request for reconsideration of decisions ofdenial of certificate if has valid reason

Group certification

Majority of agriculture practitioners worldwide are small holders and are oftenlocated in remote areas with long travel times from one place to another. Furthermore,the overall revenue from their agricultural production is usually too small to allowaviable farm inspection by external inspection body for each farmer. Based on these observations, an idea was generated to develop Grower Group Certification (CGC) where group certification refers to the certification of a group ofproducers who are in close proximity to one another, whose farms are uniform inmost ways and who are organized under one management and marketing system(IFOAM)

According to NPOP, certification of an organized group of producers,processors and exports with similar farming and production systems and which are ingeographical proximity.

Internal control system(ICS)

As per IFOAM and NPOP definition, Internal Control System (ICS) is adocumented quality assurance system that allows the external certification bodies todelegate the inspection of individual group members to a body identified from withinthe operators of the group. This can be legally recognised farmers association,cooperatives, NGOs or exporters, project or a farmer group. This means, in practicethat a growers group basically controls all

farmers for compliance with organicproduction rules according to defined procedures. The organic certification body thenmainly evaluates whether the ICS is working well and efficiently. The ICS guards theintegrity of the organic quality of the products particularly in smallholder projects. It is a system in which all persons dealing with products (grower, buyers and storekeepers) are identified, registered, instructed on the requirement of organiccertification and where large number of farmers are to be inspected by a foreign certification body, the involved cost can be very high. In such cases, smallholdergroup certification can be done. It is done for defined groups (may be upto 900 to 1000 farmers) with similar farming and production system located in geographicalproximity. Only one application is required to certify the entire farms of such smallholder group and the certification and inspection fee is shared by each individualfarmer / operator. Contracted to ensure compliance. The activities of these persons are monitored byregular visits and documentary control. Besides this, the persons involved are madeaware of their common responsibilities for the products, which imply certain socialcontrol.

Developing countries are increasingly looking for ways to reduce certificationcosts and procedures in order to make certification more feasible for small farmers. ICS is an alternative scheme, which help group of small farmers to reduce costs andsimplify procedures of internal inspection and certification.

The external auditor can invariablyinsist on physical inspection of allindividual holdings extending up to and above four hectares. It may also be noted thatthe number of units above four hectares should not exceed 50 per cent of the total area under the group certification.

Category	**Land holding**
Marginal farmers	< 1 ha
Small farmers	1 to 1.99 ha
Semi-medium farmers	2 to 3.99 ha
Medium farmers	4 to 9.99 ha
Large farmers	> 10 ha

International inspection and certification agencies

1. Argencert - first Argentine agency accredited by IFOAM.

2. California Certified Organic Farmers(CCOF) -to promote and support organic agriculture in California and elsewhere.

3. International Federation of Organic Agriculture Movements (IFOAM) - coordinating the network of the organic movement around the world.

4. The Ecological Farming Association - promotes ecologically sound agriculture.

5. Organic FarmingResearch Foundation (OFRF) - Sponsors research related to orgamc farming practices,

6.Organic trade Association- representing the organic industry in Canada and the United States..

7. Community Alliance with Family Farmers - scale agriculture that cares for the land, sustains local economics, and promotes social justice.

8. Institutefor Marketecology (IMO) - one of the first and most renowned international agencies for inspection, certification and quality assurance of ecofriendly products.

9. SKAL - operates worldwide in Western and Eastern Europe, South America and Southern Asia.

10.ECOCERT INTERNATIONAL- an inspection and certification body accredited to verify the conformity of organic products against the organic regulations of Europe, Japan and the United States.

11. DEMETER - The Bio Dynamic Farming and Gardening Association is the certifier in New Zealand.

Certification Agencies in India

Ecocert (G) (Aurangabad), Skal (N) (Bangalore), IMO (S) (Bangalore), SGS (S) (Gurgaon), Naturland (G) (Gurgaon), Lacon (G) (Cochin), Indocert (S) (Cochin), APOF (Bangalore), ISCOP (Coimbatore), IRFT (Mumbai), Bioinspectra (S) (Cochin)

The Certification Process:

(a) All procedural steps in processing the application, until final certification

(b) Licensed Operators and their production be indicated throughout the certification process

(c) The procedures for extension and updating certification, including certification of individual products

(d) The certification decisions to be recorded and clearly communicated to thelicensed operator

(e) Reason certification is denied shall be clearly stated

(f) Authorized Inspection and Certification Agency shall be able to impose conditions and restrictions. Mechanisms for monitoring compliance with such conditions and restrictions shall be in place.

(g) The criteria for the acceptance of applicants, formerly certified by other certification system shall be documented.

(h) Processing of inspection reports and certification decision shall be done in a timely manner.

(i) Processing of any issue related to violations shall be done with highest priority.

Certification Records

1. Integrity of the System: The certification system shall be based on written agreements, with clear responsibilities of all parties involved in the chain of operations for production of a certified product.

2. Disciplinary measures: The Authorized Inspection and Certification Agencies shall have a documented range of disciplinary measures (sanctions) including measures to deal with minor infringements of the standards. A clear procedure for imposing such measures shall be in place. The disciplinary measures applied shall be effective.

3. Withdrawal of certification: Where an infringement that affects the organic integrity is found, the Authorized Inspection and Certification Agencies shall ensure that the indication of certification is removed from the entire lot of the production run which is affected by the infringement concerned. Where a violation is made by the Licensed Operator, the Authorized Inspection and Certification Agencies shall withdraw certification from the Licensed Operator for a specified period and inform about their decision to Agricultural Marketing Adviser

4. Marks and Certificates: The Authorized Inspection and Certification Agencies shall exercise proper control over the use of its Licenses, certificates and certification marks. Incorrect references to the certification system or misleading use of licenses, certificates or marks shall be dealt with by suitable disciplinary actions. The Authorized Inspection and Certification Agencies shall have documented procedures for withdrawal and cancellation of contracts, certificates and certification marks.

5. Certificate of registration / Scope Certificate: The Certificate of Registration shall include:

i) Name and address of the Licensed Operator

ii) Name and address of the Inspection and Certification Agency and authorization number

iii) Reference to the applicable standards

iv) Products or product categories

v) Date of issue

vi) Validity

6. Transaction certificate: The Authorized Inspection and Certification Agencies may issue transaction certificates only if the seller provides all the required details. The Authorized Inspection and Certification Agencies shall take reasonable measures to verify that the information provided is correct before issuing the certificate. The Authorized Inspection and Certification Agencies shall take adequate measures, including ensuring that certificates contain sufficient information, to prevent fraudulent usage. Copies of all export certificates and transaction certificates issued shall be sent to Agricultural Marketing Adviser for records and will also be stored in a manner that enables easy retrieval and inspection of information on each Licensed Operator.

7. Product certificate: The certifier on request of the Licensed Operator will issueproduct certificate to the buyer, at a cost to be borne either by the Licensed Operator or the buyer. Product certificate shall contain the following.

i) The name and description of the seller

ii) The name and description of the buyer

iii) The date of delivery of the product

iv) The date of issuing of certificate

v) A clear indication of the product, its quantity and wherever applicable, the quality and season thereof

vi) Lot numbers and other identification (marks) of the products

vii) Reference to invoice or bill of lading

viii) An indication of the Authorized Inspection and Certification Agency and the applicable standard

ix) A statement from the certifier that the product is produced according to the applicable standards

Mandatory checks during Inspection:-

For conversion to organic cultivation

(a) The conversion plan and the measures referred by the Licensed Operator are to be checked by the inspector before approval by the Authorized Inspection and Certification Agency. Approval must be confirmed every year after the initiation of the conversion plan.

(b) The inspector should check that the above conditions are met in the case of production of seeds, vegetative propagating material and transplants. For the production of perennial crop products (fruit growing, vines) the following should be checked for meeting the conditions for compliance:

i) Whether production area is a part of the conversion plan

ii) To check that whether the last part of the area in the farm would be converted to organic production in the shortest possible period (not to exceed a maximum of 5 years).

iii) Appropriate measures are taken by the Licensed Operator for the separation of the products obtained from organic and non-organic units.

(c) The conversion plan proposed by the Licensed Operator should be signed with an undertaking that he / she shall perform all the operations, shall comply with these rules.

Plants and plant products at production site

(a) Shall ensure that the Licensed Operator shall inform in advance the Authorized Inspection and Certification Agency regarding harvest of every crop in the farm.

(b) Shall also ensure that after completion of the harvesting process, the Licensed Operator shall inform the Inspection and Certification Agency about the total quantity harvested on the units concerned. Any distinguishing features observed for demarcation of organic produce from non-organic produce (such as quality, colour, average weight, etc.) shall also be informed.

(c) In case an Licensed Operator runs several production units in the same area, with non-organic crops or crop products, the inspector should also inspect the storage area of non-organic input products (such as fertilizers, plant protection products, seeds etc.).

(d) Whether the farms and storage units are separated from non-organic farms and non-organic storage units. Processing and packaging units of organic products can be considered as part of the production unit, when processing & packaging activities take place at the production site

(e) On suspicion of use of unauthorized products, samples for testing of products(Products as well as inputs used) must be taken.

For collection of wild products:

(a) When the Licensed Operator's activity is limited to only collection of wild products, the Inspection and Certification Agency must take all the precautionary measures to ensure compliance with the provisions under Organic Agricultural Produce Grading and Marking Rules.

(b) The inspector should countercheck to ensure that the guarantees given by the producer meet the provisions.

(c) Measures taken by the Licensed Operator for compliance with these rules must be checked and noted.

Grant of Certification

If the Authorized Inspection and Certification Agency, after a preliminary inquiry and thorough inspection is satisfied that the applicant having regard to requisite skill, resources, production, processing previous performance and antecedents relevant to the issuance of the License is fit to use the Certification Mark, the Authorized Inspection and Certification Agency shall grant a License in prescribed form authorizing the use of the Certification Mark in respect of the product or class of products manufactured by the applicant in respect of the process employed in any production, manufacture or work, subject to such terms and conditions as specified in these Rules. The Authorized Inspection and Certification Agency shall intimate the applicant about grant of License.

The Applicant shall be entitled to use the certification or grade designation markas provided under these Rules and restrict his use thereof to goods or services, which will meet the norms and standard specification of the products. The certification mark may be affixed to the products and/or used on packaging or promotional material or in the context of advertising activities.

In the event of a withdrawal of the right to use the aforesaid Mark the License shall be returned to the Authorized Inspection and Certification Agency. The right to use the grade designation mark expires at the same time without giving rise to any indemnification claim against NAB and/or Authorized Inspection and Certification Agency.

The Applicant is entitled to the aforesaid Mark but shall be answerable for the safety of their products themselves. They shall furnish proof of holding sufficient product liability insurance in respect thereof, if required by the Authorized Inspection and Certification Agency. No liability whatsoever will be accepted by the Authorized Inspection and Certification Agency or NAB.

A License shall be granted on prescribed form for a period of one year and a declaration by Licensed Operator shall be given on prescribed form. The Authorized Inspection and Certification Agency during the period of the validity of the License may alter by giving one month's notice to a Licensed Operator any terms and conditions subject to which the License has been granted. Where Authorized Inspection and Certification Agency, after a preliminary inquiry, is of the opinion that a License should not be granted, the Authorized Inspection and Certification Agency shall give a reasonable opportunity to the applicant of being heard, either in person or through a representative authorized by him on his behalf, and may take into consideration any fact or explanation urged on behalf of the

applicant before rejecting the application. A License shall expire at the end of the period for which it is granted

Particulars of all Licenses issued by Authorized Inspection and CertificationAgency under these regulations in connection with the use of the Certification/Grade Designation Mark shall be entered in a register which will be inspected and evaluated by NAB.

Product labeling

In some countries, organic standards are formulated and overseen by the government. The United States, the European Union, Canada and Japan have comprehensive organic legislation and theterm "organic" may be used only by certified producers. Being able to put the word "organic" on afoodproduct is a valuable marketing advantage in today's consumer market, but does not guaranteethe product is legitimately organic.Certification is intended to protect consumers from misuse ofthe term, and make buying organics easy. However, the organic labeling made possible by certification itself usuallyrequires explanation. In countries without organic laws, government guidelines may or may not exist, whilecertification is handled by non-profit organizations and private companies.

Products madeentirely with certified organic ingredients and methods can be labelled "100% organic". Products with 95% organic ingredients can use the word "organic". Both may also display organic seal. A third category, containing a minimum of 70% organic ingredients, can be labelled "made with organic ingredients". In addition,products may also display the logo of the certification body that approved them.Products made with less than 70% organic ingredients cannot advertise thisinformation to consumers and can only mention this fact in the product's ingredientstatement.

Internationally, equivalency negotiations are underway and some agreements are already in place to harmonizecertification between countries, facilitating international trade. There are also international certification bodies,including members of the International Federation of Organic Agriculture Movements (IFOAM) working onharmonization efforts. Where formal agreements do not exist between countries, organic product for export is oftencertified by agencies from the importing countries, who may establish permanent foreign offices for this purpose. In the US, the National Organic Program (NOP) was enacted as federal legislation in Oct. 2002. It restricts the useof the term "organic" to certified organic producers (excepting growers selling under $5,000 a year, who must stillcomply and submit to a records audit if requested, but do not have to formally apply). Certification is handled bystate, non-profit and private agencies that have been approved by the US Department of

Agriculture (USDA). In Japan, the Japanese Agricultural Standard (JAS) was fully implemented as law in April, 2001. This was revised in November 2005 and all JAS certifiers were required to be re-accredited by the Ministry of Agriculture

In India, APEDA regulates the certification of organic products as per National Standards for Organic Production.'The NPOP standards for production and accreditation system have been recognized by European Commission andSwitzerland as equivalent to their country standards. Similarly, USDA has recognized NPOP conformity assessmentprocedures of accreditation as equivalent to that of US.

With these recognitions, Indian organic products duly certified by the accredited certification bodies of India areaccepted by the importing countries." In March 2000, the Ministry of Commerce launched NPOP (NationalProgramme for Organic Production) design to establish national standards for organic products which could then besold under the logo India Organic. For proper implementation of NPOP, NAPP (National Accreditation Policy and Programme) has been formulated, with Accreditation Regulations announced in May 2001. These make itmandatory that all certification bodies whether internal or foreign operating in the country must be accredited by an Accreditation Agency. The regulations make provision for export, import and local trade of organic products. However, currently only the exports of organic products come under government regulations. Thus an agricultural product can only be exported as an organic product if it is certified by a certification body dulyaccredited by APEDA. Organic crop production, organic animal production, organic processing operations, forestryand wild products are the categories of products covered under accreditation.

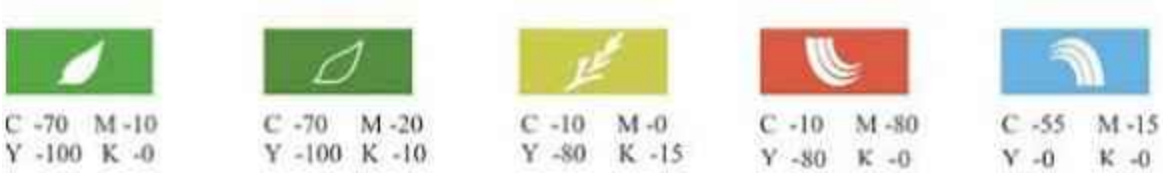
In China, the China Green Food Development Center awards two Standards: A and AA; while the former standarddoes permit some use of synthetic agricultural chemicals, the latter is more stringent.

Organic logo

A trademark – "India Organic" will be granted on the basis of compliance with the National Standards for Organic Production (NSOP). Communicating the genuineness as well as the origin of the product, this trademark is ownedby the Government of India. Only such exporters, manufacturers and processors whose products are duly certifiedby the accredited inspection and certification agencies, will be granted the license to use of the logo

which would begoverned by a set of regulations.

Specifications



Symbolizing the rhythm of cosmic and earth forces represented by the blue and brown waves of force and energy,'India Organic' logo celebrates the essence of nature. These forces work in harmony upon the earth's environmentand this rhythm is reinforced and supported by the green plant growth. The colours used have a special significancein the logo concept. The cosmic force in blue symbolizes universal purity. Richness of soil, nourished with naturalingredients in organic farming, is symbolized by the earth forces in golden brown. The plant in green uses the colourof nature and natural products untouched by chemicals. The blue background is symbolic of earth's environmentthat is congenial for life to thrive in and is also free of pollution and harmful chemicals. India Organic etched overthe surface authenticates the carrier as "Organic" and also establishes the Indian connection for all the carriers of themark. Beautifully synthesizing all the elements of our environment, the logo also communicates total adherence tothe National Organic Standards.

Certification issues

Organic certification is not without its critics. Some of the staunchest opponents of chemical-based farming andfactory farming practices also oppose formal certification. They see it as a way to drive independent organic farmersout of business and to undermine the quality of organic food. Other organizations such as the Organic Trade Association work within the organic community to foster awareness of legislative and other related issues, andenable the influence and participation of organic proponents.

Obstacle to small independents

Originally, in the 1960s through the 1980s, the organic food industry was composed of mainly small, independentfarmers, selling locally. Organic "certification" was a matter of trust, based on a direct relationship betweenfarmersand consumer. Critics view regulatory certification as a potential barrier to entry for small producers, by burdeningthem with increased costs, paperwork, and bureaucracy.

Manipulation of regulations

Critics of formal certification also fear an erosion of organic standards. Provided with a legal framework withinwhich to operate, lobbyists can push for amendments and exceptions favourable to large-scale production, resulting in"legally organic" products produced in ways similar to current conventional food. Combined with the fact thatorganic products are now sold predominantly through high volume distribution channels such as supermarkets, theconcern is that the market is evolving to favour the biggest producers, and this could result in the small organicfarmer being squeezed out

Manipulation of certification regulations as a way to mislead or outright dupe the public is a very real concern.Some examples are creating exceptions (allowing non-organic inputs to be used without loss of certification status)and creative interpretation of standards to meet the letter, but not the intention, of particular rules. For example, acomplaint filed with the USDA in February 2004 against Bayliss Ranch, a food ingredient producer and itscertifying agent, charged that tap water had been certified organic and advertised for use in a variety of water-basedbody care and food products, in order to label them "organic" under US law. Steam-distilled plant extracts,consisting mainly of tap water introduced during the distilling process, were certified organic and promoted as anorganic base that could then be used in a claim of organic content. The case was dismissed by the USDA, as theproducts had been actually used only in personal care products, over which the department at the time extended nolabeling control. The company subsequently adjusted its marketing by removing reference to use of the extracts infood products. Several months later, the USDA extended its organic labeling to personal care products; thiscomplaint has not been refilled.

In December 2005, the 2006 agricultural appropriations bill was passed with a rider allowing 38 syntheticingredients to be used in organic foods. Among the ingredients are food colorings, starches, sausage and hot-dogcasings, hops, fish oil, chipotle chili pepper, and gelatin. This allowed Anheuser-Busch in 2007 to have its Wild HopLager ‘certified organic’ even though hops were grown with chemical fertilizers and sprayed with pesticides."

Misrepresentation of the term *organic*

The word *organic* is central to the certification (and organic food marketing) process, and this is also questioned by some. Where organic laws exist, producers cannot use the term legally without certification. To bypass this legal requirement for certification, various alternative certification approaches, using currently undefined terms like "authentic" and "natural" instead of "organic", are emerging. In the US, motivated by the cost and legal requirements of certification (as of Oct. 2002), the private farmer-to-farmer association,

Certified Naturally Grown, offers a "non profit alternative eco-labelling program for small farms that grow using USDA Organic methods but are not a part of the USDA Certified Organic program." In the UK, the interests of smaller-scale growers who use 'natural' growing methods are represented by the Wholesome Food Association, which issues a symbol based largely on trust and peer-to-peer inspection. A related concern holds that certification is replacing consumer education, and this goes against the essential,holistic nature of organic farming. By reducing complex issues and regulations to a simple, convenient *certified organic* label, consumers may more easily ignore the principles and practices behind organics, leaving the definition of organic farming and organic food open to manipulation.

Accreditation

The act of accrediting or the state of being accredited, especially the granting of approval to an institution of learning by an official review board after the school has met specific requirements or A process of formal recognition of a school or institution attesting to the required ability and performance in an area of education, training, or practice or The act of granting credit or recognition (especially with respect to educational institution that maintains suitable standards.

Accreditation is a process in which certification of competency, authority, or credibility is presented. Organizations that issue credentials or certify third parties against official standards are themselves formally accredited by accreditation bodies (such as UKAS); hence they are sometimes known as "accredited certification bodies". The accreditation process ensures that their certification practices are acceptable, typically meaning that they are competent to test and certify third parties, behave ethically and employ suitable quality assurance. One example of accreditation is the accreditation of testing laboratories and certification specialists that are permitted to issue official certificates of compliance with established standards, such as physical, chemical, forensic, quality, and security standards.

In March 2000, the Ministry of Commerce launched NPOP (National Programme for Organic Production) design to establish National Standards for Organic Products, which could then be sold under the logo ‘India Organic’. For proper implementation of NPOP, NAPP (National Accreditation Policy and Programme) has been formulated with Accreditation regulation announced in May 2001. These make it mandatory that all certification bodies whether internal or foreign operating in the country must be accredited by

an Accreditation agency. An agricultural product can only be exported as organic product if acertification body duly accredited by APEDA certifies it.

Position of Accreditation in India

As per the National Programme for Organic Production (NPOP) an accreditation refers registration by the accreditation agency for certifying agency for certifying organic farms, products and processes as per the guidelines of the National Accreditation Policy and Programme for Organic Product. NPOP programme in context of Indian accreditation scenario, defined the function of accreditation agencies like

- Prescribe the package of practices for organic products in their respective schedule
- Undertake accreditation of inspection and certifying agencies who will conduct inspection and certify products as having been produced in accordance with NPOP
- Monitor inspection made by the accredited inspection agencies
- Lay down inspection procedures
- Advise the National Steering Committee on Organic Production
- Accept Accredited certification programmes if such programme confirm to National Standard
- Accreditation Agencies shall evolve accreditation criteria for inspection and/or certifying agencies and programme drawn up by such agencies for their respective area of operation and products
- Accreditation agencies shall prepare an operating manual to assist accredited agencies to abide by such a manual must contain appropriate directions, documentation formats and basic agency and farm records for monitoring and authentication of adherence to the organic production programme
- Eligible inspection and certification agencies implementing certification programmes will be identified by the Accreditation Agency

Accreditation agencies

1. APEDA
2. Coffee Board
3. Tea Board
4. Spice Board
5. Coconut Board
6. Cocoa and Cashew nut Development Board

In turn, these accreditation agencies have inspection and certification agencies. These agencies directly contact farmers, organic input producers, processors and marketing agencies.

Organic Agriculture Statistics (2018-19)

Area

Cultivated Area (Organic + In-conversion)	1938,220.79 ha
Wild Harvest Collection Area	1490,417.98 ha
Total Area(Cultivated + Wild Harvest)	34,28,638 ha **(3.4 m ha)**

Production

Farm Production	2607385.00 tons
Wild Harvest Production	37930.67 tons
Total Production	**2645315.67 tons(2.7 m t)**

Operators

Individual farm producers	2371 No.
ICS Groups	3488 (11.47 lakh farms)
Total Processors	1452
Total Trader	777
WH Operators	90 (Total collectors 29384)
Total Operators	**8301 No.**
Total Farmers	**11,49,371**

Lecture-17

Crop production standards- NPOP guidelines- Principles, recommendations and standards - Quality considerations- Assessment methods – Premium and export opportunities

Organic standards

The standards are indicated as directions of sustainability and should be looked upon at global level. The standards acknowledge measure of comparison of qualitative or quantitative value for degree or level of requirement of excellence and rules of production.

The organic standards were defined as minimum production practices and requirements, which must be followed strictly if the agricultural products to be labeled as organic. The standards may be of

1. Global standards : i. Mandatory Standards
   ii. Voluntary standards
2. Regional standards
3. National standards
4. Certification standards

1. Global standards

i. Mandatory standards

Abide by law and regulations and passed by government or regulatory bodies.

eg. CODEX standards – are set by world level organizations like FAO and WHO at global level. These are reference standards for any dispute or problems.

ii. Voluntary standards

These standards influence the organic farming at world level. Eg. IFOAM standard. The major advantages are

- It can influence the mandatory standard
- It includes new techniques and various types of innovations.
- It recognizes the traditional standards in a continuous manner.

2. Regional standards

- Developed by mandatory standards

  eg. EU regulations

3. National standards

Technical barriers of trade (TBT) specially provide guidelines to prepare national standards and stimulate nations to develop their own standards. In India, the Ministry of Commerce brought out the standard *viz*., National Standard for Organic Products (NSOP) on May 8th, 2000 as a regulation and not as a legislation.

National Programme for Organic Products (NPOP, 2000)

It provides a means for certification programme in our country to establish their equivalence to regulatory requirements worldwide in the organic production, processing and trade. The main aim is to develop National Standards to prepare specific standards, to institute a logo "Indian Organic", and to approve certification bodies.

4. Certification standards

Standards and regulations - 2009 witnessed several major developments in the field of standards and regulations. The new EU regulation on organic production came into force as well as the Canadian organic standard. Furthermore, the Australian domestic organic standard was implemented. Canada and the U.S. concluded the world's first fully reciprocal agreement between regulated organic systems, and the EU introduced procedures for approving certification bodies from outside the EU. It is expected that these developments will ease trade in organic products and foster the future growth of the sector. The number of countries with organic standards has increased to 73, and there are 16 countries that are in the process of drafting legislation. In 2009, FAO, IFOAM and UNCTAD started the Global Organic Market Access (GOMA) project. The aim of GOMA is to facilitate equivalence, harmonization and other types of cooperation in order to simplify the process for trade flow of products among the various organic guarantee systems.

There has been modest growth in the number of certification bodies. The total is 488, up from 481 in 2008. Most certification bodies are in the European Union, the United States, Japan, South Korea, China, Canada, and Brazil. A growing number of organic producers are certified through Participatory Guarantee Systems (PGS) across the world. PGS are locally focused quality assurance systems. It is estimated that around 10'000 small operators are involved in PGS world-wide. The leading countries with regards to PGS are located in the global South. Several organic standard setters have also developed draft standards for climate "add-ons" for organic certification, and it is expected that the use of carbon labeling by retailers will grow considerably in the future. It is developed by certifying bodies through committee meeting and each agency has its own standards. Organic standards were defined on the minimum production practices and handling requirements which must be followed for the resulting agricultural products to be labeled as organic.

There are many standards *viz.*, Soil Association Standards (SA), European Commission regulations (EU), CODEX Standards IFOAM Standards etc. Each standard has its own production practices and all have similarities and dissimilarities.

Requirements for organic certification

The requirements for organic certification are outlined in much detail in the legislation that is used as the basis for certification. Summarized briefly, the main requirements of Regulation for the producers of agricultural crops are:

1. Soil fertility has to be maintained by means of crop rotation, adapted cultivation techniques and nutrient cycles. Pest attacks need to be minimized by means of a healthy soil, natural enemies and adapted crop varieties. Only those farm inputs (fertilizers, pesticides, etc.) that are listed in the regulation may be used in organic farming.

2. Only certified organic seeds should be used.
3. All farm or processing activities needs to be carefully documented on every level, thus ensuring a full traceability of the product flow.
4. If there is also a conventional unit, organic unit has to be clearly separated from the inorganic unit and the same product may not be produced in both units. Conventional and organic products may not be mixed at any stage.
5. Farms that are converting to organic farming have to undergo 2 (annual crops) or 3 years (perennial crops) of transition period. After the first 12 months, the products can be marketed as "organic in conversion".
6. Organic products need to be labeled as "organic" or "organic in conversion" throughout the whole chain of harvest, transport, storage, processing and export.
7. Specific requirements apply for organic livestock and honey production.
8. Every farm, processor or exporter producing or handling organic produce needs to be inspected and certified once in a year by an accredited certification agency.

In simplified terms, the NOP regulations require:

For crop farms –

1. 3 years with no application of prohibited materials (no synthetic fertilizers, pesticides, or GMOs) prior to harvest of the first certified organic crop;
2. Implementation of an Organic System Plan, with proactive fertility systems;
3. conservation measures; environmentally sound manure, weed, disease, and pest management practices; and soil building crop rotation systems;
4. Use of natural inputs and/or approved synthetic substances on the National List;
5. No use of prohibited substances while certified;
6. No use of genetically engineered organisms, (GMOs) defined in the rule as ”excluded methods”;
7. No sewage sludge or irradiation;
8. Use of organic seeds, when commercially available;
9. Use of organic seedlings for annual crops;
10. Restrictions on use of raw manure and compost;
11. Maintenance of buffer zones, depending on risk of contamination; and
12. No residues of prohibited substances exceeding 5% of the EPA tolerance.

For livestock operations –

1. Implementation of an Organic Livestock Plan;
2. Mandatory outdoor access, when seasonally appropriate;
3. Access to pasture for ruminants;
4. No antibiotics, growth hormones, slaughter by-products, or GMOs;
5. 100% organic feed and approved feed supplements;
6. Sound animal husbandry and preventative health care;

For processing operations –

1. No commingling or contamination of organic products during processing;
2. Implementation of an Organic Handling Plan;
3. No use of GMOs or irradiation;
4. Proactive sanitation and facility pest management practices;
5. Use of organic agricultural ingredients in "organic" products, when commercially available; and
6. Use of approved label claims for "100% organic", "organic" (at least 95% organic ingredients), "Made with organic ingredients" (at least 70% organic ingredients), and proper use of the word "organic" in ingredient list (less than 70% organic ingredients).

Guidelines for organic production and processing:

Standards for organic production and processing established by the accreditation agencies for specific crops in accordance with the national standards for organic products.

Habitat:

The area in which a plant or animal species naturally exists.

Hazard analysis critical control point (haccp):

A systematic process that identifies food safety hazards, critical control points, critical Limits, corrective actions and documentation and integrates monitoring procedures to Ensure food safety.

Homeopathy:

Homeopathy is a system of medicine based on the principle of *"similia, similibus, Curentur* (let likes be treated by likes)".

Homeopathic treatment:

Treatment of disease based on administration of remedies prepared through successive Dilution of a substance that in larger amounts produces symptoms in healthy subjects Similar to those of the disease itself.

Ingredient:

Shall mean any substance, including a food additive, used in the manufacture or Preparation of a food and present in the final product although possibly in a modified Form. Those items, the use of which is prohibited in organic farming.

Input manufacturing:

Shall mean the manufacturing of organic production or processing inputs.

Inputs permitted:

Those items that can be used in organic farming.

Inputs restricted:

Those items that are allowed in organic farming, in a restricted manner, after a careful Assessment of contamination risk, natural imbalance and other factors arising out of their Use. Farmers should consult the certifying agency.

Inspection:

Shall include the site visit to verify that the performance of an operation is in accordance With the production or processing standards.

Inspection and certification agency:

Shall be the organization responsible for inspection and certification.

Inspection agency:

Shall mean the agency that performs inspection services as per the national accreditation Policy and programme.

Inspector:

Shall be the person appointed by the inspection and certification agency to undertake the inspection of an operator.

Internal control system:

A documented quality assurance system that allows the external certification body to Delegate the inspection of individual group members to a body identified from within the Operators of the group.

Internal review:

Shall mean an assessment of the objectives and performance of a programme by the Certification or the accreditation agency itself.

Irradiation:

High energy emissions from substances for the purpose of controlling microbial, Pathogens, parasites and pests in food, preserving the food or inhibiting physiological Processes such as sprouting or ripening.

Labelling:

Means any written, printed or graphic matter that is present on the label, accompanies the

Food, or is displayed near the food, including that for the purpose of promoting its sale or Disposal.

Licence:

Shall be the accreditation contract that grants a certifier the rights associated with its Accredited status in line with the national program for organic production.

Livestock:

Shall mean any domestic or domesticated animal including bovine (including buffalo and Bison), bovine, porcine, caprine, equine, poultry and bees raised for food or in the production of food. The products obtained by hunting or fishing of wild animals shall not be considered as part of this definition.

Marketing:

Means holding for sale or displaying for sale, offering for sale, selling, delivering or placing on the market in any other form.

Multiplication:

The growing of seed / stock / plant material to supply for future production

National accreditation body (nab):

Shall be the agency set up by the steering committee for national programme for organic production for accrediting inspection and certification agencies.

Natural fibre:

A filament of plant or animal origin.

Operator:

Shall mean an individual or a business enterprise practicing organic farming or organic processing.

Organic:

Refers to a particular farming system as described in these standards and not to the term used in chemistry.

Organic agriculture:

It is a system of farm design and management to create an eco-system, which can achieve sustainable productivity without the use of artificial external inputs such as chemical fertilizers and pesticides.

Organic production unit:

Shall mean a unit / holding or stock farm complying with the rules of NPOP regulations.

Organically-produced feeding stuffs / feed materials:

Shall mean feeding stuffs / feed material produced in accordance with the rules of production laid down in NPOP regulations.

Organic seeds and planting material:

Seed and planting material produced under certified organic system.

Package of practices:

Guidelines for organic production and processing established by the accreditation agencies for specific crops, specific to the region.

Parallel production:

Shall mean any production where the same unit is growing, breeding, handling or processing the same products both in a certified organic and a non-certified organic system. Similarly a situation with "organic" and "in conversion" production of the same product is also parallel production.

Part conversion:

Shall be when part of a conventional farm or unit has already been converted to organic production or processing and a part is in the process of conversion.

Plant protection product:

Shall mean any substance intended for preventing, destroying, attracting, repelling, or controlling any pest or disease including unwanted species of plants or animals during the production, storage, transport, distribution and processing of food, agricultural commodities, or animal feeds.

Preparation:

Shall mean the operations of slaughtering, processing, preserving and packaging of agricultural and animal products and also alterations made to the labeling concerning the presentation of the organic production method.

Processing aids:

A substance or material not consumed as a food ingredient by itself but used in the processing of raw materials, food or its ingredients to fulfil a certain technological purpose during treatment or processing and which may result in unintentional but unavoidable presence of residues or derivatives in the final product.

Quality system:

Documented procedures, which are established, implemented, and periodically audited to ensure that production, processing, handling, management, certification, accreditation and other systems meet the specified requirements and outcomes by following standardized protocols.

Raw materials:

All ingredients other than food additives.

Sanitize:

To adequately treat the produce or food-contact surfaces by a process that is effective in destroying or substantially reducing the numbers of vegetative cells of microorganisms of public health concern, and other undesirable microorganisms, but without adversely affecting the safety and quality of the product.

Split production:

Where only part of the farm or processing unit is certified as organic. The remainder of the property can be (a) non-organic, (b) in conversion or (c) organic but not certified. Also see parallel production.

Standards:

Shall mean the standards for national organic products established by the steering committee for national programme for organic production.

Surveillance:

The measures undertaken to provide monitoring of an operator's / certification body's compliance with the standards / criteria for meeting the certification / accreditation requirements.

Transaction / import certificate:

Document issued by a certification body declaring that the specified lot or consignment of goods is derived from production and / or processing system that has been certified.

Use of GMO and GMOs derivatives:

A plant, animal, microbe or their derivatives that are transformed through genetic engineering.

Veterinary drug:

Means any substance applied or administered to any food-producing animal, such as meat or milk-producing animals, poultry, fish or bees, whether used for therapeutic, prophylactic or diagnostic purposes or for modification of physiological functions or behaviour.

Market opportunities and challenges for Indian organic products

Certified Farms

By February 2002

Certified organic farms in India = 1426

Area=2775 hectares

During 2018 February

Certified organic farms in India = 835,000 (30%) to the world (2.7 million producers)

Area=1.5m ha (2.59 %) to the world area (57.8 m ha)

Major Products produced in India by organic farming

Type	Products
Commodity	Tea, Coffee, Rice, Wheat
Spices	Cardamom, Black pepper, White pepper, Ginger, Turmeric, Vanilla, Mustard, Tamarind, Clove, Cinnamon, Nutmeg, Mace, Chili
Pulses	Red Gram, Black Gram
Fruits	Mango, Banana, Pineapple, Passion fruit, Orange, Cashew nut, Walnut
Vegetables	Okra, Brinjal, Garlic, Onion, Tomato, Potato
Oil seeds	Sesame, Castor, Sunflower
Others	Cotton, Sugarcane, Herbal extracts

Organic Products Produced and Exported as per ranking

Rank	Product	Rank	Product
1	Tea	6	Coffee
2	Spices	7	Cashew
3	Fruits	8	Oil seeds
4	Vegetables	9	Wheat
5	Rice	10	Pulses

Products for which Indian Production has a comparative advantage

Product	States	Major Locations
Tea	Assam, West Bengal, Uttranchal	Darjeeling, Guwahati, Dehradun
Spices	Kerala, Tamil Nadu, Karnataka	Cochin, Coimbatore, Idduki, Coorg
Coffee	Kerala, Tamil Nadu, Karnataka	Coimbatore, Coorg, Wayanadu, Peeremade
Rice	Punjab, Haryana, Assam, Maharastra, Tamil Nadu	Amritsar, Jalandar, Darrang, Ratnagiri, Kanchipuram, Thiruvallur
Wheat	Punjab, Haryana, Uttar Pradesh	Ambala, Patiala, Bhatinda, Faridkot
Fruits	All India	Various Locations
Cotton	Maharashtra, Gujarat, Madhya Pradesh	Akola, Amravati, Amreli, Kheda, Indore
Vegetables	All India	Various Locations

Major Constraints affecting organic certification (as per ranking by Producers and Exporters)

1. Cost
2. Quality
3. Lengthy procedure
4. Availability of Services

5. International Validity
6. Complicated
7. Certification is Unnecessary

Frame Conditions of the Organic Market in India

Category	Present Problems
Govt. authorities	➢ So far low involvement/interest ➢ No subsidies provided to organic producers ➢ Lots of paper work (eg. Duty) ➢ No initiatives to educate farmers ➢ Delay on policy implementation ➢ Target only export market ➢ No marketing support
Certification Organizations	➢ High cost of certification ➢ No standardization in terms of regulations
NGOs	➢ Lack of funds ➢ Not completely linked up with certification agencies ➢ Lack of proper infrastructure ➢ Unorganized
Traders – domestic	➢ Low level of awareness ➢ Low domestic market ➢ Low price realization ➢ Lack of proper marketing network
Process industry	➢ Lack of organic raw material ➢ No standardization in terms of process ➢ No dependable supply of organic products ➢ Availability of upgraded processing technologies

Organization of organic producers	➢ Not organized ➢ Some do not provide any benefits to farmers associated ➢ Over expectations ➢ Lack of organic raw materials ➢ Lack of effort to target domestic market ➢ Lack of modern technology

Export Market

- Organic Tea is produced and Exported directly
- Other organic products are exported through exporters
- Countries of imports in the order of priority:
- Europe : Netherlands, United Kingdom, Germany, Belgium, Sweden, Switzerland, France, Italy, Spain
- Americas : USA, Canada
- Middle East : Saudi Arabia, UAE
- Asia : Japan, Singapore
- Australia
- Africa : South Africa

Indian organic exported products

Product	Players	Quantity (tons/yr)
Tea	Arya Tea Co. Ltd, Kolkata; Bombay Burmah Trading Corporation, Coimbatore; Chamong Tea Pvt Ltd., Kolkatta; Hindustan Lever Ltd, Nilgiri; Tea promoters India, Kolkata	3500
Coffee	Arogya Organic Coffee Cultivators, Chikmaglur Bombay Burmah Trading Corporation, Coimbatore; Peeremade Development Society, Idduki poabs; Organic Products Pvt. Ltd., Pallakad Yellikodge Estate, Chikmaglur	600

Spices	Accelerated Freeze Drying Co. Ltd., Cochin; Cochin Spices Ltd., Cochin; Lotus Spices Ltd., Cochin; Peeremade Development Society, Idduki; Unicorn Natural Products Ltd., Hyderabad	700
Rice	Indian Organic Food, Delhi; Ion Exchange Enviro Farms, Pune; Grewals Organic Agriculture Farms, Sirsa; Picric Ltd, Sonepat; Sunstar Overseas Ltd., Bahalgarh	3500
Wheat	Giraff International, Hissar; Grewals Organic Agriculture Farms, Sirsa; LT Overseas Ltd., Bahalgarh; Sunstar Overseas Ltd., Bahalgarh	1400
Pulses	Grewals Organic Agriculture Farms,Sirsa; Ion Exchange Enviro farms, Pune; Pratibha Syntex Ltd., Indore	400
Oilseeds	Enfield Agrobase Pvt Ltd., Chennai; Grewals Organic Agriculture Farms, Sirsa; Ion Exchange Enviro Farms, Pune	100
Fruits & Vegetables	Grewals Organic Agriculture Farms, Sirsa; Mahesh Agri Exim Pvt Ltd., Surat; Ion Exchange Enviro Farms, Pune; IQF Foods Ltd., Banglore; Namadharis Fresh, Banglore; Picric Ltd., Sonepat	2500
Cashew nut	Narayan Ganesh Prabhu Zantye & Co. Goa, Trading Organic Association, Goa	375
Cotton, Medical herbs & extracts, Aloe vera,	Pratibha Syntex Ltd., Indore; Maikaal bioRe Pvt. Ltd., Mhow; Rohini Herbal, Chennai; Khadinge Plants and Herbas Pvt. Ltd;	Cotton 1200, Herbs 250

Country and imports

- Germany and Netherlands – Spices, Fruits, Tea and Nuts
- Netherlands – Sesame
- Switzerland – Tea, Cotton and Spices

Demand for organic products in selected markets (t / year)

- Banana – 6410
- Wheat – 1000
- Soybean – 1000
- Pineapple – 900
- Mango – 650
- Ginger – 280
- Hazel Nut – 250
- Oranges – 175
- Spices – 150
- Coconut -100

Price premiums obtained in the selected export markets

Product	Price premium %	Product	Price premium %
Tea	47	Mango	25
Rice	44	Pineapple	28
Protein grains	44	Bananas	31
Spices	30	Nuts	40
Vanilla	53		

Obstacles for the low import of Indian organic products

Aspect	Obstacle
Price	- Too high expectations
Quality	- Low consistency, Contamination
Availability	- Reliability of Exporters
Logistics	- Slow shipment, Import restrictions in other countries
Certification	- Lack of national certification and accreditation
Export authorities	- Time consuming, Complicated paper work
Information	- Lack of information on certification
Administration	- Inconvenient modus of payment
Customer service	- "Poor" from Indian traders after sales

---------- GOOD LUCK ----------


VERMICOMPOST
GREEN LEAF MANURES
CROP ROTATION
ORGANIC FARMING
MANURES
BIOLOGICAL MANAGEMENT
BIOFERTILIZERS
ANIMAL HUSBANDRY


India Organic
एपीडा
APEDA
INDIAN SOCIETY FOR CERTIFICATION OF ORGANIC PRODUCTS
ORGANIC
ISCOP
राष्ट्रीय जैविक खेती परियोजना
NATIONAL PROJECT ON ORGANIC FARMING
TAMILNADU ORGANIC


USDA
ORGANIC
India Organic
ORGANIC
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